

THE UNIVERSITY OF SYDNEY

ELEC5620

Model Based Software Engineering

Stage 1 Report

AI-Powered One-Person Beauty Commerce and Content Company

Mon 10-12 Group 6

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ELEC5620 Stage 1 Report

Requirements Architecture and System Modelling

AI-Powered One-Person Beauty Commerce and Content Company

No.	Registered name	SID	Owned capability
1	Danyang Wang	redacted	BA Beauty Advisor
2	Siyuan Lei	redacted	CS Customer Service and Order
3	Zhengdu Wei	redacted	MC Marketing Content
4	Jiaxuan Ren	redacted	AP Approval and Governance
5	Guiyu Liu	redacted	CO Coordination and Recovery

Design status: PROPOSED. Formal requirement approval, affected-owner acceptance and Stage 2 implementation/testing remain separate. Runtime values are synthetic; acceptance cases are planned, not executed.

1 Executive Summary

The proposed One-Person AI Company modernises a cosmetics/skincare WeChat commerce prototype. One Human Founder / Operator supervises three AI business roles: Beauty Advisor (BA), Customer Service and Order Agent (CS), and Marketing Content Agent (MC). They provide grounded recommendations, bounded after-sales handling and reviewable promotional-video drafts. Approval and Governance (AP) and Coordination and Recovery (CO) are shared controls.

Five requirement/use-case chains connect features, structure, runtime and behaviour. Deterministic evidence, authority and persistence checks constrain large language model (LLM) proposals; clarification, failure, uncertain execution and pending review are explicit outcomes. The proof of concept (PoC) uses controlled mocks and excludes production settlement and autonomous public posting.

2 Project Background and Scope

2.1 Problem and meaningful improvement

A small beauty retailer needs product advice, routine after-sales support and marketing material under one accountable operator. The existing mini-program supplies feasibility evidence, but does not establish requirements or demonstrate the proposed agent behaviour.

The improvement combines specialised agents, traceable hand-offs and recovery. Agents interpret requests and generate explanations/drafts; controlled services enforce evidence, authority and human checkpoints. This coordinated workflow provides the meaningful improvement over the legacy shop.

2.2 Scope and demonstration boundary

In scope	Excluded from this proof of concept
Clarification, authorised product/stock retrieval and explainable recommendations	Medical diagnosis, treatment advice and virtual try-on platform
Order enquiry and policy-supported return/exchange/refund-request handling	Production payment or bank settlement; complete logistics platform
Campaign brief, script, storyboard and draft promotional media with founder review	Autonomous public posting; training a foundation/video model
Traceable multi-agent hand-offs, approval, uncertainty and safe recovery	Modelling every legacy administrator page or commerce feature

Demonstrations cover consultation, after-sales handling and marketing drafts. GR-S-01 combines existing BA/CS/AP/CO goals. Labelled controlled adapters may replace LLM, media and commerce services; live availability, policy thresholds and production security are not assumed.

3 Stakeholders and Responsibilities

Role or capability	Goal and authority
Customer	Provides consultation/after-sales intent, clarifies constraints and confirms a current proposed action. Receives grounded advice, recorded outcomes and remaining obligations.
Human Founder / Operator	Owns company accountability; initiates campaigns/coordination and makes exact-scope consequential decisions. Human approval cannot create missing stock, policy eligibility or clinical authority.
AI Beauty Advisor	Clarifies beauty needs and explains authorised available products. May write consultation evidence; cannot create orders, change prices, settle payments or diagnose.
AI Customer Service and Order Agent	Verifies order association and policy evidence, proposes supported resolutions and executes only permitted PoC service actions after current confirmation and required approval.
AI Marketing Content Agent	Interprets authorised campaign context, prepares script/storyboard/media drafts and seeks founder review. No autonomous external publication or unsupported factual claims.
Approval and Governance	Controls risk-based human review, committed decisions, audit and delivery. It never executes the requesting role's business action.
Coordination and Recovery	Transfers authorised minimal context, records work/hand-offs and manages safe recovery. It adds no authority to a receiving role and is not an AI business role.

4 Requirements Specification

4.1 Agreement and common requirements

These proposed common requirements apply to all five chains. Agreement is a requirement classification, not evidence of formal approval.

ID	Common requirement	Acceptance basis
GR-C-01	One accountable Human Founder / Operator and three distinct AI business roles; AP/CO remain controls.	Inspect role, feature and composite models; GR-T-01.
GR-C-02	All factual advice/content uses authorised evidence and identifies missing, conflicting or unsupported data.	BA/CS/MC evidence and invalid-output scenarios.
GR-C-03	Keep role authority, Customer confirmation, human approval and successful execution distinct.	AP-T-02/03/07; CS-T-05; GR-T-02.
GR-C-04	Transfer only authorised decision-relevant context with origin/revision correlation.	Payload inspection and CO-T-01/02; GR-T-02.
GR-C-05	Expose failure and uncertain outcomes; reconcile before potentially duplicate execution.	CS-T-06, CO-T-06/08 and GR-T-03.
GR-C-06	Mocks are labelled; optional features do not control mandatory success; no production settlement or autonomous posting.	Optional-OFF tests and inspection of service calls.

4.2 Mandatory and optional capability catalogue

Mandatory ID	Owned goal	Core features	Optional feature
BA-M-01	Explainable personalised product recommendation	BA-F-01 / BA-F-02	BA-F-03
CS-M-01	Routine return, exchange or refund request	CS-F-01 / CS-F-02	CS-F-03
MC-M-01	Approval-controlled promotional-video draft	MC-F-01 / MC-F-02	MC-F-03
AP-M-01	Founder control of consequential agent actions	AP-F-01 / AP-F-02	AP-F-03
CO-M-01	Traceable cross-agent hand-offs and recovery	CO-F-01 / CO-F-02	CO-F-03

Each chain has two core features and one optional feature. All five optional features may be OFF while core success and safe incomplete outcomes remain usable. Revision, rejection and safe recovery remain core obligations.

4.3 Nonfunctional requirements

ID	Requirement and verification
BA-NFR-01	Evidence integrity: Each displayed recommendation retains product ID, catalogue reference/version, inventory observation and addressed Customer constraint. Verify every factual claim against the saved references.
BA-NFR-02	Least authority: BA has read-only catalogue/inventory access, with no order/payment/refund/publication capability. Test prohibited requests for zero corresponding writes; consultation/review records remain permitted.

ID	Requirement and verification
BA-NFR-03	Failure transparency: Failed/missing/conflicting required service results produce explicit incomplete outcomes, with no invented facts or available-stock claim. Inject each failure; inspect response and saved outcome.
BA-NFR-04	Data minimisation: Forward only consultation-relevant Customer input to LLM/review; exclude unrelated profile fields and credentials. Inspect allowlisted payloads. Retention, production access and performance targets remain unagreed.
CS-NFR-01	Evidence integrity: Saved eligibility/resolution records retain request, authorised order and applicable policy references. Verify against these sources.
CS-NFR-02	Least authority/privacy: Failed identity/order verification blocks protected disclosure. LLM/review receive only required permitted context. Inspect payloads and absence of prohibited calls.
CS-NFR-03	Failure/duplicate protection: Expose uncertain writes; reconcile by actionRef before another execution. Inject post-write timeout; inspect traces for duplicates.
CS-NFR-04	Audit/optional independence: Distinguish confirmed execution from evidence-save failure. Inspect state/result with notification ON, OFF and failed; notification cannot reverse the core result.
MC-NFR-01	Claim integrity: Every factual claim in script, storyboard and content draft traces to authorised product information/source references. Verify generated claims against recorded evidence.
MC-NFR-02	Authority separation: LLM/media generation and checks cannot establish approval or publication readiness. Exercise successful generation without a valid Founder decision through AP; the draft remains unapproved.
MC-NFR-03	Failure transparency: Explicitly represent missing information, unsupported content, media unavailability, failed persistence and pending/failed review. Inject each; inspect state and visible outcome for no false completion.
MC-NFR-04	Traceability: Retain campaign identity, authorised sources, draft revision, generation and review outcomes. Inspect stored evidence to reconstruct the source/revision/decision path.
CO-NFR-01	Traceability: Hand-offs and significant status changes retain the work request, participating roles and recorded outcome.
CO-NFR-02	Authority: Coordination cannot expand a business role's authority; restricted actions remain blocked until valid Founder approval is recorded.
CO-NFR-03	Transparency: Explicitly represent failed, unavailable and uncertain coordination; never report these as completion.
CO-NFR-04	Recovery: Avoid uncontrolled duplicate execution; reconcile uncertain prior outcomes before a potentially duplicative retry.
AP-NFR-01	Decision traceability: Retain origin/request, exact action/context revision, evidence, authorised decision maker, decision, reason and recording outcome. Retention/access remain unagreed; no production tamper-proof guarantee.
AP-NFR-02	Authority: Invalid/pending/absent/rejected/stale/unrecorded decisions grant no restricted authority. Unknown risk blocks action for clarification/review.
AP-NFR-03	Transparency: Distinguish receipt, notification, recording and delivery failures. Failed delivery preserves a committed decision; failed recording remains unapproved.
AP-NFR-04	Duplicate delivery: Reuse committed decisionId and exact scope. Receivers reject duplicate/stale correlation; repeated approval delivery cannot repeat business execution.

Security/access, retention, stock freshness, latency, review expiry and external reliability values need agreement. Explicit fixtures and failure injection replace unagreed production thresholds; Section 13 supplies observable checks.

4.4 Individual ad hoc requirements

4.4.1 BA-M-01 Danyang Wang

Owner Danyang Wang redacted . Mandatory; priority Must. Status PROPOSED.

Informal requirement The system shall provide the Customer with an explainable product recommendation that reflects their stated beauty needs and constraints and is supported by authorised product and availability evidence.

Rationale Routine consultation should help the Customer make an informed choice while reducing the Human Founder's need to handle every enquiry personally.

Stakeholder need Customer product advice and scalable routine consultation. Related use case BA-UC-01; feature definitions are in Section 5.2.

Criterion	Observable acceptance condition
BA-AC-01	Given a request missing information needed to distinguish suitable products, ask a focused clarification question and do not silently infer the missing constraint.
BA-AC-02	Given sufficient information and matching products, present at least one authorised, available product with a reason linked to the Customer's input and catalogue evidence.
BA-AC-03	Given no suitable product, unavailable stock or unknown availability, state that no reliable available match can be offered; never label an unknown or unavailable item immediately purchasable.
BA-AC-04	Given an unsupported generated claim or conflicting evidence, withhold that claim, explain the limitation and offer clarification or human review where appropriate.
BA-AC-05	Given a medical-diagnosis or treatment request, stop that part of the consultation and state the scope limit. Founder review cannot authorise medical advice.
BA-AC-06	Given cancellation, service failure or human review pending, expose that outcome and retain a traceable outcome record without executing a purchase or claiming review is complete.

Detailed refinements BA-FR-01 clarification; BA-FR-02 catalogue/stock retrieval; BA-FR-03 validation/explanation; BA-FR-04 incomplete outcomes. Optional BA-O-01 adds verified alternatives. BA-T-01–07 cover these refinements.

4.4.2 CS-M-01 Siyuan Lei

Owner Siyuan Lei redacted . Mandatory; priority Must. Status PROPOSED.

Informal requirement The system shall process a Customer's return, exchange or refund request against authorised order evidence and applicable after-sales policy, providing a recorded resolution or a traceable escalation when resolution is unsupported or outside the agent's authority.

Rationale Routine after-sales handling should reduce the Human Founder / Operator's workload while preserving Customer visibility and human control over disputed or high-impact decisions.

Stakeholder need Supported after-sales outcomes with bounded founder workload. Related use case CS-UC-01; feature definitions are in Section 5.2.

Criterion	Observable acceptance condition
CS-AC-01	Given an identity/order mismatch, withhold protected order details and do not execute a resolution; request correction or offer an authorised human-review path.
CS-AC-02	Sufficient authorised order/policy evidence produces referenced eligibility and relevant order facts. Missing/conflicting policy leaves an unresolved outcome; invent no window, threshold or rule.

Criterion	Observable acceptance condition
CS-AC-03	With routine policy authority and Customer confirmation, execute only the permitted action, save its service-confirmed outcome and explain remaining steps. Label simulation; acknowledgement is not settlement/physical completion.
CS-AC-04	Given an unavailable or unknown replacement, do not confirm a successful exchange; explain the constraint and request Customer selection of a policy-supported alternative or escalate.
CS-AC-05	Disputed, exceptional, high-risk, financially significant or approval-controlled actions need a valid recorded human decision. Absent approval, rejection and review-service failure remain distinct.
CS-AC-06	Timeout/uncertain writes require visible uncertainty and outcome lookup before duplicative retry. Failed evidence saving cannot be reported as fully recorded success.
CS-AC-07	With optional status notifications disabled or delivery failing, the core request remains accessible through its recorded status; notification failure does not reverse a confirmed resolution.

4.4.3 MC-M-01 Zhengdu Wei

Owner Zhengdu Wei redacted . Mandatory; priority Must. Status PROPOSED.

Informal requirement The system shall enable the AI Marketing Content Agent to produce a promotional-video draft from an authorised campaign brief and product information, while ensuring that the resulting promotional content is reviewed by the Human Founder / Operator before any external publication.

Rationale Marketing-content preparation can require substantial manual effort from the Human Founder / Operator. Automating routine campaign interpretation, script and storyboard preparation, and draft-media generation can reduce this workload while preserving human control over externally published promotional content.

Stakeholder need Scalable content preparation with human publication control. Related use case MC-UC-01; feature definitions are in Section 5.2.

Criterion	Observable acceptance condition
MC-AC-01	With sufficient authorised campaign/product input, save the brief and produce script/storyboard linked to authorised evidence.
MC-AC-02	Missing/conflicting/unsupported information requires clarification, available authorised retrieval or an incomplete outcome; invent no claims.
MC-AC-03	An available media service may generate draft media from validated script/storyboard; record its result.
MC-AC-04	Expose media unavailability/failure. Clearly label any permitted PoC mock/fallback; never call it real external generation success.
MC-AC-05	Failed content/quality checks block approval-ready completion; record the issue for revision.
MC-AC-06	Save Founder rejection/revision decisions and route to the appropriate revision path; do not mark approved.
MC-AC-07	No external publication or publication-approved claim without recorded valid Founder approval; this PoC remains draft-only.

4.4.4 AP-M-01 Jiaxuan Ren

Owner Jiaxuan Ren redacted . Mandatory; priority Must. Status PROPOSED.

Informal requirement The system shall require a recorded, valid Human Founder / Operator decision before a requesting role executes an agent action classified as high-impact, externally

published, financially significant or disputed. Review acceptance is not approval. A decision applies only to its recorded action/context revision and cannot override other business eligibility checks.

Rationale The founder must control consequential actions while the three business agents handle bounded work. AP manages approval requests, decisions and audit/delivery; it does not execute exchanges, settle payments or publish content. Risk categories/thresholds and decision expiry are proposed policies requiring human agreement; the PoC may use declared fixtures.

Stakeholder need Founder control and traceable consequential decisions. Related use case AP-UC-01; feature definitions are in Section 5.2.

Criterion	Observable acceptance condition
AP-AC-01	A valid restricted action creates a traceable PENDING approval request and returns a request receipt only after request persistence; missing/unauthorised context or uncertain risk blocks the action.
AP-AC-02	Pending, missing, invalid, rejected or unrecorded approval never enables a restricted action; show the actual review state.
AP-AC-03	An authorised Founder decision is correlated to the exact request, action and context revision, and is committed with the request state and audit record before delivery.
AP-AC-04	Rejection stops the proposed restricted action; REQUEST_REVISION returns it for correction. A changed action/revision requires fresh review and retains the prior history.
AP-AC-05	Recording failure leaves approval unavailable and permits storage-only recovery of the same decision; it cannot cause business dispatch.
AP-AC-06	Notification/delivery failure is visible. Recovery queries or redelivers the same committed decision; duplicate/stale events cause no repeated action.
AP-AC-07	Approval cannot create stock, policy eligibility, Customer confirmation, publication authority beyond the agreed scope or a successful execution result.
AP-AC-08	Disabling AP-F-03 assisted revision/resubmission leaves all core review decisions usable. A manually corrected new request still needs fresh review.

4.4.5 CO-M-01 Guiyu Liu

Owner Guiyu Liu redacted . Mandatory; priority Must. Status PROPOSED.

Informal requirement The system shall coordinate traceable task hand-offs among authorised AI business roles when a workflow requires work to pass between roles, while preserving required context, workflow status, authority boundaries and safe failure handling.

Rationale Cross-agent workflows require a controlled mechanism for transferring work between AI business roles. Coordination should reduce the Human Founder / Operator's need to manually relay routine information while ensuring that hand-offs, failures and recovery outcomes remain observable and traceable.

Stakeholder need Collaboration without lost context, authority or workflow visibility. Related use case CO-UC-01; feature definitions are in Section 5.2.

Criterion	Observable acceptance condition
CO-AC-01	Given a valid cross-agent work request with sufficient authorised context, the system shall identify the receiving AI business role, transfer the required context and record the hand-off and resulting workflow status.
CO-AC-02	Given missing or invalid decision-relevant hand-off context, the system shall not silently infer the missing information or represent the hand-off as successfully completed; it shall request clarification or record a safe incomplete outcome.
CO-AC-03	Given an unavailable receiving AI business role, the system shall record the failed or unavailable hand-off and follow an applicable recovery, escalation or cancellation path rather than report successful completion.

Criterion	Observable acceptance condition
CO-AC-04	Given a supporting tool or service failure, the system shall expose and record the failure or uncertainty and shall not treat the dependent task as successfully completed.
CO-AC-05	Given a retry-eligible failure when controlled automatic retry is enabled, the system may retry the failed operation in a bounded and traceable manner. Exhausted or unavailable retry shall lead to safe escalation or cancellation.
CO-AC-06	Given an action requiring Human Founder / Operator approval, the restricted action shall not proceed while valid approval is absent. Pending, rejected and unavailable approval outcomes shall remain distinct from approval.
CO-AC-07	Given a duplicate or uncertain execution outcome, the system shall reconcile the existing recorded outcome before a retry that could duplicate the action or before marking the workflow Completed.

Detailed refinements CO-FR-01 context validation; CO-FR-02 authorised transfer; CO-FR-03 status/failure recording; CO-FR-04 safe recovery. Optional CO-O-01 adds bounded retry. CO-T-01–08 cover these refinements.

4.5 User requirement view

GR-RQ-01 maps Customer and Founder needs to the five mandatory requirements and their observable responses. The view connects stakeholder needs to BA/CS/MC services and AP/CO controls.

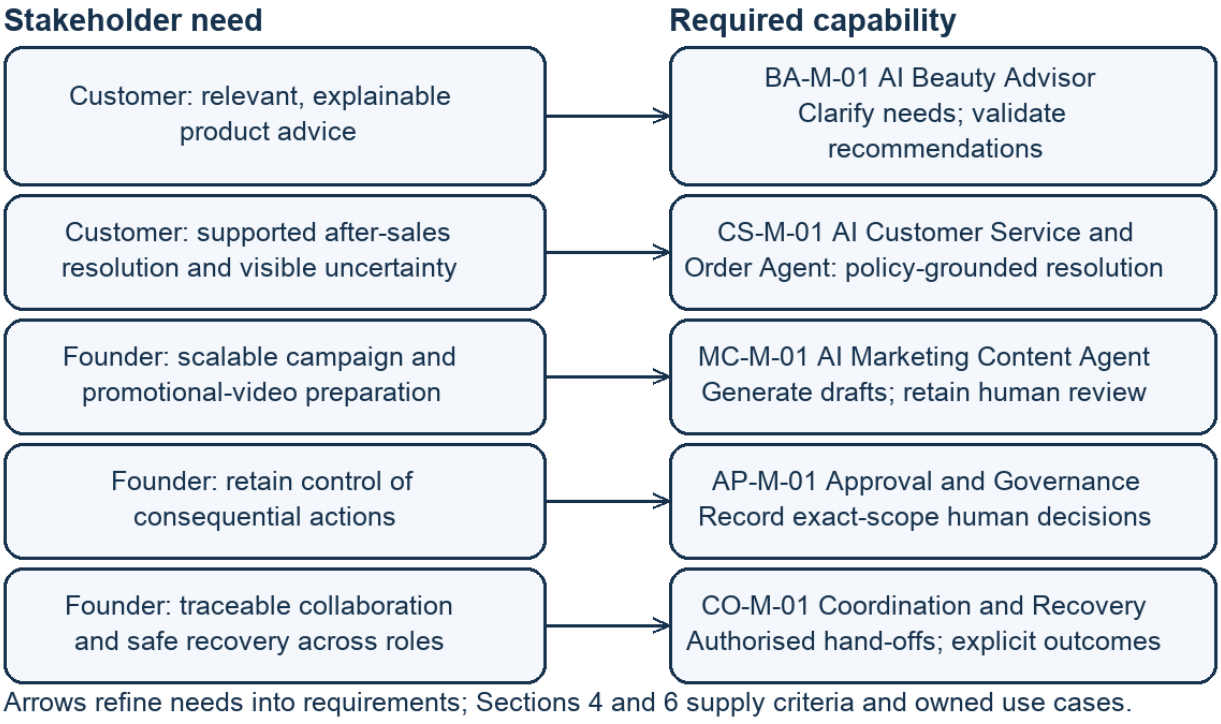


Figure 1 GR-RQ-01 Customer and Founder needs refine into three business capabilities and two shared controls.

5 Feature Modelling

5.1 Comprehensive feature model

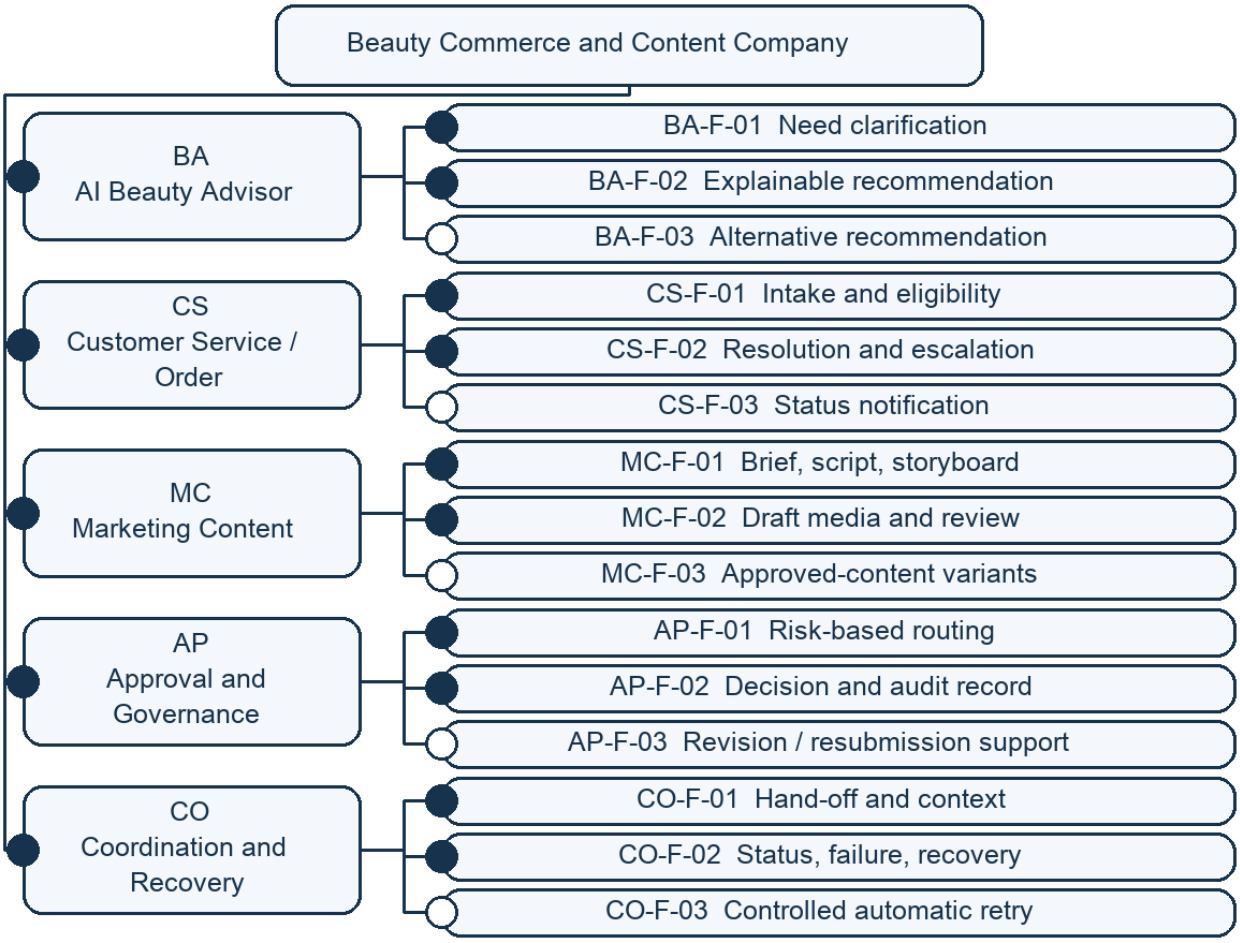


Figure 2 GR-FD-01 All five mandatory capability branches and fifteen features; filled circles are mandatory and open circles optional.

Constraints: BA-F-02 requires BA-F-01; CS-F-02 requires CS-F-01; MC-F-02 requires MC-F-01; CO-F-02 requires CO-F-01. Each corresponding F-03 requires its F-02 and is optional. AP-F-02 requires AP-F-01; AP-F-03 requires AP-F-02. Restricted actions in CS, MC or CO require a valid AP-F-02 decision. AP-F-03 provides assisted linked resubmission; REQUEST_REVISION itself remains a core AP-F-02 decision. No XOR/OR feature group or unregistered fourth business agent is introduced.

5.2 Feature descriptions and constraints

ID and type	Feature and user value	Constraints and planned evidence
BA-F-01 Core	Customer Need Clarification	Capture and correct decision-relevant needs; ask focused questions, avoid diagnosis/unrelated data and permit cancellation. Evidence: BA-FR-01; BA-NFR-04; BA-T-01/07
BA-F-02 Core	Explainable Product Recommendation	Validate authorised product IDs, explicit needs, stock and factual reasons; record and deliver grounded choices or an explicit incomplete result. Requires BA-F-01. Evidence: BA-FR-02/03/04; BA-T-02/03/04/05

ID and type	Feature and user value	Constraints and planned evidence
BA-F-03 Optional	Alternative Product Recommendation	On Customer request, show verified available alternatives and differences; never relax a constraint silently. Requires BA-F-02; no-match remains safe when OFF. Evidence: BA-O-01; BA-T-06
CS-F-01 Core	After-Sales Request Intake and Eligibility Check	Verify identity/order association before protected disclosure; retrieve policy evidence and clarify without inventing rules. Evidence: CS-AC-01/02; CS-T-01/02
CS-F-02 Core	Routine Resolution and Exception Escalation	Requires eligibility, current Customer confirmation and any exact-scope approval; recheck stock/policy, persist actionRef, execute and reconcile/save actual outcome. Evidence: CS-AC-03/04/05/06; CS-T-03/04/05/06
CS-F-03 Optional	After-Sales Status Notification	Notify only an authorised recipient about recorded status. Delivery failure or OFF leaves core resolution accessible; channel/permission policy is configurable. Evidence: CS-AC-07; CS-T-07
MC-F-01 Core	Campaign Brief Script and Storyboard Generation	Interpret authorised objective/product evidence and validate generated claims. Missing/conflicting information leads to clarification; script/storyboard remain drafts. Evidence: MC-AC-01/02; MC-T-01/02/03
MC-F-02 Core	Draft Media Generation and Founder Review	Generate and check real or labelled mock draft media, save exact revision/evidence and obtain a committed applicable Founder decision. Generation or checking cannot grant approval. Evidence: MC-AC-03 through MC-AC-07; MC-T-04/05/06
MC-F-03 Optional	Approved Content Variant Generation	Generate a traceable variant from an approved campaign concept. Each new variant remains draft and receives fresh review; inheritance is not assumed. Mandatory workflow works when OFF. Evidence: MC-T-07
AP-F-01 Core	Risk-Based Approval Routing	Validate authorised action/context and declared risk policy. Persist required review before receipt; unknown classification blocks action for clarification/review. Evidence: AP-AC-01/02; AP-T-01/02
AP-F-02 Core	Approval Decision and Audit Record	Founder selects APPROVE, REJECT or REQUEST_REVISION; commit exact decision/request/audit before correlated delivery. Separate recording, delivery and execution outcomes. Evidence: AP-AC-02 through AP-AC-07; AP-T-02 through AP-T-07
AP-F-03 Optional	Revision Request and Resubmission	Assist a linked successor request preserving history; fresh review is required. With OFF, core REQUEST_REVISION and manual corrected submission remain available. Evidence: AP-AC-04/08; AP-T-04/08
CO-F-01 Core	Agent Task Hand-off and Context Transfer	Validate target/context, transfer only authorised fields and retain intent/actionRef and acknowledgement. A receipt is not completed work; target authority remains unchanged. Evidence: CO-FR-01/02; CO-T-01/02/03
CO-F-02 Core	Workflow Status Failure Recording and Recovery	Record actual outcome, failure and escalation; reconcile uncertainty before repeat. Save failure and unapproved work cannot become COMPLETED. Evidence: CO-FR-03/04; CO-T-03/04/06/07/08

ID and type	Feature and user value	Constraints and planned evidence
CO-F-03 Optional	Controlled Automatic Retry	Retry only a declared eligible, bounded, authorised read or proven-not-executed operation. Retain attempt history; OFF/exhaustion still permits escalation or safe cancellation. Evidence: CO-O-01; CO-T-05/06

6 Use Case Modelling

6.1 Overall use case diagram

GR-UCD-01 bounds the complete company application. Customer initiates BA-UC-01/CS-UC-01; Human Founder / Operator initiates MC-UC-01/AP-UC-01/CO-UC-01. Internal agents, repositories and adapters are not external actors. Sections 6.2.1–5 specify these five goals; approval/coordination are conditional dependencies, not unconditional include relationships.

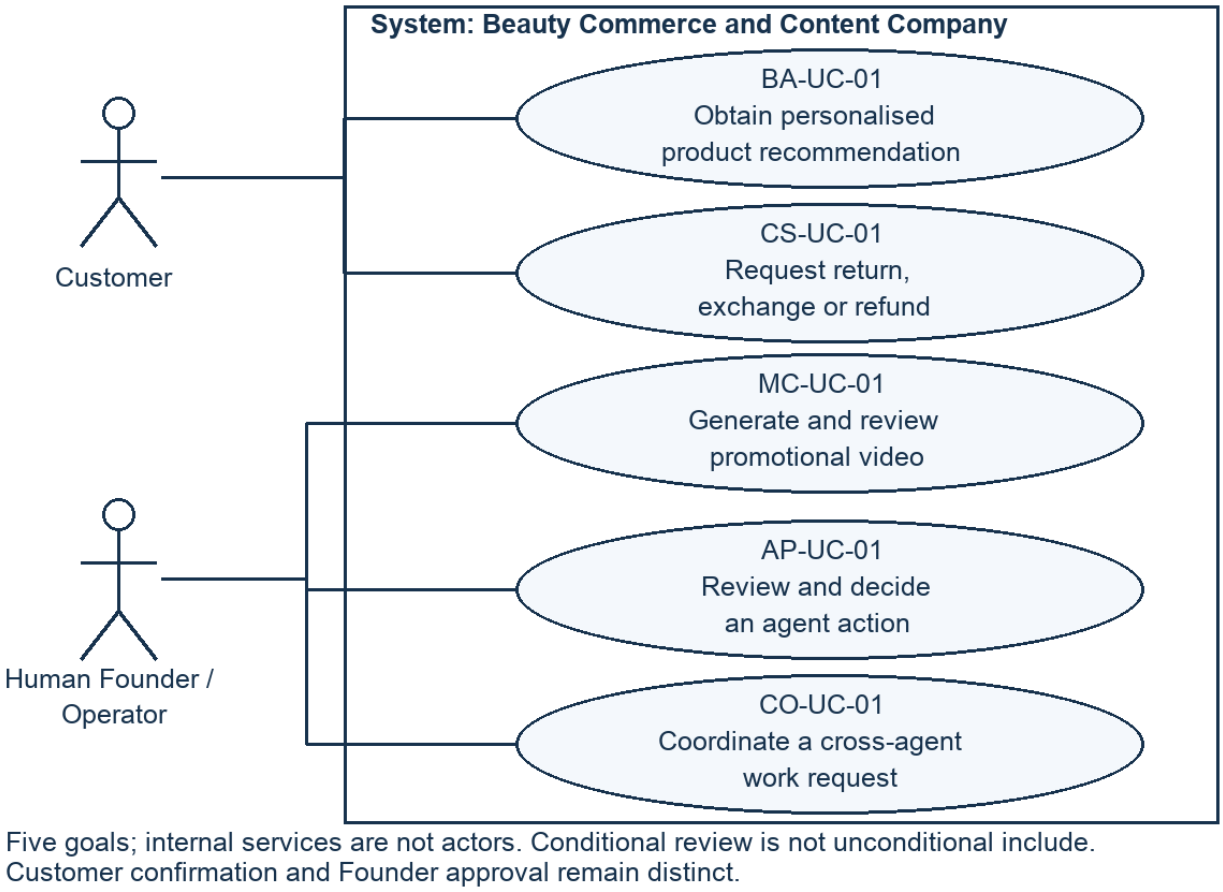


Figure 3 GR-UCD-01 Whole-system actor and five-goal view; internal business roles and adapters remain inside the application boundary.

6.2 Five principal use case specifications

6.2.1 BA-UC-01 Obtain Personalised Product Recommendation

Owner and status Danyang Wang redacted . PROPOSED; formal approval/submission remains pending.

Goal: Obtain an understandable product recommendation reflecting the Customer's stated needs and supported availability.

Level: User-goal level

Primary actor: Customer

Supporting actors: Founder reviews only escalated advice. Beauty Advisor, Catalogue, Inventory, LLM and Approval services are internal participants.

Related requirements: BA-M-01; BA-FR-01/02/03/04; BA-NFR-01/02/03/04; optional BA-O-01.

Authority: Read authorised catalogue/stock and relevant Customer input; record consultation and minimal audit/review context; advise the requesting Customer. No orders, payments, refunds, inventory/price changes or public posting. Product selection is not purchase authorisation.

Preconditions: Customer access and service interfaces/labelled mocks are configured. Successful service responses are checked during the flow.

Trigger: The Customer submits a beauty consultation request containing a goal, preference or constraint.

Main success scenario

1. The Customer submits the request.
2. Create/resume the consultation and record relevant input.
3. Extract needs; clarify missing information through A1.
4. Retrieve authorised catalogue evidence.
5. Check availability against the retrieved product IDs.
6. Propose candidates with reasons and supporting evidence.
7. Validate product IDs, constraints, facts and availability; unsupported output follows E3.
8. Save the validated result; present at least one available recommendation with evidence-backed reasons and limitations.
9. Customer reviews the advice; consultation success ends here. Purchasing requires a separate Customer-initiated workflow.

Alternative flows

A1 at step 3: Ask a focused question and return to step 3. If Customer declines, record an incomplete outcome without guessing.

A2 at steps 4–5: If no product matches, invite revised needs and return to step 3; otherwise record a safe incomplete outcome.

A3 at step 5: With BA-F-03 ON and Customer requesting alternatives, retrieve fresh evidence, validate alternatives and explain differences; otherwise follow A2.

A4: Customer correction invalidates the prior recommendation and requires re-evaluation. Cancellation stops processing, suppresses late output and is recorded.

Failure and uncertainty flows

E1: Catalogue/stock error, timeout or missing availability produces an explicit incomplete outcome. Do not invent stock; Customer-initiated retry refreshes both sources.

E2: Stop diagnosis/treatment advice; Founder review cannot override this medical boundary.

E3: Withhold failed, malformed, unsupported or contradictory LLM output, or advice not justified by evidence. Clarify, seek human review or remain incomplete; do not invent a confidence threshold.

E4: With Customer consent, submit minimal context and the issue for Founder review. REVIEW_PENDING requires an acknowledged receipt; unavailable review is not submitted. Rejection or no decision grants no approval. Later advice repeats fresh validation; pending review is not success.

E5: Required recording failure remains incomplete. Deliberate retry creates no financial effect.

Success postcondition: Available, validated advice is delivered; input, evidence, stock observation and outcome are recorded.

Failure postcondition: Unsupported, cancelled or pending advice is not success; no order/unrelated write occurs. Record the reason when storage is available; otherwise expose recording failure.

Data used or changed: Relevant input, optional authorised profile subset, product/stock evidence, recommendation, outcome and minimal audit/review context. Retention remains unagreed.

Tool/service dependencies: Product Catalogue Service, Inventory Service, LLM gateway, consultation repository and Founder Approval Service for the conditional review branch; implementations or controlled mocks are interchangeable through explicit interfaces.

Stop or hand-off: Routine advice only; review cannot override medical/unsupported-output boundaries. GR-S-01 transfers minimal context without purchase authority.

6.2.2 CS-UC-01 Request Return Exchange or Refund

Owner and status Siyuan Lei redacted . PROPOSED; formal approval/submission remains pending.

Goal: Obtain a supported after-sales resolution or a traceable referral for human decision.

Level: User-goal level

Primary actor: Customer

Supporting actors: Founder handles exceptions; internal CS, identity/order, after-sales, stock, approval, LLM and audit services remain within the overall system boundary.

Related requirements: CS-M-01; CS-AC-01 through CS-AC-07.

Authority: Read authorised Customer/order, policy and stock evidence; record the request, eligibility, routine outcome and audit. Restricted actions require human review. No policy changes, unrelated-account access, public posting, purchasing or real settlement. LLM classification grants no authority.

Preconditions: Customer access and service interfaces/labelled mocks are configured. Identity, eligibility and successful replies are verified during the flow.

Trigger: The Customer submits a return, exchange or refund request.

Main success scenario

1. The Customer supplies the order reference, requested resolution and reason.
2. Create/resume the request; verify Customer/order association before protected disclosure.
3. Clarify missing details and retrieve authorised order/policy evidence.
4. Check eligibility and routine authority; unknown rules follow E2/E4.
5. Check exchange availability; missing/unavailable stock follows A2.
6. Explain the resolution and obligations; obtain current Customer confirmation.
7. Recheck evidence and applicable approval before invoking the permitted service action.
8. Confirm the service outcome; record evidence and actionRef.
9. Show the outcome and next steps. Optional notifications do not control core success.

Alternative flows

A1: Clarify missing/corrected details, then repeat verification and eligibility. If Customer declines, remain unresolved without guessing.

A2: For unavailable/unknown exchange stock, offer policy-supported alternatives. Customer selection repeats eligibility/authority checks; otherwise seek review or remain unresolved.

A3: Record pre-action cancellation when storage permits. After known/unknown execution, report actual status; cancellation does not imply rollback.

A4: CS-F-03 requires recipient/channel permission and recorded delivery status. OFF or failed notification does not block the core outcome.

Failure and uncertainty flows

E1: Identity mismatch/unavailability blocks disclosure and action; use correction/support with minimal context.

E2: Missing/conflicting/unknown policy, ineligibility, dispute, low confidence or unsupported LLM output blocks automation. Clarify or seek review; invent no threshold.

E3: Read failure remains unresolved. After write timeout, look up the same actionRef before retry; if still unknown, stop execution and escalate. Do not assume no effect.

E4: Send AP minimal evidence, proposed action and issue. Pending status requires receipt; service failure is not confirmed submission. Applicable approval permits exact-scope re-evaluation only; rejection stops, revision requires clarification, and absent/invalid approval remains pending. Member 4 owns the review contract.

E5: Recording failure retains known receipt via an available permitted mechanism. Do not claim rollback or repeat execution before reconciliation.

Success postcondition: Record the resolution, order/policy evidence and Customer/AP decisions; show next steps. Physical-return acknowledgement or mock refund does not prove settlement.

Failure postcondition: Preserve unresolved, rejected, cancelled or pending status, reasons/references when storage permits, and visible uncertainty.

Data used or changed: Reason/resolution, authorised identity/order association, order/policy/stock evidence, Customer confirmation, applicable review/decision, service reference/outcome and audit. Retention/access remain unagreed.

Tool or service dependencies: Identity/order verification, after-sales, Inventory, Founder Approval, LLM and audit storage. Production APIs/schemas remain unagreed; labelled mocks imply no financial execution.

Stop or hand-off: Reverify Customer/order after transfer. Restricted action requires current confirmation and exact-scope decision; unknown execution remains unresolved.

6.2.3 MC-UC-01 Generate and Review Promotional Video

Owner and status Zhengdu Wei redacted . PROPOSED. MC-M-01; MC-F-01/02 and optional MC-F-03.

Goal and actors Founder obtains a grounded, reviewable promotional-video draft and recorded exact-revision decision. Internal participants: MarketingContentService, authorised catalogue, LLM/media adapters, ReviewGateway/AP and MarketingRepository.

Authority Read authorised campaign/product context; generate, revise, store and submit drafts/evidence. No product-fact changes, inherited approval or autonomous publication.

Preconditions and trigger Founder has workflow access and submits a campaign objective through configured interfaces/labelled mocks. Evidence sufficiency, successful responses and valid approval are checked during the flow.

Main success scenario

1. Founder supplies the brief, objectives, constraints and products.
2. Save campaign/context and identify its revision.
3. Retrieve authorised product evidence; clarify missing facts.
4. Propose an evidence-linked script/storyboard with the LLM.
5. Validate IDs, facts and criteria; withhold invalid content.
6. Generate media via the real adapter or a permitted labelled mock.
7. Validate media; save the exact draft revision and generation evidence.
8. Submit that revision for review; retain its request/receipt ID.
9. Founder reviews the draft and selects APPROVE.
10. AP validates current scope/Founder authority and commits decision, request state and audit before delivery.
11. MC validates correlation and saves its local APPROVED view.
12. Return the approved draft and recorded decision; do not publish.

Alternatives A1: Founder correction repeats evidence/claim checks. A2: REQUEST_REVISION records reasons and revises the draft for fresh generation, validation and review. A3: REJECT records a stop. A4: Optional variants retain campaign traceability and receive individual exact-revision review; OFF preserves the core flow.

Failure and uncertainty E1: Missing/conflicting evidence → clarify or INCOMPLETE. E2: Invalid LLM output → withhold, revise/regenerate or CONTENT_REJECTED. E3: Media failure → permitted

labelled fallback or GENERATION_FAILED. E4: Failed/undefined criterion → record and revalidate; invent no pass threshold. E5: Receipt without decision → REVIEW_PENDING. E6: Rejection/revision → A3/A2. E7: Unconfirmed/failed review grants no approval. E8: Draft/evidence/local-decision save failure → RECORDING_FAILED; recover the same record without a new decision. Suppress stale/duplicate callbacks and cancelled output.

Success postconditions The grounded exact draft revision and authorised Founder decision are recorded and returned. No publication has occurred; an approved draft is not evidence of external execution.

Minimum guarantee and data Label unapproved/incomplete content accurately. Retain campaign/brief, product sources, script/storyboard, media/mock outcome, revision and review/committed-decision references. Expose storage failure; retention/access remain open.

6.2.4 AP-UC-01 Review and Decide an Agent Action

Owner: Jiaxuan Ren redacted . Status PROPOSED. AP-M-01; core AP-F-01/02; optional AP-F-03. Integration attribution is in Section 15.

System boundary: Beauty Commerce and Content Company. Primary actor: Human Founder / Operator. Requesting role/coordination adapter, risk policy, audit, notification and delivery services are internal participants.

Goal: Record a consequential-action decision for the exact current scope. Pending review is not execution authority.

Trigger: A role submits minimal action context; Founder opens the review request.

Preconditions: Authenticated/authorised Founder; identifiable action/request and risk evidence; permission to transfer minimal context; consequential action not yet executed. Unknown risk grants no permission.

Authority: Read review context; persist requests, decisions, audit and delivery status. No business execution, policy rewrite or expanded rights.

Main success scenario

1. The requesting role submits origin ID, proposed action, context revision, evidence references and applicable actionRef through its local review interface.

2. Validate context/authority/risk; persist PENDING without business dispatch.

3. Return receipt; notify Founder and report ReviewPending to the requester.

4. Founder examines the action, evidence, reason and conditions.

5. Founder chooses APPROVE, REJECT or REQUEST_REVISION with a reason under the declared fixture policy.

6. Recheck authority/current revision; atomically commit decision, request state and audit.

7. Record delivery and return the correlated decision. Delivery failure does not erase the commit.

8. Requester checks fresh eligibility/scope: APPROVE permits re-evaluation, not execution by AP; rejection stops; revision requires correction.

Success postcondition: Exact-scope decision/audit committed; delivery status distinguished. Business success is not implied.

Minimum guarantee: Without a committed applicable decision, restricted action remains blocked. Report accepted/failed submission truthfully; retain identifiers where stored.

A1: A clearly unrestricted action may use existing authority without AP; invent no approval. Unknown/disputed authority remains blocked for review or clarification.

A2: REQUEST_REVISION grants no permission. Optional AP-F-03 assists a linked successor request; otherwise correction is manual. Both require fresh review and retained history.

A3: No decision remains pending. Eligible pre-decision cancellation/expiry suppresses late results; expiry duration remains unagreed.

E1: Missing/unauthorised context or risk-check failure prevents acceptance or requires clarification; restricted action remains blocked.

E2: Reject stale, conflicting or unauthorised decisions; approval is not carried forward.

E3: Commit failure → RECORDING_FAILED, without authority. Retry only storage of the same decision.

E4: Notification failure leaves the persisted request pending. Delivery failure preserves the commit; retry/lookup the same decision ID. Duplicate/stale delivery never executes business work.

Models: AP-CL-01 A/B, AP-OBJ-01, AP-ACT-01, AP-SEQ-01 A/B and AP-STM-01. REQUEST_REVISION is core; assisted resubmission is optional.

6.2.5 CO-UC-01 Coordinate a Cross Agent Work Request

Owner and status Guiyu Liu redacted . PROPOSED. CO-M-01; CO-FR-01 through CO-FR-04; CO-NFR-01 through CO-NFR-04; optional CO-O-01.

Goal and actors: Founder initiates/monitors authorised BA/CS/MC work. Internal participants: coordinator, role adapters, persistence and conditional ReviewGateway/AP.

Authority: Route minimal authorised context; record status/failures and recover safely. No new role authority or approval.

Preconditions and trigger Founder initiates cross-agent work through configured controlled adapters. Context validity, target availability and successful results are checked during the flow.

Main success scenario

1. Founder starts the work request.
2. Save CrossAgentWorkRequest and its context.
3. Validate context and allowed scope.
4. Select the target role.
5. Check role authority and conditional AP review.
6. Prepare minimal permitted context.
7. Persist intent/actionRef before transfer.
8. Receive acknowledgement; the role works within its own authority. Receipt is not success.
9. Reconcile unknown effects by actionRef; record confirmed result and trace.
10. Coordinate the next required authorised, traceable hand-off with the same control checks.
11. Mark COMPLETED only after all required work succeeds and outcomes are saved.
12. Founder observes completed workflow status and trace.

Alternative flows A1: Missing/invalid context → CLARIFICATION_REQUIRED; correction repeats validation, otherwise remain safely incomplete/cancelled. A2: Cancellation stops further hand-offs without hiding pending/unknown effects or reversing known execution. A3: Optional CO-F-03 permits bounded eligible retry only for read-only work or proven no prior effect.

Failure and uncertainty flows E1: Unavailable target → record failure and recover/escalate. E2: Supporting-service failure → truthful incomplete outcome. E3: Review receipt → AWAITING_APPROVAL; current APPROVE repeats validation, REQUEST_REVISION needs correction/fresh review, REJECT stops dispatch. E4: Retry OFF/exhausted/unsafe → ESCALATED or safe CANCELLED. E5: Unknown effect → RECONCILING by actionRef; stillUnknown escalates, confirmed results are recorded, notExecuted permits reassessment. E6: Save failure → RECORDING_FAILED; retain actual result and retry storage only.

Success postconditions All required cross-agent work has confirmed successful results and the relevant hand-offs/context/outcome evidence is saved. Only then is the request COMPLETED and its trace visible to the Founder.

Minimum guarantee and data Never mark failed, uncertain, unapproved or cancelled work completed. Retain context/revision/scope, roles, intent/actionRef, hand-off/outcome/failure/retry/escalation and applicable AP references. AP owns authoritative decisions; CO holds ApprovalReference views.

6.3 Integrated cross agent scenario

Scenario GR-S-01 combines existing use cases. Customer requests an authorised exchange and product advice; Founder initiates coordination. CS verifies order/policy and sends only permitted needs/product context through CO to BA. BA-F-03 must be ON for optional alternatives; the core path allows verified advice or a safe no-match outcome.

BA returns grounded advice; CO records the hand-off without declaring CS resolution. CS rechecks policy, stock and Customer confirmation. Consequential/disputed exchange enters AP-UC-01: receipt grants no authority; applicable recorded approval returns to verification/confirmation. CS saves intent/actionRef before AfterSalesPort execution, records actual outcome and looks up uncertain effects before repeating. Storage recovery never repeats execution.

GR-S-01 covers BA/CS/AP/CO. MC is an independent Founder campaign path with shared approval rules. GR-OBJ-01 is a partial synthetic pre-decision snapshot; GR-COL-01 shows numbered messages; GR-CMP-01 contains all six typed parts, including MC. No executed result is claimed.

7 Architecture Analysis and Design

7.1 Drivers principles and viewpoints

The event-driven design separates three specialised business services through ports/adapters. Deterministic logic checks identity, evidence, policy, scope and persistence; LLMs propose language/content. Adapters distinguish live/mock results and expose failures. Revisions suppress stale results; human review is durable.

Viewpoint	Evidence and design question
Logical	GR-CL-01 and complete member class slices: which entities, ownership, associations and receiver contracts realise each requirement?
Process and runtime	GR-S-01, object/composite/communication and individual behaviour models: how do work, decisions, uncertainty and recording progress?
Development	Presentation, application/domain services and provider adapters: how can live and controlled mock components be replaced without changing authority?
Scenario	Five use cases and GR-S-01 expose both success and safe incomplete paths; Section 13 links acceptance evidence.
Physical deployment	Not selected as a formal deployment model. Hosting technology and production availability/access guarantees remain open.

4+1 organises the selected views; full physical coverage is not claimed. BA: evidence/minimal data. CS: eligibility/actionRef/reconciliation. MC: grounded media/exact-draft review. AP: commit versus delivery. CO: acknowledgement versus outcome/recovery.

7.2 Alternatives and tradeoffs

Discarded choice	Reason and retained tradeoff
Unrestricted single LLM with direct commerce/publication writes	Conflicts with role differentiation, evidence checks and human control. Bounded services require more orchestration and contracts.
Peer-to-peer autonomous negotiation for every workflow	Weakens shared trace/status/recovery and adds complexity to routine single-role tasks. Central coordination is used when a handoff is required.
Form-only/catalogue-filter system or fully manual content creation	Useful deterministic baseline, but inadequate natural-language clarification and draft-production value. Deterministic validation remains alongside LLM assistance.
Direct coupling to one media provider	Availability/cost/schema dependence would weaken demonstration. Adapters permit labelled mocks but cannot promise production performance.

Discarded choice	Reason and retained tradeoff
Automatic approval after successful generation/quality check	Capability and authority differ. Founder review adds delay, while preserving explicit exact-scope control.
Blind retry or assumed distributed rollback	Could duplicate actions or misrepresent service effects. Persist references, perform read-only lookup and recover storage/delivery separately.
Approval service executes requester actions or one globally renamed ReviewPort	Mixes business eligibility/recovery with governance and conceals incompatible local signatures. Explicit typed adapters add integration work.

The narrow use cases do not require unrestricted ReAct or Planner-Executor loops. Future autonomous planning needs justified use cases and revised authority, tool, cancellation and duplicate-effect analysis.

7.3 Supporting architecture slices

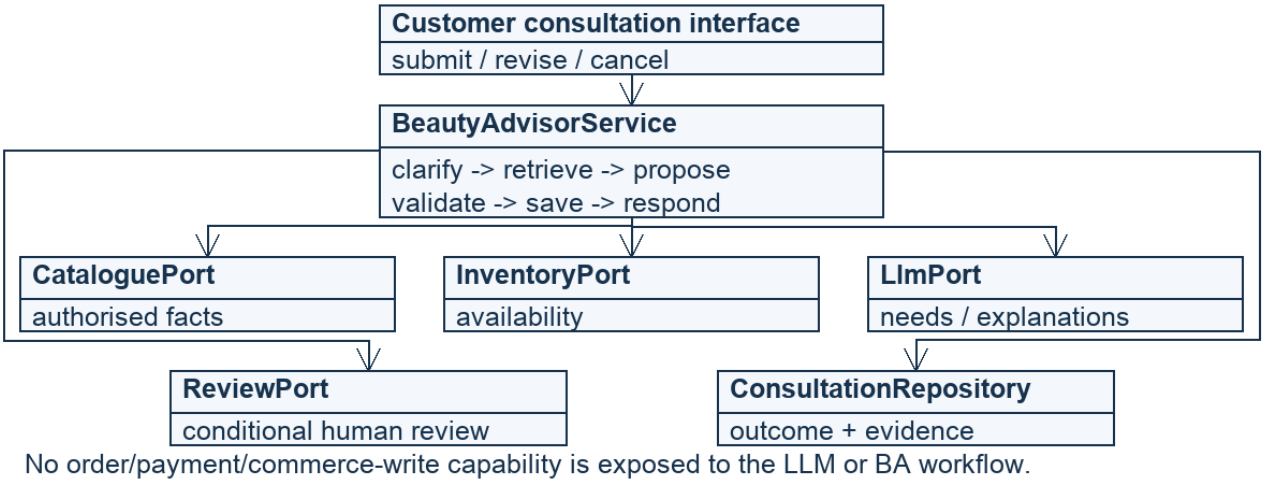


Figure 4 BA-ARCH-01 BA control and tool boundaries; no commerce execution capability.

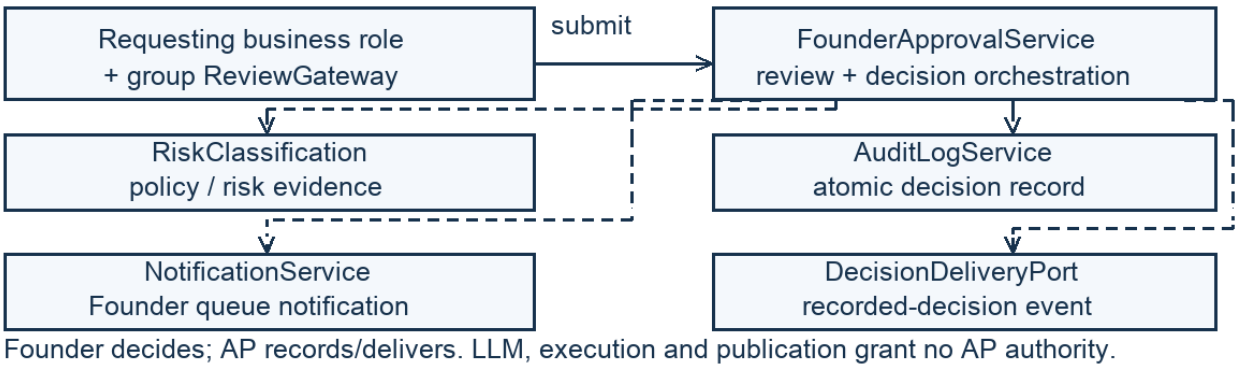


Figure 5 AP-ARCH-01 Review routing, committed decision and delivery boundaries; execution remains with the requester.

GR-CMP-01 (Section 9.2) shows overall typed parts. The BA/AP slices explain local control, without claiming deployment or replacing individual behaviour models.

8 Elementary Structure Modelling

8.1 Integrated and complete local class models

The Unified Modeling Language (UML) class views define entities, service operations, interfaces and relationships. GR-CL-01 A–D combines the five local models; BA::/CS::/MC::/AP::/CO:: qualify same-named types. Multiplicities constrain links. Composition denotes exclusive ownership and shared model lifetime; retained history uses associations without prescribing database deletion.

Model IDs use the same BA/CS/MC/AP/CO prefixes as their requirement and use case: for example, BA-ACT-01 traces to BA-M-01 and BA-UC-01. CL denotes classes, OBJ objects, CMP composite structure, COL communication, ACT activity, SEQ sequence and STM state machine. Section 13.1 maps all five chains; Section 15 records attribution.

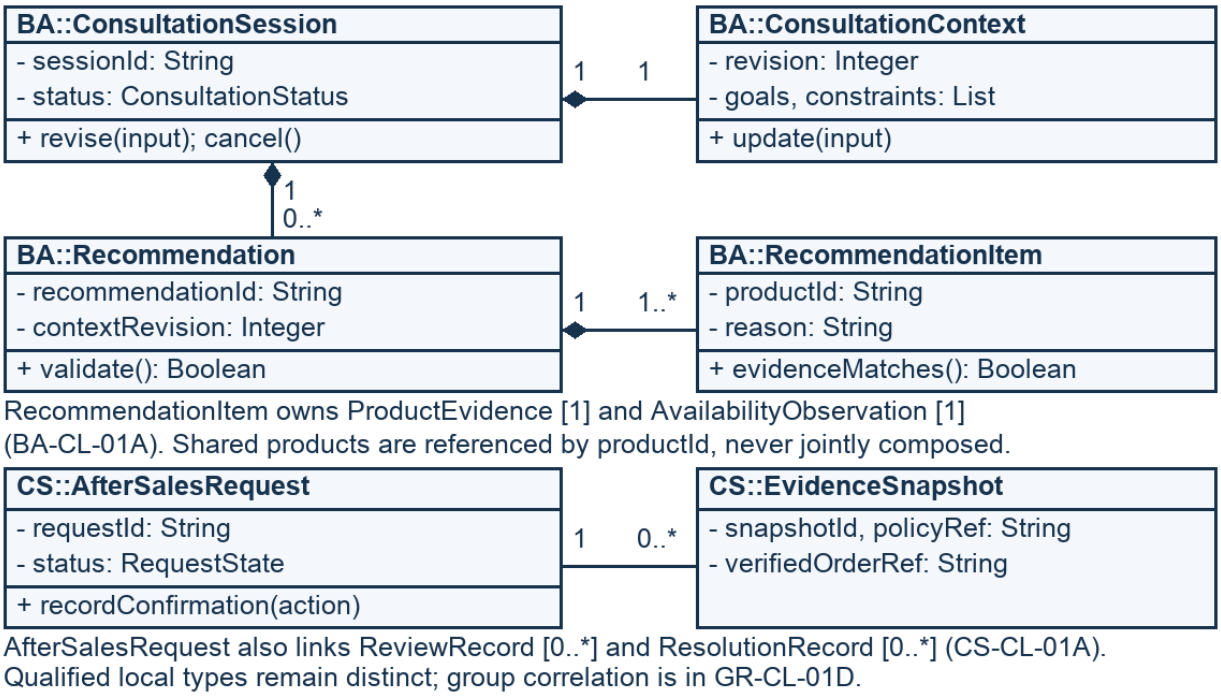


Figure 6 GR-CL-01A BA and CS domain ownership and retained evidence.

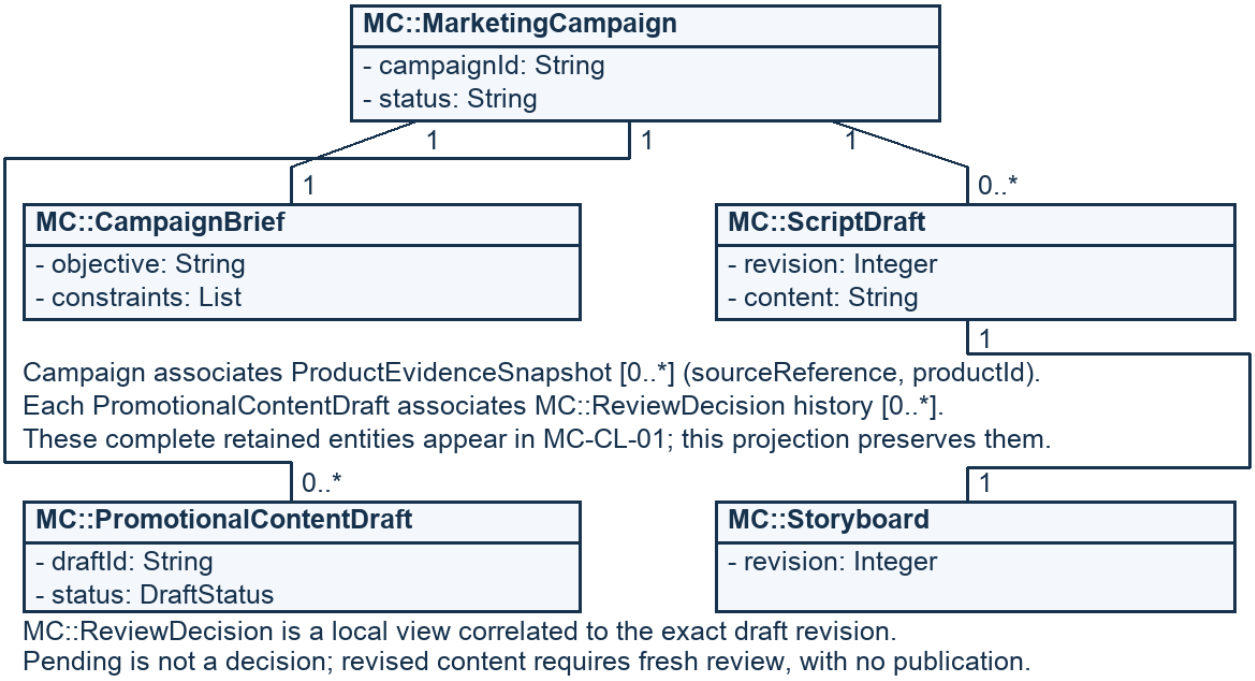
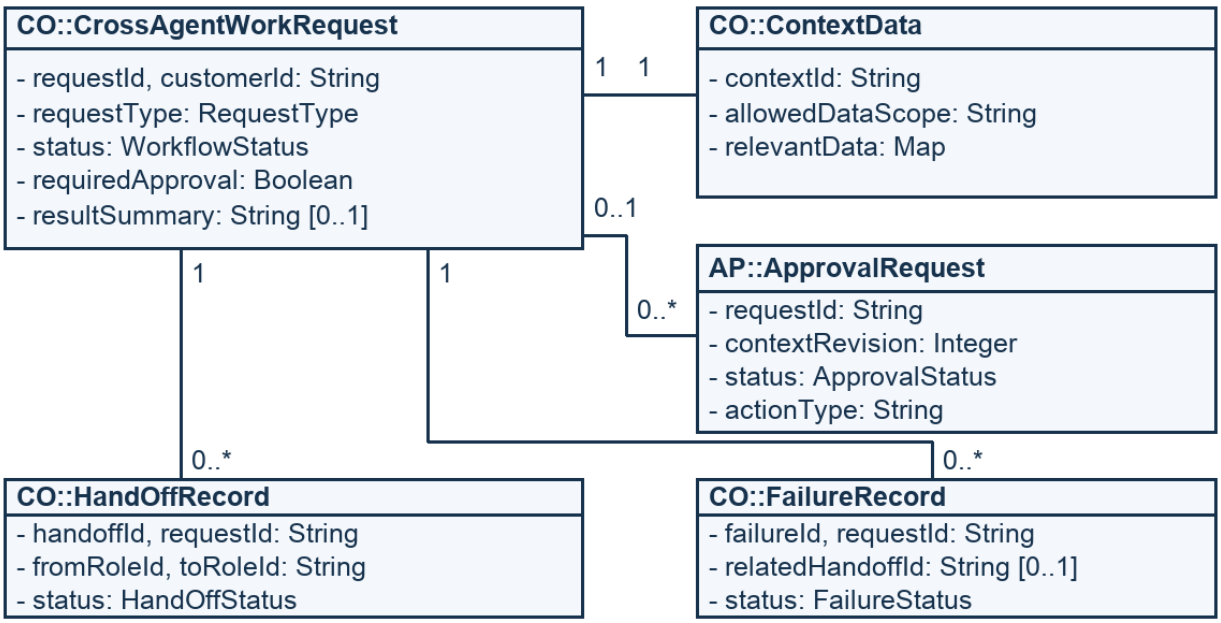
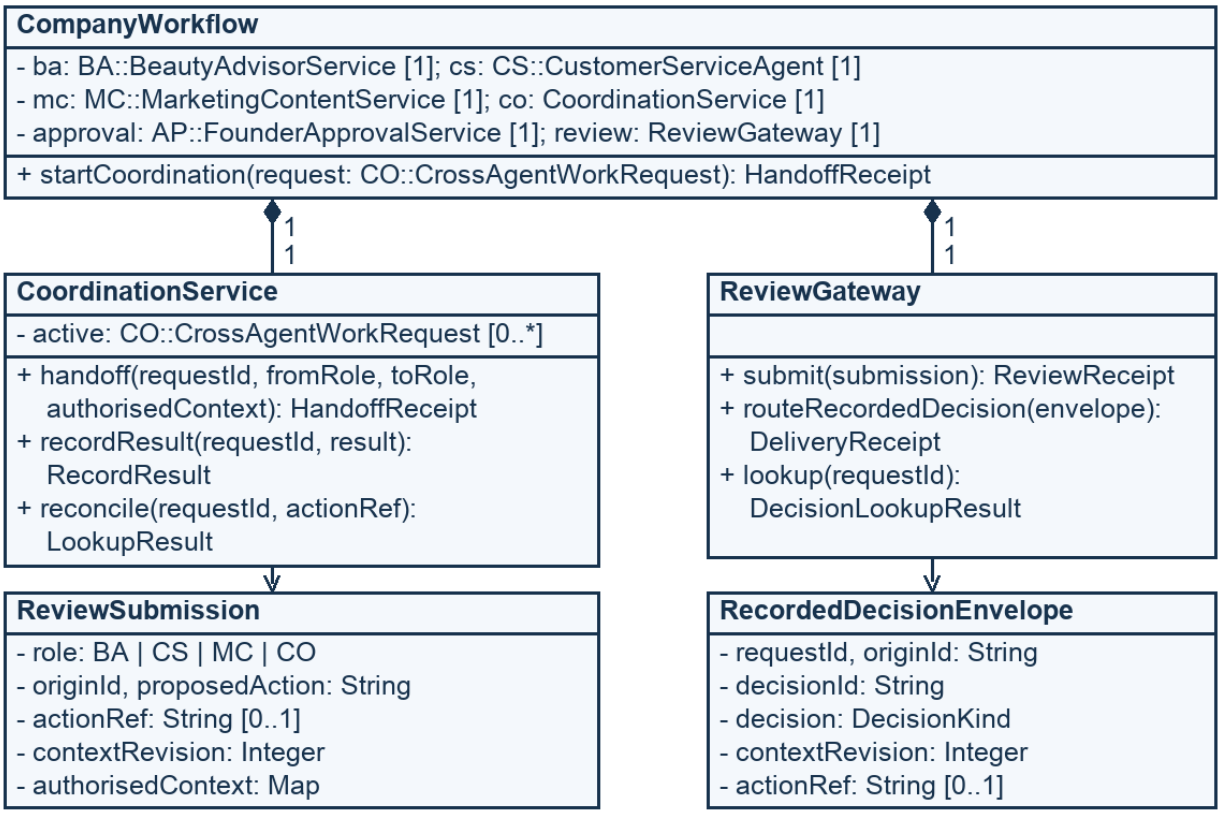


Figure 7 GR-CL-01B MC campaign/draft ownership and role-local committed review views.



CrossAgentWorkRequest also retains RetryRecord [0..*] (optional) and EscalationRecord [0..*]. Eligible FailureRecord links RetryRecord [0..*].
AP request associates ApprovalDecision [0..1] and AuditRecord [0..*] (AP-CL-01A).
A revised action may create another AP request; each AP request belongs to at most one CO workflow in this group projection. Non-coordinated requests have no CO link.
CO ApprovalReference observes authoritative AP decisions; ordinary links retain history.
This projection adds no execution authority.

Figure 8 GR-CL-01C Coordination records associate with AP requests; the local CO model explicitly uses ApprovalReference.



Facades adapt local contracts; receipts acknowledge only. Result variants: Section 8.2.

Figure 9 GR-CL-01D CompanyWorkflow parts, group facades, submission and recorded-decision envelope contracts.

8.1.1 BA-CL-01 Complete local class model

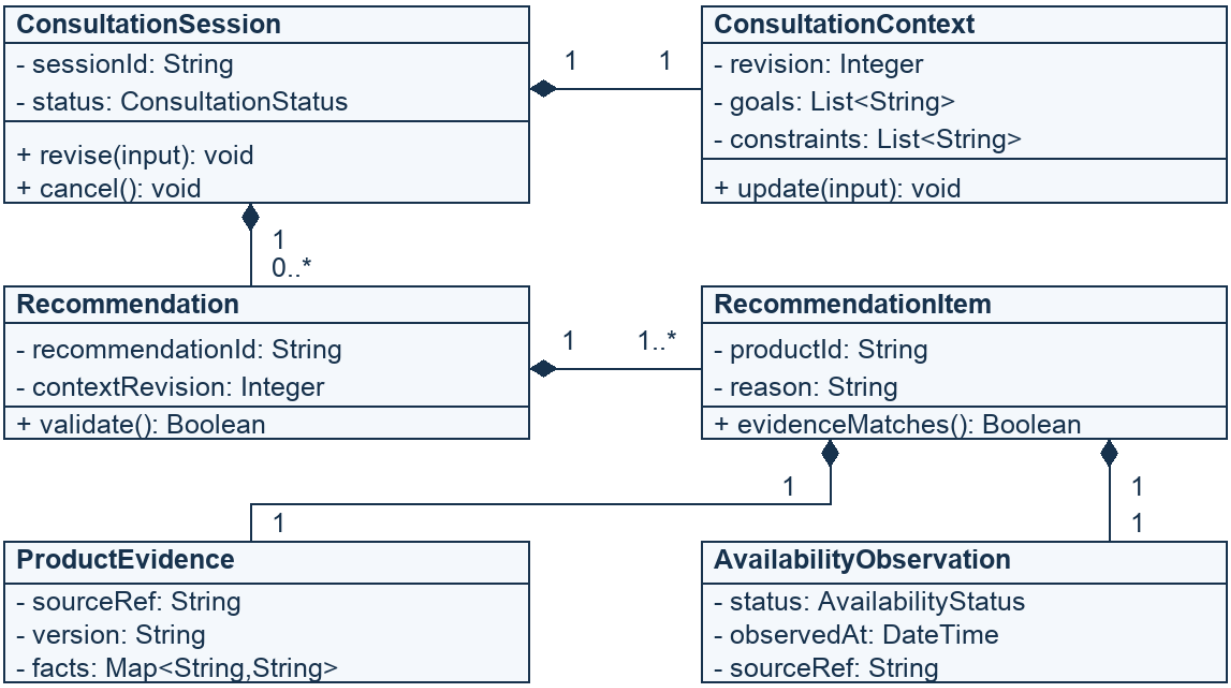


Figure 10 BA-CL-01A Consultation domain, composition and evidence snapshots.

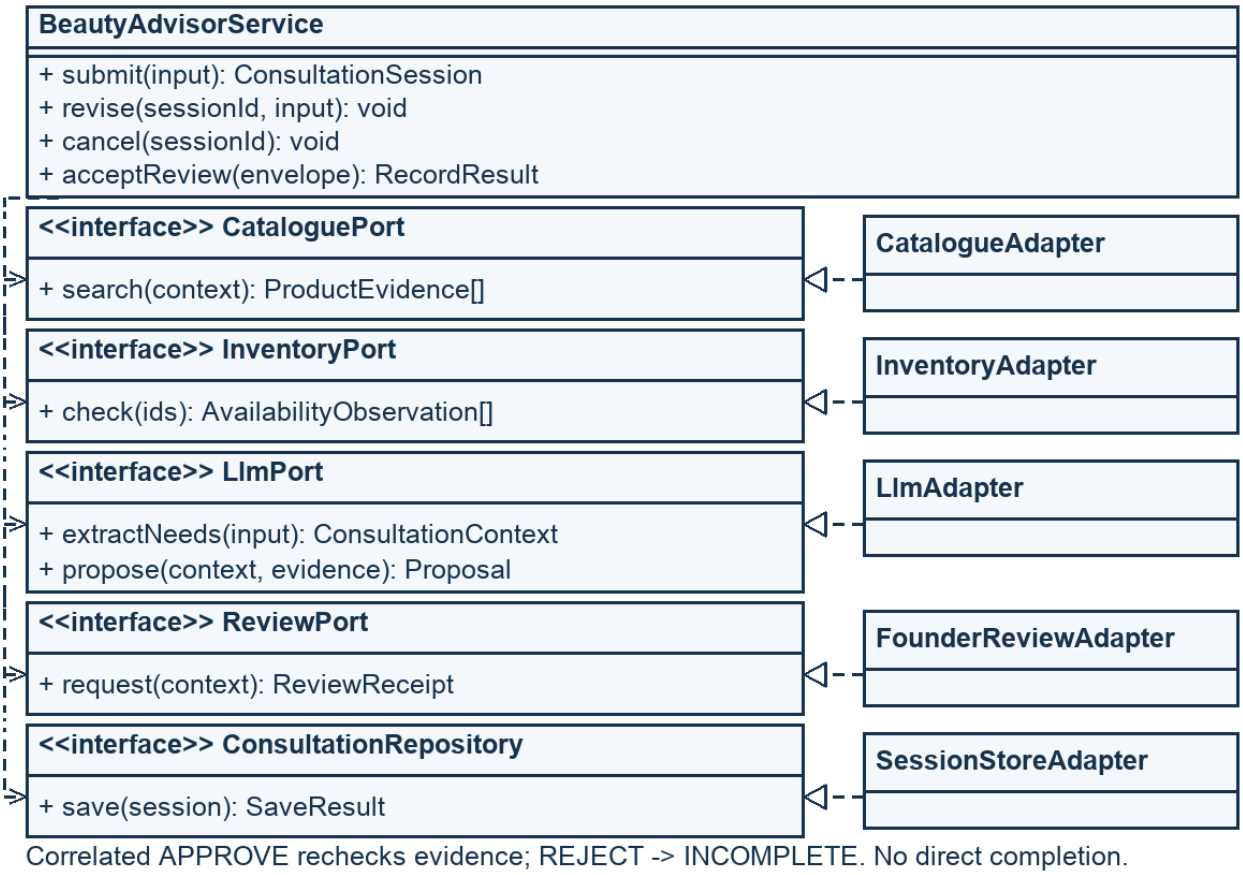
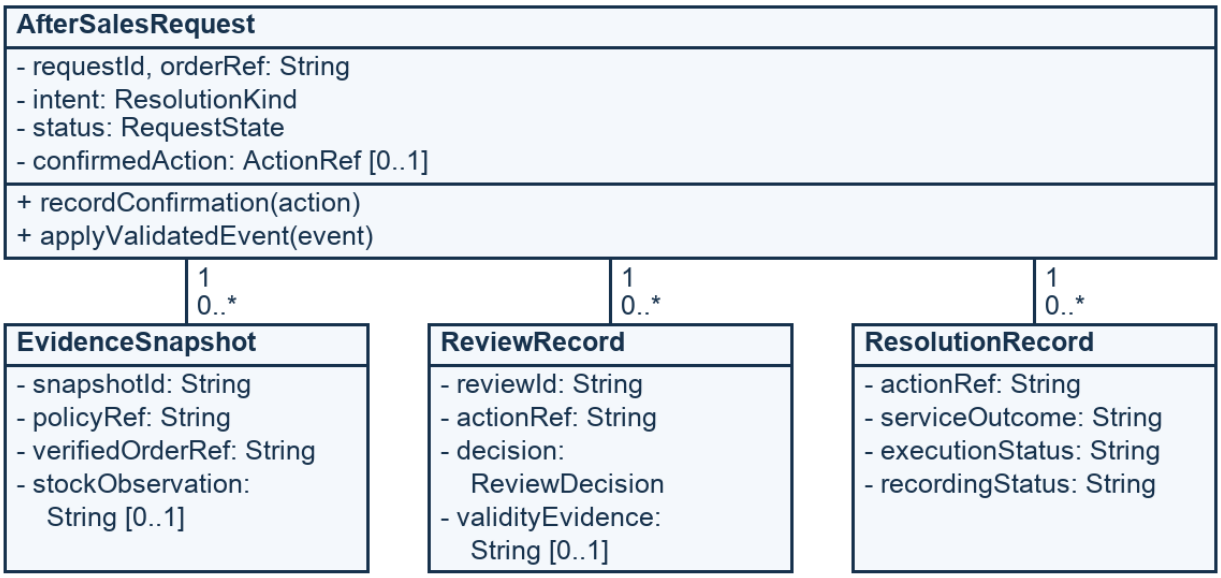


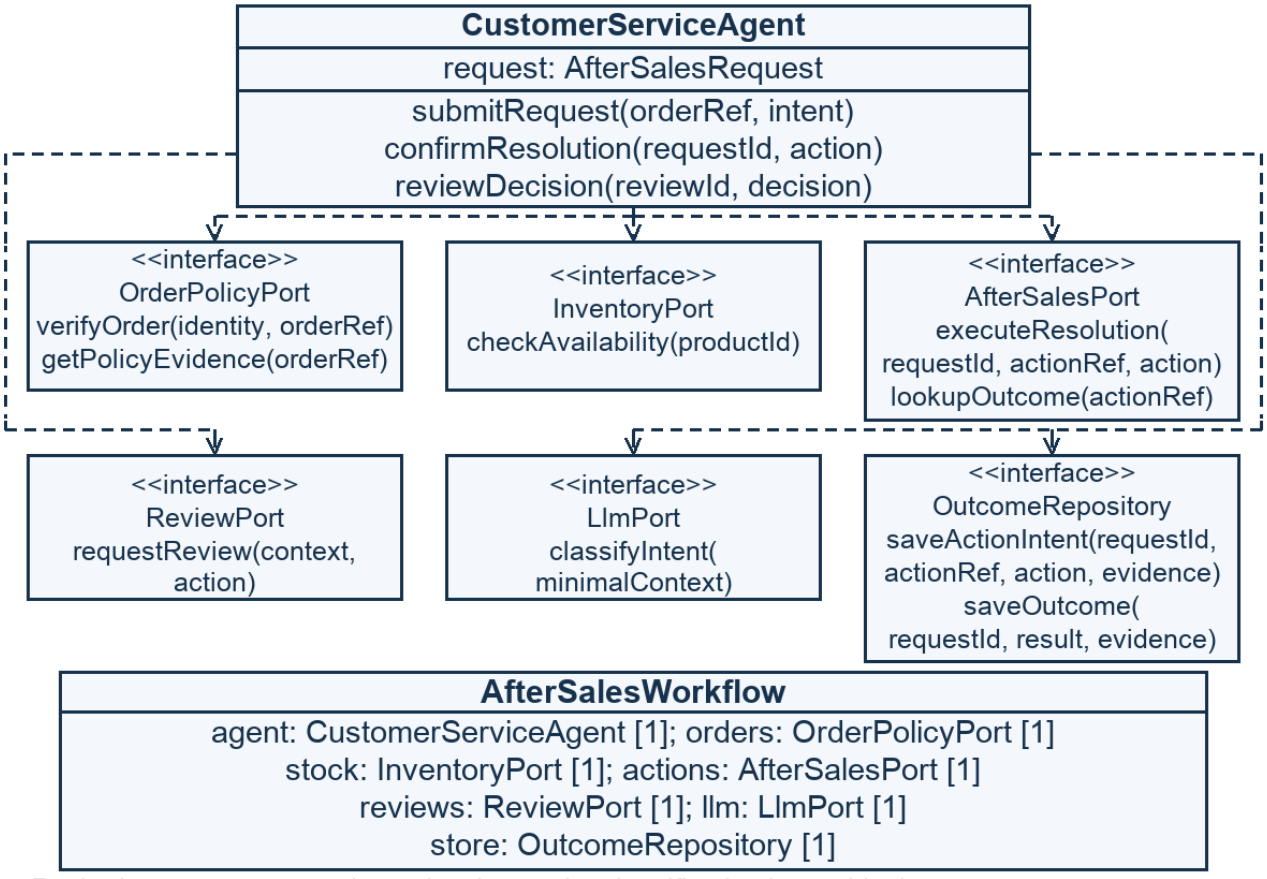
Figure 11 BA-CL-01B BA-local interfaces and adapter realisations with hollow triangle notation.

8.1.2 CS-CL-01 Complete local class model



History links are ordinary associations. PENDING is local review tracking, not DecisionKind.
ActionRef is a String action identity; CS state values and adapters: Section 8.2.

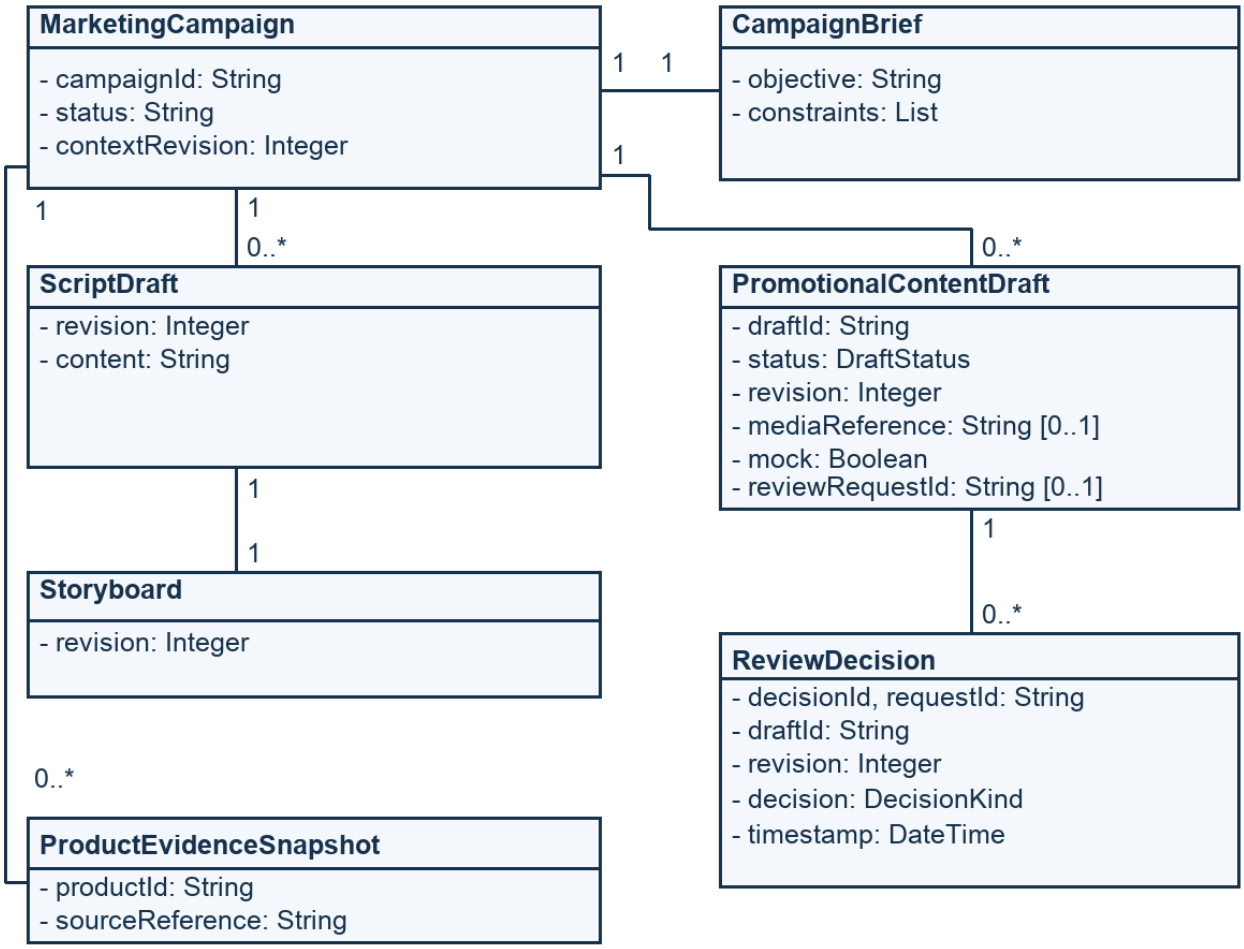
Figure 12 CS-CL-01A Request-linked evidence, approval and resolution history.



Dashed arrows are agent dependencies; optional notification is outside the core.

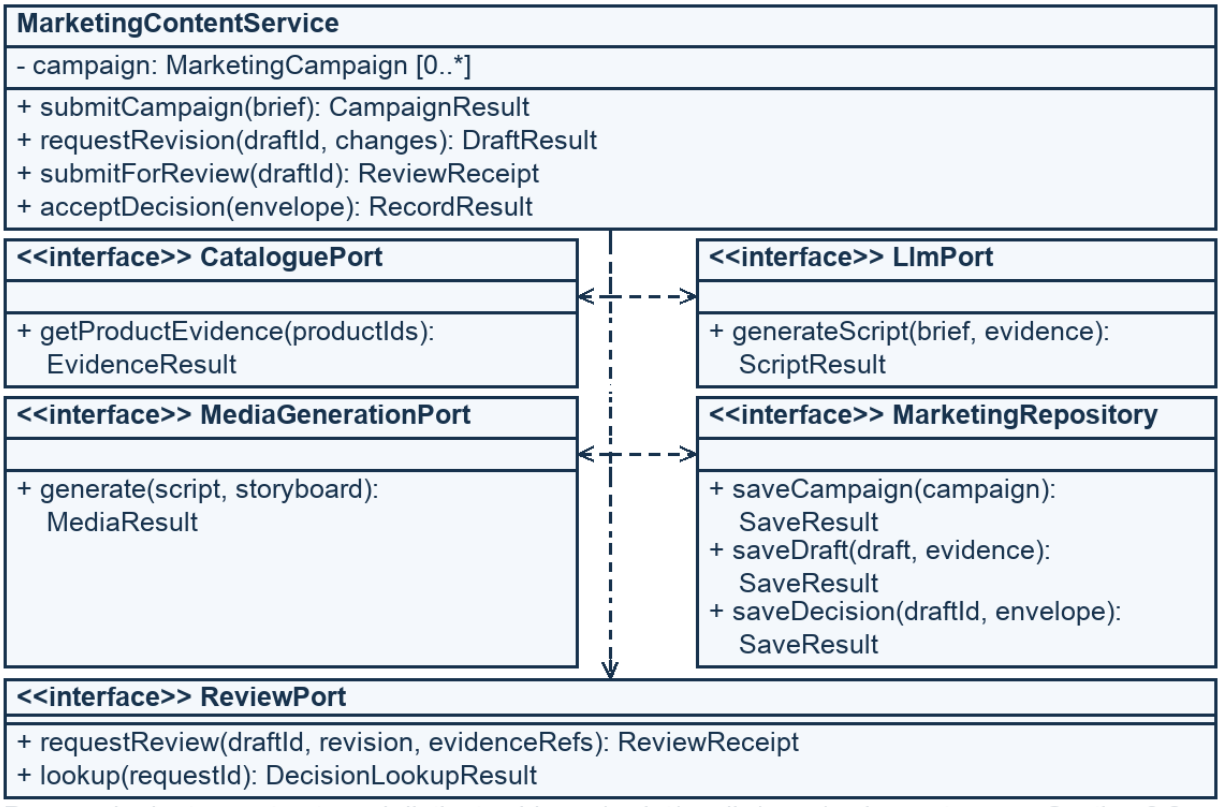
Figure 13 CS-CL-01B CS service operations, adapter interfaces and result contracts.

8.1.3 MC-CL-01 Complete local class model



Pending has reviewRequestId, no decision. ReviewDecision is a committed AP local view.
Retain history; each revised draft needs fresh exact-scope review.

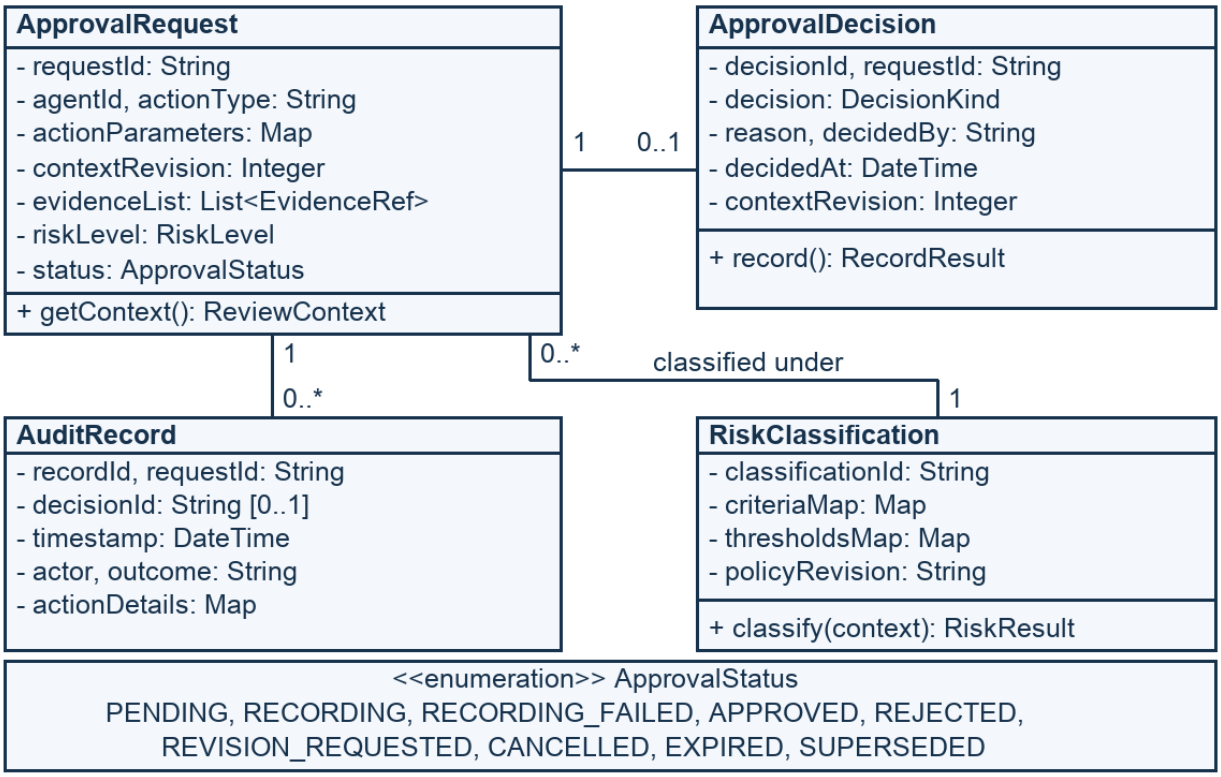
Figure 14 MC-CL-01A Campaign, draft revisions and committed decision history; pending has no decision.



Proposed adapter contracts and distinct evidence/script/media/save/review outcomes: Section 8.2.

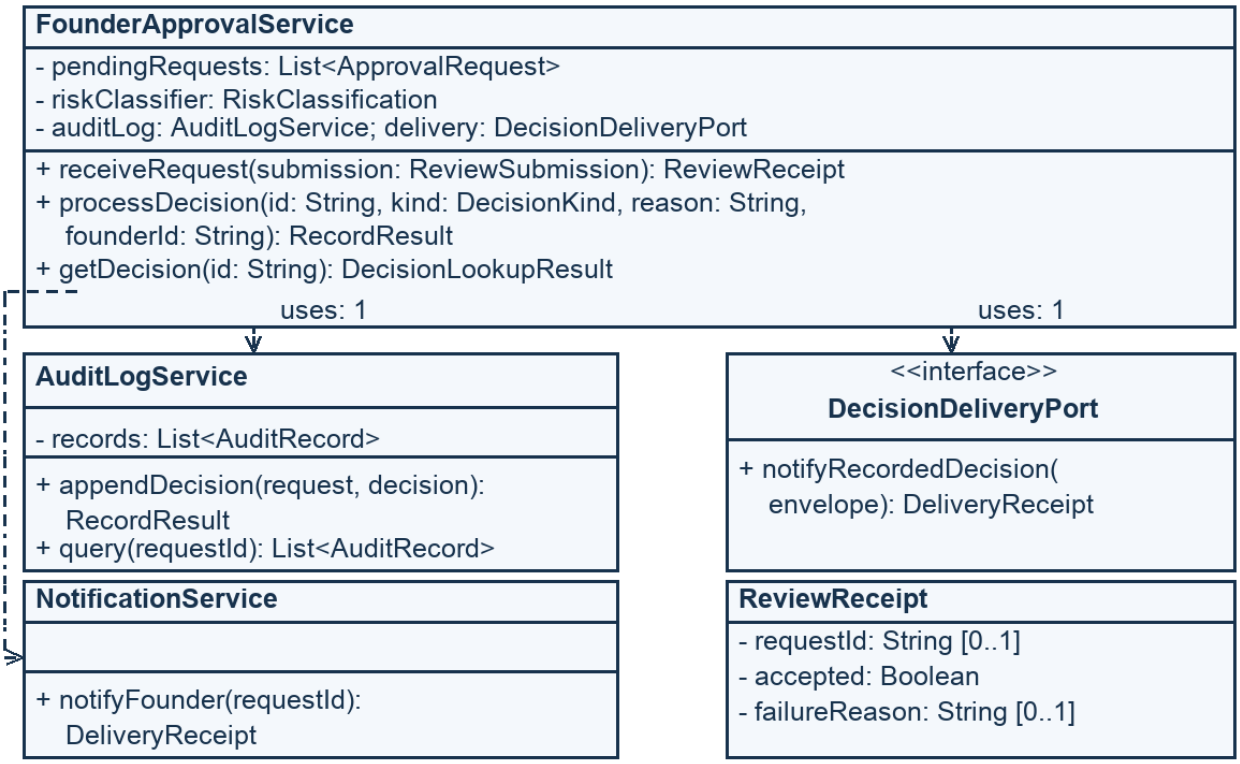
Figure 15 MC-CL-01B Concrete proposed MC-side receiver contracts and explicit result types.

8.1.4 AP-CL-01 Complete local class model



RiskLevel = LOW | MEDIUM | HIGH | CRITICAL; unknown classification blocks action.
DecisionKind and request/decision invariants are defined in Section 8.2.

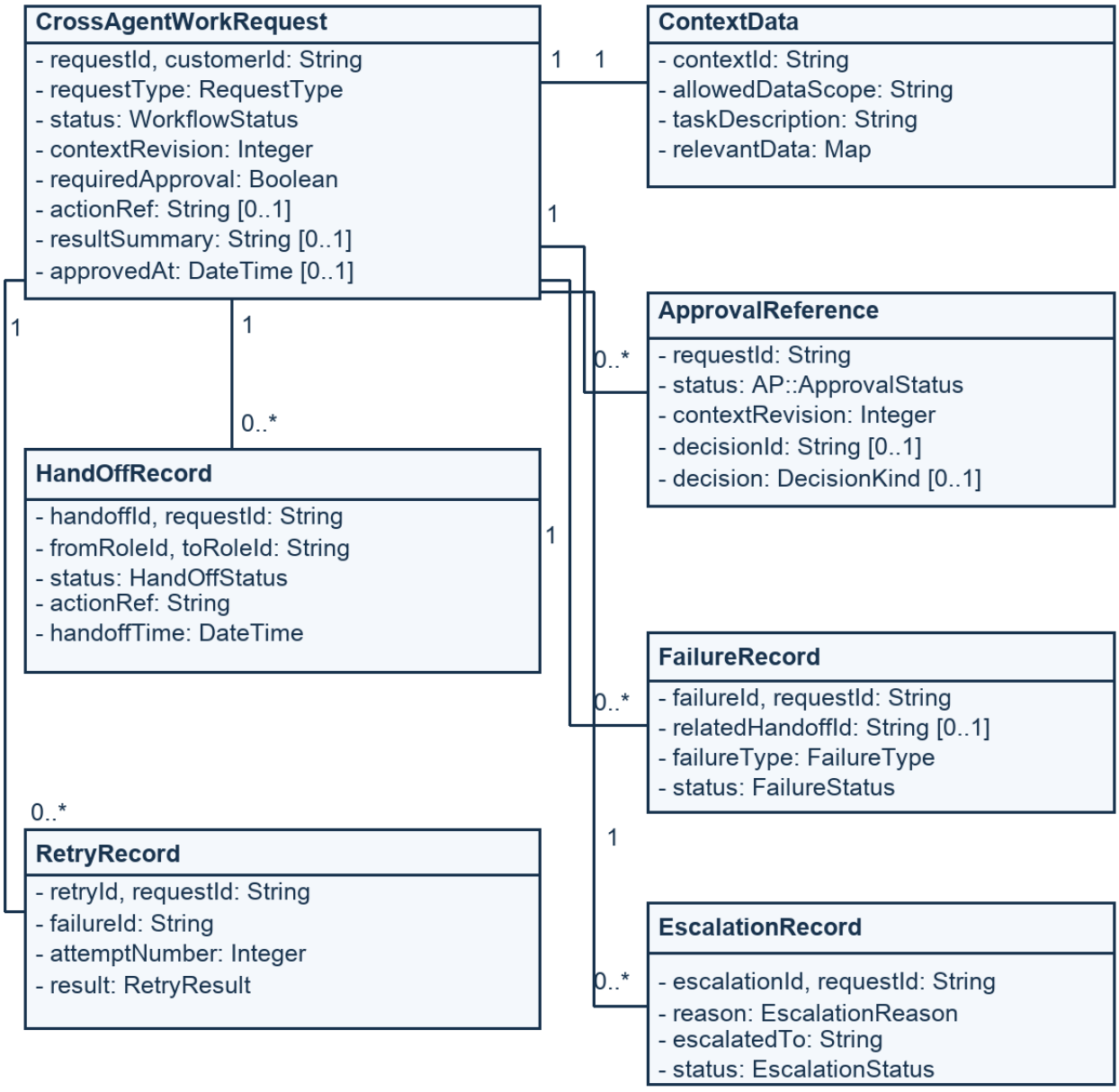
Figure 16 AP-CL-01A Approval request, optional committed decision, audit and risk policy.



Proposed atomic decision/request/audit commit precedes delivery. Failure grants no authority.
Result types and adapters: Section 8.2. Provider guarantees require implementation evidence.

Figure 17 AP-CL-01B Founder review, commit, lookup and recorded delivery contracts.

8.1.5 CO-CL-01 Complete local class model



ApprovalReference observes AP; pending has no decisionId/decision.
Retained trace history, failure/retry links and values: Section 8.2.

Figure 18 CO-CL-01A Work, context and trace records; ApprovalReference is a view of authoritative AP state.

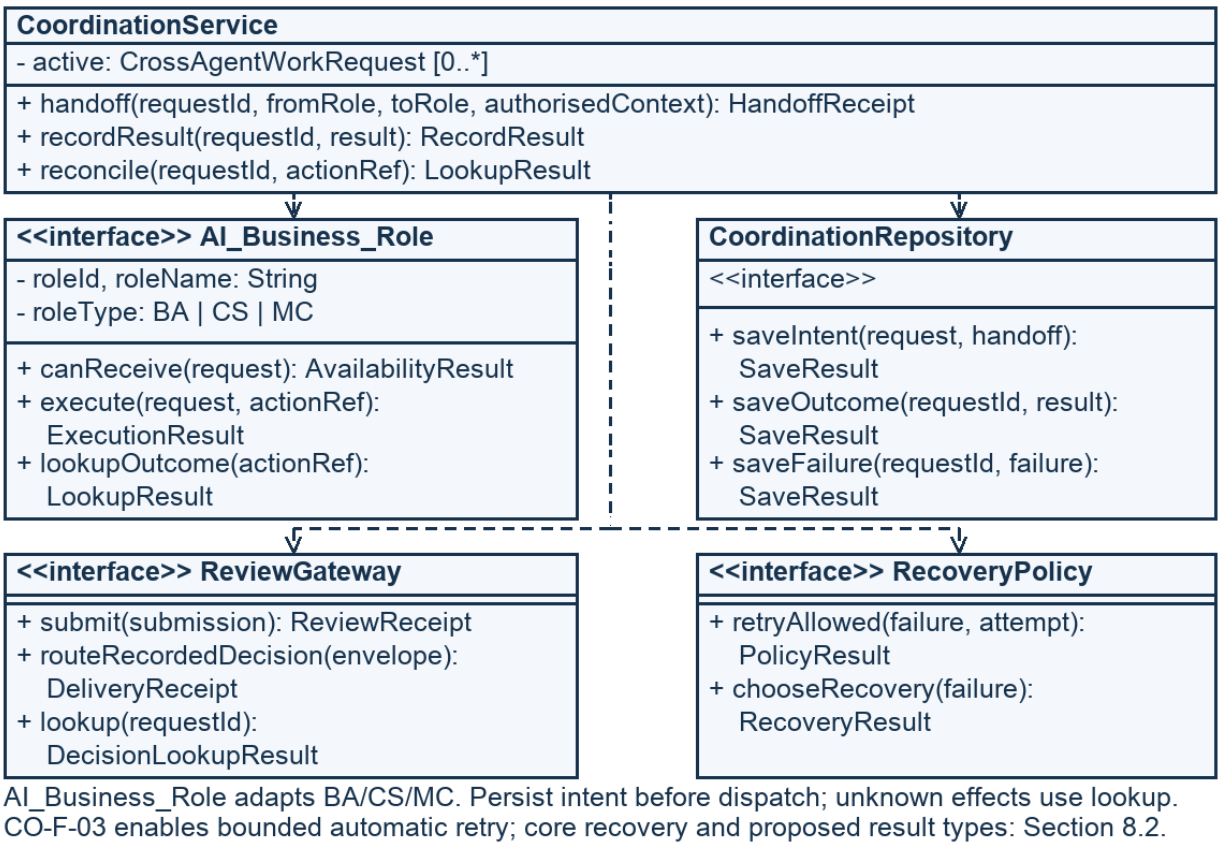


Figure 19 CO-CL-01B Coordination facade and proposed role, repository, review and recovery contracts.

8.2 Responsibilities and shared contracts

CompanyWorkflow has exactly one ba, cs, mc, co, approval and review part. BA/CS/MC are business roles; CoordinationService and ReviewGateway are proposed integration facades. Local signatures retain their distinct meanings; a shared provider may realise several adapters without making those interfaces identical.

Role	Receiver operations and result meaning
BA	BeautyAdvisorService submit/revise/cancel/acceptReview; CataloguePort.search(context); InventoryPort.check(productIds); LlmPort.propose(context,evidence); ReviewPort.request(context); ConsultationRepository.save(session). Proposal is validated, stock UNKNOWN is not available, receipt is not approval.
CS	CustomerServiceAgent submitRequest/confirmResolution/reviewDecision; OrderPolicyPort verifyOrder/getPolicyEvidence; InventoryPort.checkAvailability; LlmPort.classifyIntent; AfterSalesPort.executeResolution(requestId,actionRef,action)/lookupOutcome(actionRef); OutcomeRepository.saveActionIntent/saveOutcome. NotificationPort is optional.
MC	MarketingContentService submitCampaign/requestRevision/submitForReview/acceptDecision; CataloguePort.getProductEvidence; LlmPort.generateScript; MediaGenerationPort.generate; ReviewPort.requestReview/lookup; MarketingRepository.saveCampaign/saveDraft/saveDecision. These proposed adapter contracts do not establish provider API capabilities.
AP	FounderApprovalService.receiveRequest/processDecision/getDecision; AuditLogService.appendDecision/query; NotificationService alerts Founder; DecisionDeliveryPort.notifyRecordedDecision. Commit decision/request/audit before event delivery; failed commit grants no authority.
CO	CoordinationService.handoff/recordResult/reconcile; AI_Business_Role.canReceive/execute/lookupOutcome; CoordinationRepository.saveIntent/saveOutcome/saveFailure; RecoveryPolicy.retryAllowed/chooseRecovery. Role adapters preserve business-level checks and action-reference correlation.

Integration value	Fields or outcomes
ReviewSubmission	Origin role and originId; exact proposedAction; contextRevision; minimal authorisedContext/evidenceRefs; actionRef when applicable.
ReviewReceipt	accepted plus persisted requestId, or explicit failureReason. Accepted establishes a submitted request only.
RecordedDecisionEnvelope	AP requestId, originId, decisionId, DecisionKind, contextRevision and optional actionRef; retained mapping includes MC draftId/revision or CO/CS origin.
DecisionKind	APPROVE | REJECT | REQUEST_REVISION. PENDING is request state, not a committed decision.
DecisionLookupResult	pending, committed(envelope), missing, unavailable or invalid. AP getDecision and ReviewGateway.lookup return this type; querying creates no approval.
RecordResult / SaveResult	RecordResult: committed | invalid | recordingFailure. SaveResult: saved | invalid | recordingFailure. A committed failed business result is still failure.
HandoffReceipt / DeliveryReceipt	acknowledged | unavailable. Neither means completed execution. Role canReceive additionally exposes invalid/unauthorised inputs.
ExecutionResult / LookupResult	confirmed(result), notExecuted or stillUnknown/unknown. A confirmed result explicitly distinguishes successful and failed work; lookup performs no execution.
MC generator results	EvidenceResult returns authorised references/facts or missing/unavailable/invalid. ScriptResult returns proposed script/storyboard or failure. MediaResult distinguishes real success, labelled permitted mock and failure.
RecoveryPolicy results	PolicyResult explicitly permits or blocks bounded retry; RecoveryResult chooses clarification, read-only reconciliation, storage recovery, escalation or safe cancellation. No numeric retry/expiry threshold is implied.
CS action identity	ActionRef is a String alias for correlated action identity. confirmedAction retains the selected action reference; review and execution records retain the same identity.

8.2.1 Role adapter mappings

CS::ReviewPort.requestReview maps its exact request/action/revision into ReviewSubmission and retains the AP requestId as its reviewId correlation. Committed APPROVE/REJECT/REQUEST_REVISION map to CS APPROVED/REJECTED/REVISION_REQUESTED only after scope validation. CS independently rechecks order/policy, stock and Customer confirmation, then persists action intent before its own after-sales execution.

BA::ReviewPort.request retains the session and context revision. The adapter invokes BeautyAdvisorService.acceptReview(envelope), validating origin/session/context revision and decisionId. Current APPROVE resumes ACTIVE evidence validation, REJECT records INCOMPLETE, and REQUEST_REVISION requires correction. Failed recording is exposed; absent, stale or duplicate results grant no authority. The callback never jumps directly to COMPLETED. The existing BA CataloguePort, InventoryPort and LlmPort signatures remain unchanged.

MC::ReviewPort.requestReview maps campaign/draftId and exact revision to AP submission. The adapter retains origin-to-AP mapping and invokes acceptDecision only for committed correlated envelopes. MC saves ReviewDecision as a local view before APPROVED. New/revised variants get fresh review; stale approval is retained historically but cannot cover new content.

CO ApprovalReference history observes authoritative AP requests/decisions. Current approval resumes target/context/authority checks, never direct completion. CO::ReviewGateway is the local contract implemented by the group ReviewGateway facade. AI_Business_Role adapters preserve

BA/CS/MC checks and CO-to-role request/actionRef correlation; CO gains no settlement/publication access.

ReviewGateway.submit invokes AP receiveRequest; lookup forwards AP getDecision as DecisionLookupResult. routeRecordedDecision validates origin/revision mapping and delivers committed envelopes only. Stale/mismatched/duplicate decisionId grants no authority or repeated dispatch; AP retains authoritative decisions.

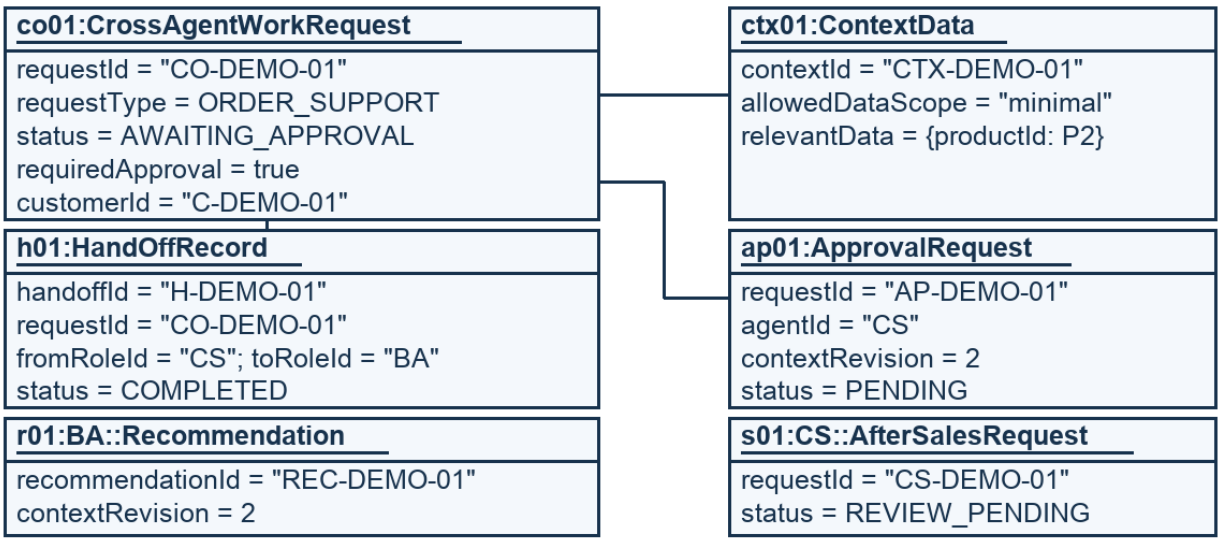
8.2.2 State and value vocabularies

Type	Declared values and invariants
BA ConsultationStatus	ACTIVE, CLARIFYING, REVIEW_PENDING, COMPLETED, INCOMPLETE, CANCELLED. Inventory AvailabilityStatus: AVAILABLE, UNAVAILABLE, UNKNOWN.
CS RequestState	CS-STM-01 A/B define one lifecycle covering assessment, clarification, confirmation/review, execution, recording and reconciliation. RequestState = ASSESSING, NEEDS_INFORMATION, AWAITING_CONFIRMATION, REVIEW_REQUESTED, REVIEW_PENDING, EXECUTING, RECONCILING, RECORDING, RECORDING_FAILED, RESOLVED, UNRESOLVED, REJECTED, CANCELLED. CS ReviewDecision tracks PENDING, APPROVED, REJECTED or REVISION_REQUESTED; PENDING has no committed AP decision. The adapter maps committed DecisionKind values separately.
MC DraftStatus	DRAFTING, VALIDATING, GENERATING, REVIEW_PENDING, REVISION_REQUIRED, CONTENT_REJECTED, GENERATION_FAILED, INCOMPLETE, RECORDING_FAILED, APPROVED, REJECTED, CANCELLED, SUPERSEDED. MarketingCampaign.status is a string projection of the current owned workflow, not an independent approval.
AP ApprovalStatus	PENDING, RECORDING, RECORDING_FAILED, APPROVED, REJECTED, REVISION_REQUESTED, CANCELLED, EXPIRED, SUPERSEDED. One request has at most one committed decision; changed scope creates a linked new request.
CO WorkflowStatus	PENDING_VALIDATION, CLARIFICATION_REQUIRED, AWAITING_APPROVAL, HANDOFF_SENT, EXECUTING, COMPLETED, FAILED, RETRYING, RECONCILING, RECORDING_FAILED, ESCALATED, CANCELLED. RETRYING, RECONCILING and RECORDING_FAILED distinguish execution retry, outcome lookup and storage recovery.
CO record values	RequestType: CONSULTATION, ORDER_SUPPORT, MARKETING_CONTENT, OTHER. HandOffStatus: PENDING, SENT, ACKNOWLEDGED, EXECUTING, COMPLETED, FAILED. FailureType: VALIDATION_ERROR, ROLE_UNAVAILABLE, SERVICE_FAILURE, APPROVAL_UNAVAILABLE, UNCERTAIN_OUTCOME, OTHER. FailureStatus: RECORDED, RESOLVED, OPEN; RetryResult: SUCCESS, FAILED, UNKNOWN; EscalationStatus: OPEN, RESOLVED; RoleStatus: AVAILABLE, UNAVAILABLE, UNKNOWN.
CO links and optionality	Each FailureRecord may link zero/one HandOffRecord, and may have zero/more RetryRecords; each RetryRecord refers to one FailureRecord. RetryRecord may link zero/one EscalationRecord. EscalationReason includes APPROVAL_REQUIRED, RETRY_EXHAUSTED, UNCERTAIN_EXECUTION, ROLE_UNAVAILABLE, SERVICE_FAILURE, OTHER. Retry records exist only if selected/attempted; unresolved escalation is resumable.

9 Complex Structure Modelling

9.1 Runtime object snapshots

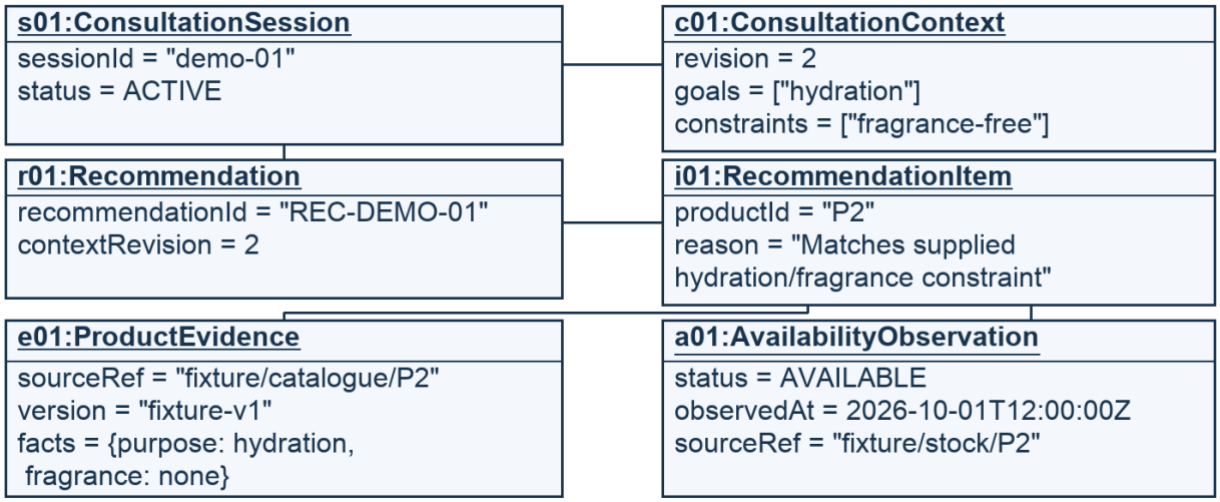
Object diagrams instantiate the class models with underlined names, typed slots and links at defined synthetic instants. BA-OBJ-01 precedes GR-OBJ-01; the other snapshots show their respective scenarios. Communication and sequence views specify message order.



Partial snapshot after BA delivery, before AP decision; completed handoff is not business completion. BA-OBJ-01 shows an earlier pre-delivery instant; correlation is retained by GR-CL-01D.

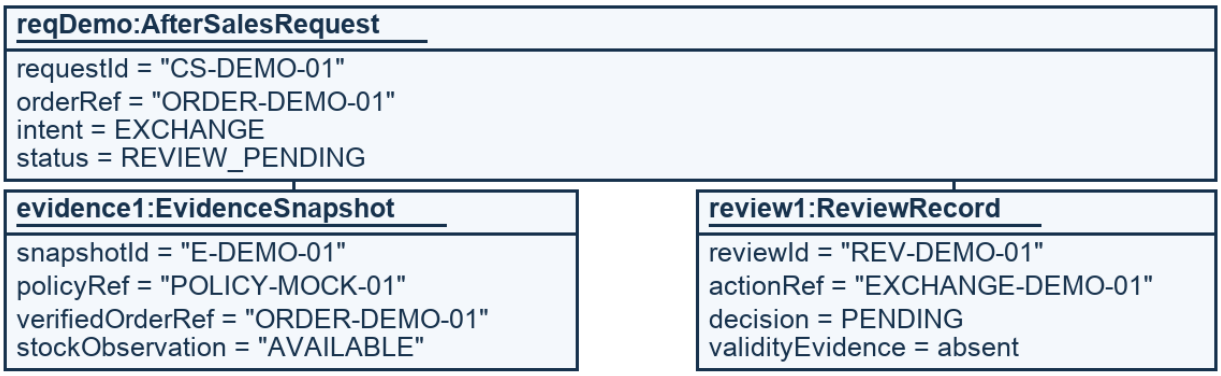
Figure 20 GR-OBJ-01 After BA advice hand-off, before Founder decision: CO AWAITING_APPROVAL, CS REVIEW_PENDING and no AP committed decision.

BA-UC-01 step 8 | after saving, before delivery | synthetic data



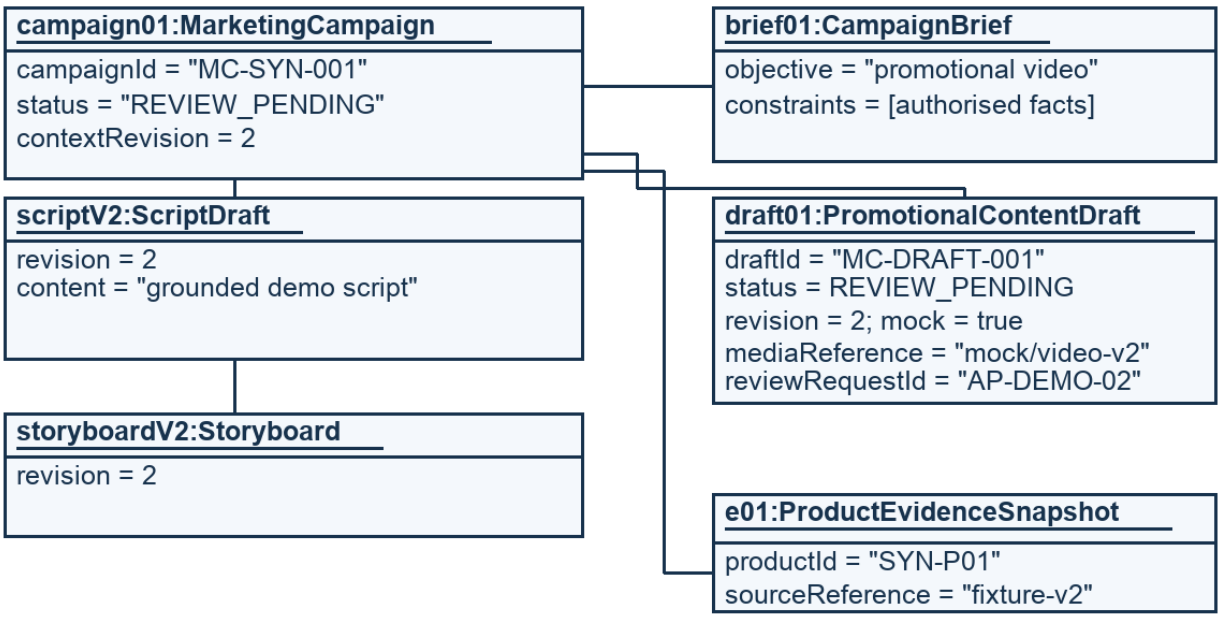
One item is shown; ACTIVE lasts until delivery. This precedes GR-OBJ-01.

Figure 21 BA-OBJ-01 Saved evidence before delivery; ACTIVE remains until the current result is delivered.



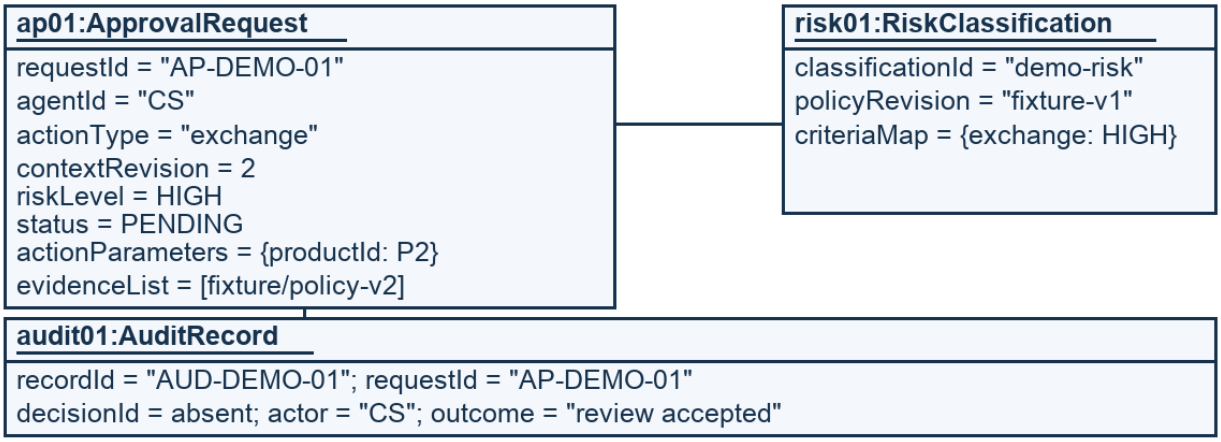
Acknowledged pending review; no ResolutionRecord or permission to execute exists.

Figure 22 CS-OBJ-01 Acknowledged pending review with evidence; no executed resolution record.



Saved mock draft with acknowledged review; no ReviewDecision exists yet.

Figure 23 MC-OBJ-01 Saved synthetic mock draft awaiting exact-revision review; no ReviewDecision exists yet.



Persisted PENDING receipt; no ApprovalDecision or executed exchange exists.

Figure 24 AP-OBJ-01 Persisted PENDING request with audit/risk fixture and zero committed decisions.

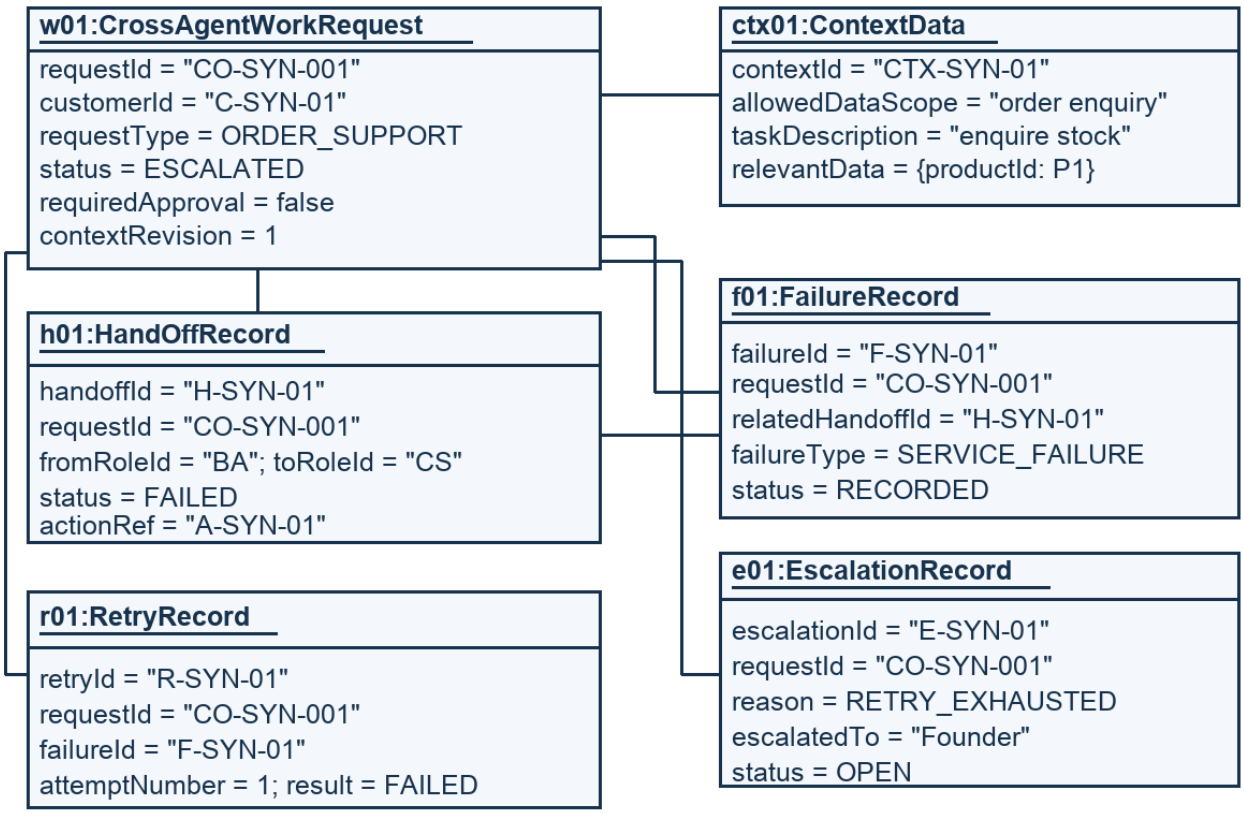


Figure 25 CO-OBJ-01 Known failed read-only handoff after optional retry and recorded escalation; no purchase or implied approval.

9.2 Composite structure and communication

Local collaborator views type concrete adapter roles by their declared contracts; interfaces are not instantiated. Binary links show collaboration; communication/sequence views supply ordering.

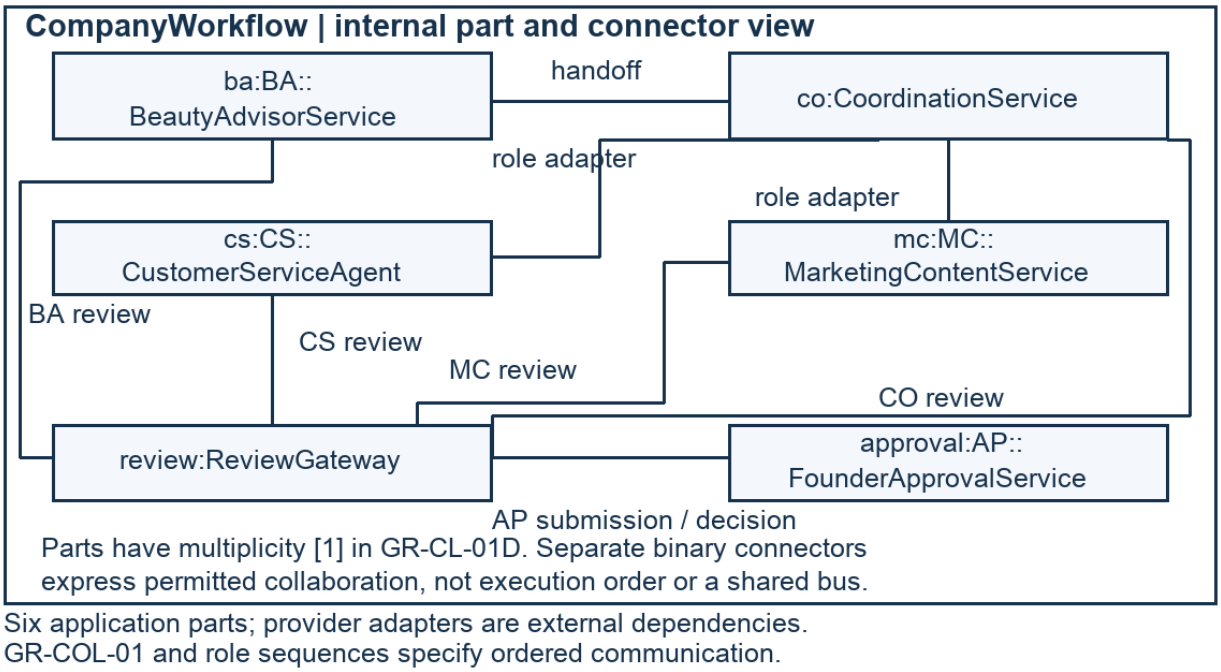
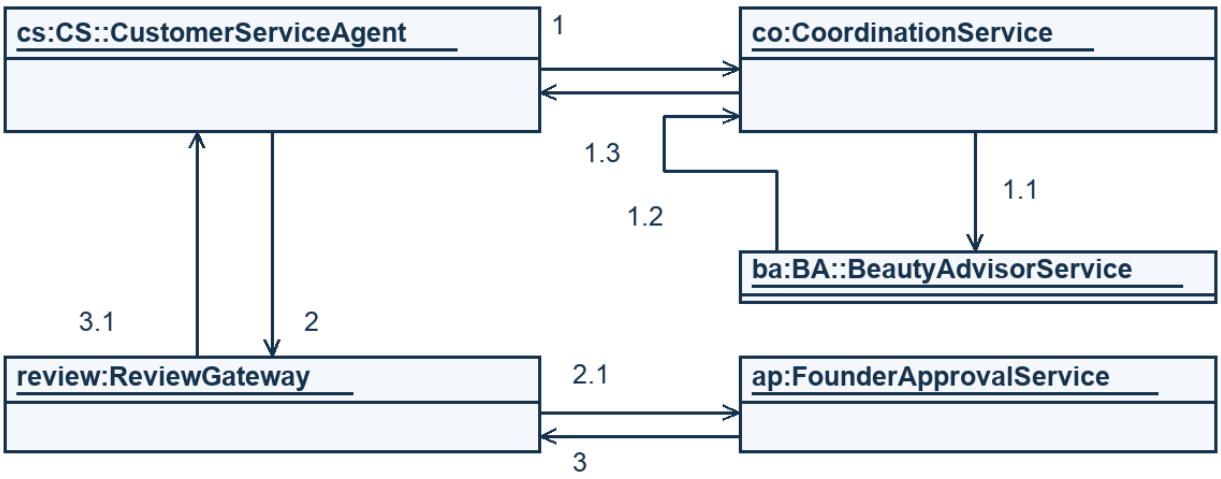


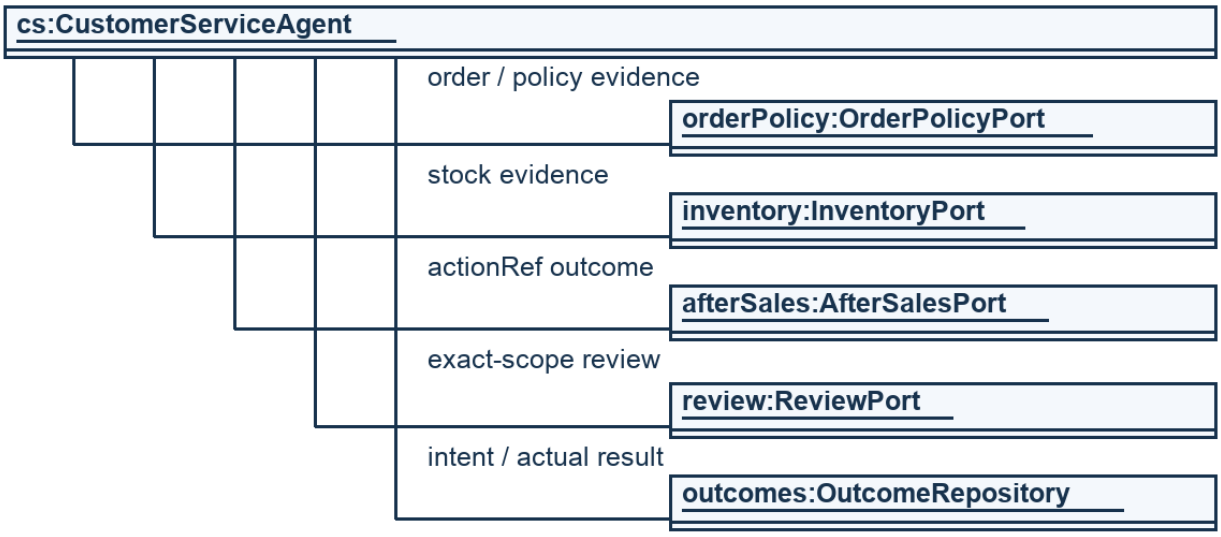
Figure 26 GR-CMP-01 Overall CompanyWorkflow with six typed parts and separate binary connectors; external adapters are outside this internal view.



1 handoff(CO id, CS, BA, authorisedContext); 1.1 submit(input) through BA adapter
1.2 recordResult(CO id, validated recommendation); 1.3 return result to CS
2 submit(ReviewSubmission for exact proposed action); 2.1 receiveRequest(submission)
3 recorded decision delivered after audit commit; 3.1 reviewDecision(CS reviewId, decision)
Replies/receipts omitted for clarity; AP-SEQ-01 and CS-SEQ-01 specify them.
Customer authorises minimal transfer; Founder coordinates the existing task. BA advises only.
CS rechecks policy, stock, confirmation and current authority before action (Section 8.2).

Figure 27 GR-COL-01 Numbered GR-S-01 messages distinguish advice, coordination and committed review.

Communication mapping: 1 calls CoordinationService.handoff; 1.1 adapts authorised Customer input to existing BA submit; 1.2 records the validated result; 1.3 returns it to CS. 2 calls ReviewGateway.submit; 2.1 calls AP receiveRequest. After Founder decision/audit commit, 3 routes the recorded event and 3.1 invokes the CS reviewDecision adapter. Absent/pending/rejected decisions cannot dispatch an exchange. MC is a distinct runtime part with its own Founder campaign path.



Partial adapter-role view; LLM and optional notification are omitted (Section 9.2).

Figure 28 CS-CMP-01 CS collaborators for evidence, review, execution and outcome recording.

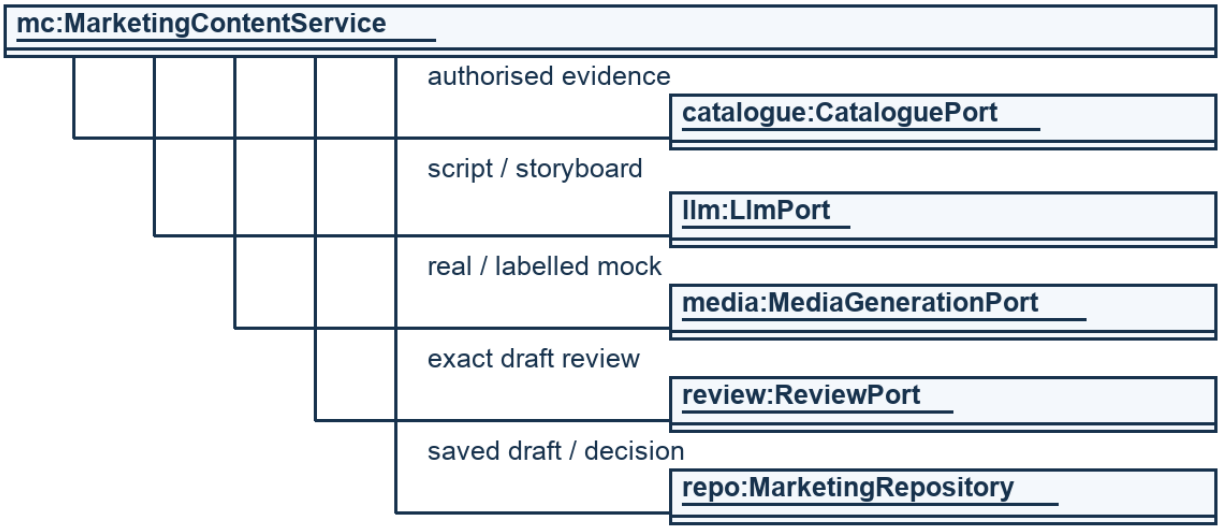


Figure 29 MC-CMP-01 Partial MC typed-instance collaboration through declared adapter boundaries.

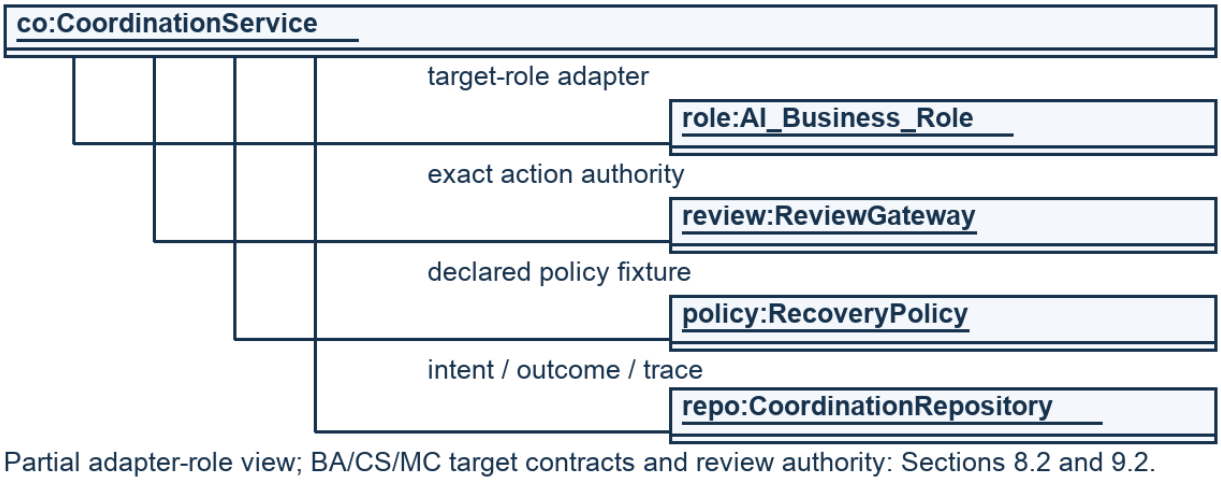


Figure 30 CO-CMP-01 CoordinationService collaborators for role dispatch, review, recovery policy and storage.

9.3 Runtime dependencies and failure boundaries

CO transfers authorised minimal context between BA and CS. BA supplies advice; CS verifies exchange eligibility, stock and Customer confirmation. MC follows an independent campaign path with review of each exact draft revision. The adapters in Section 8.2 connect these paths to AP.

Across these paths, request receipt, committed approval, business execution and recorded completion are distinct. Section 8.2 defines their authority, lookup and recovery contracts. No distributed rollback or production idempotency guarantee is presumed.

10 Behaviour Modelling

Each chain has three complementary behaviour models: activity diagrams show guarded processing paths, sequence diagrams show calls and replies, and state machines show entity lifecycles. Matching prefixes link them to the requirement/use case in Section 13.1. Labelled panels form one model; a pending/failure activity return may leave its entity resumable.

10.1 Activity models

Activity notation: diamonds select guarded alternatives; [else] is the complement. A/B circles join panels. X ends a returned-status flow; later events may resume the correlated entity. These slices require no concurrency.

10.1.1 BA-ACT-01 Consultation processing

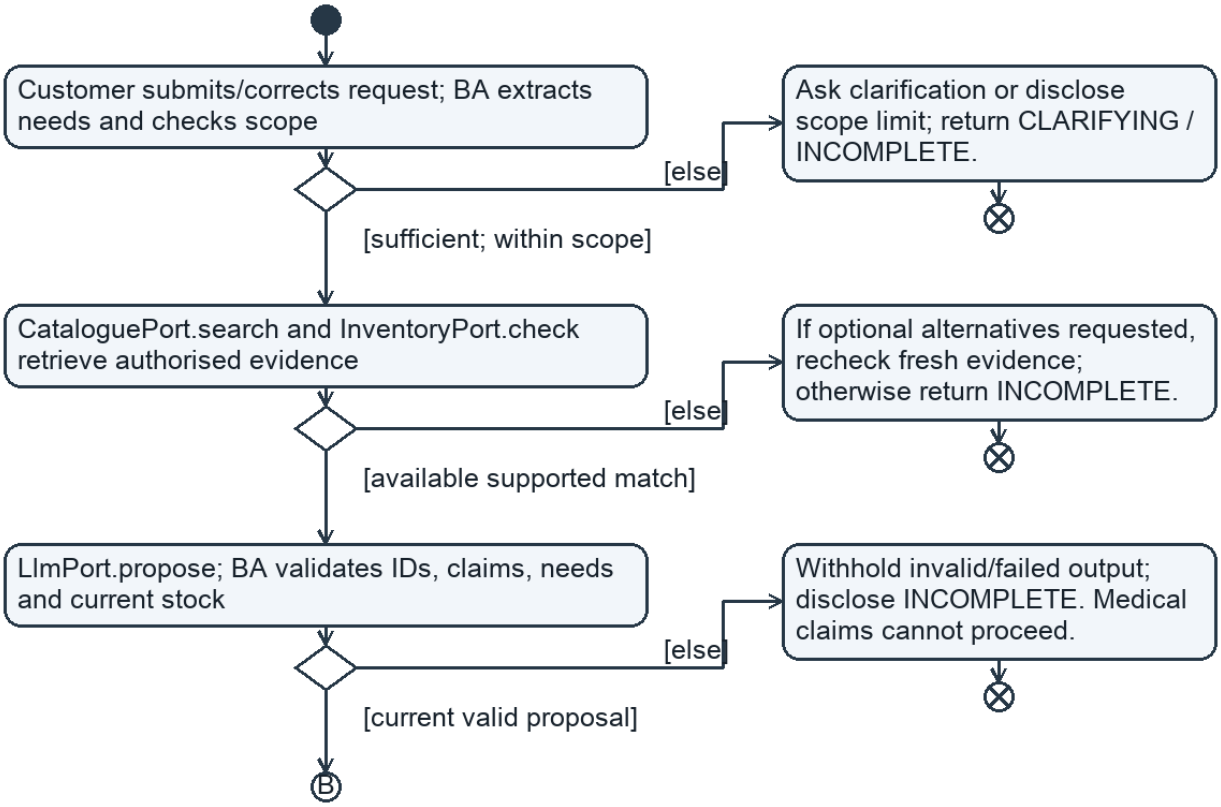


Figure 31 BA-ACT-01A Consultation assessment with guarded exits and an explicit B continuation.

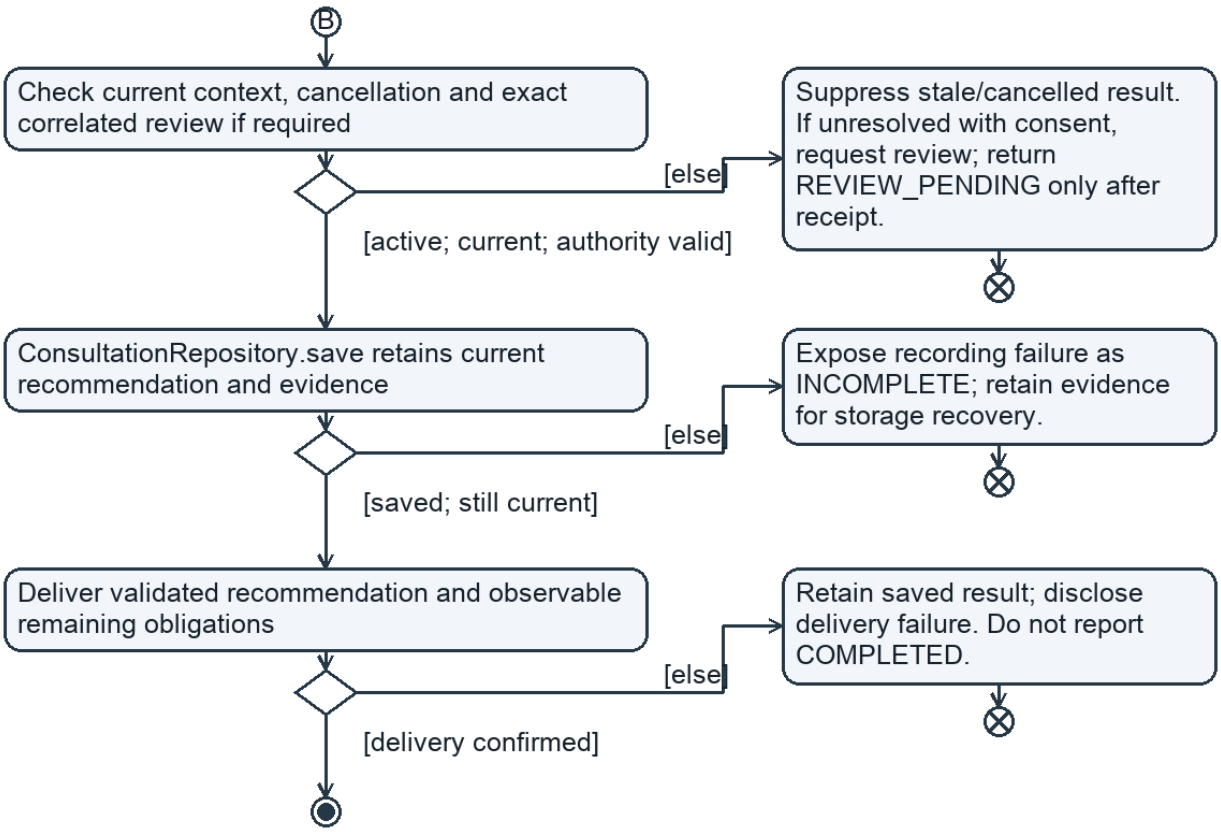


Figure 32 BA-ACT-01B B continuation: guarded current-scope, persistence and delivery outcomes.

10.1.2 CS-ACT-01 After-sales processing

A/B references continue the same CS processing activity. Clarification, review, recording and reconciliation returns preserve the request status for later events.

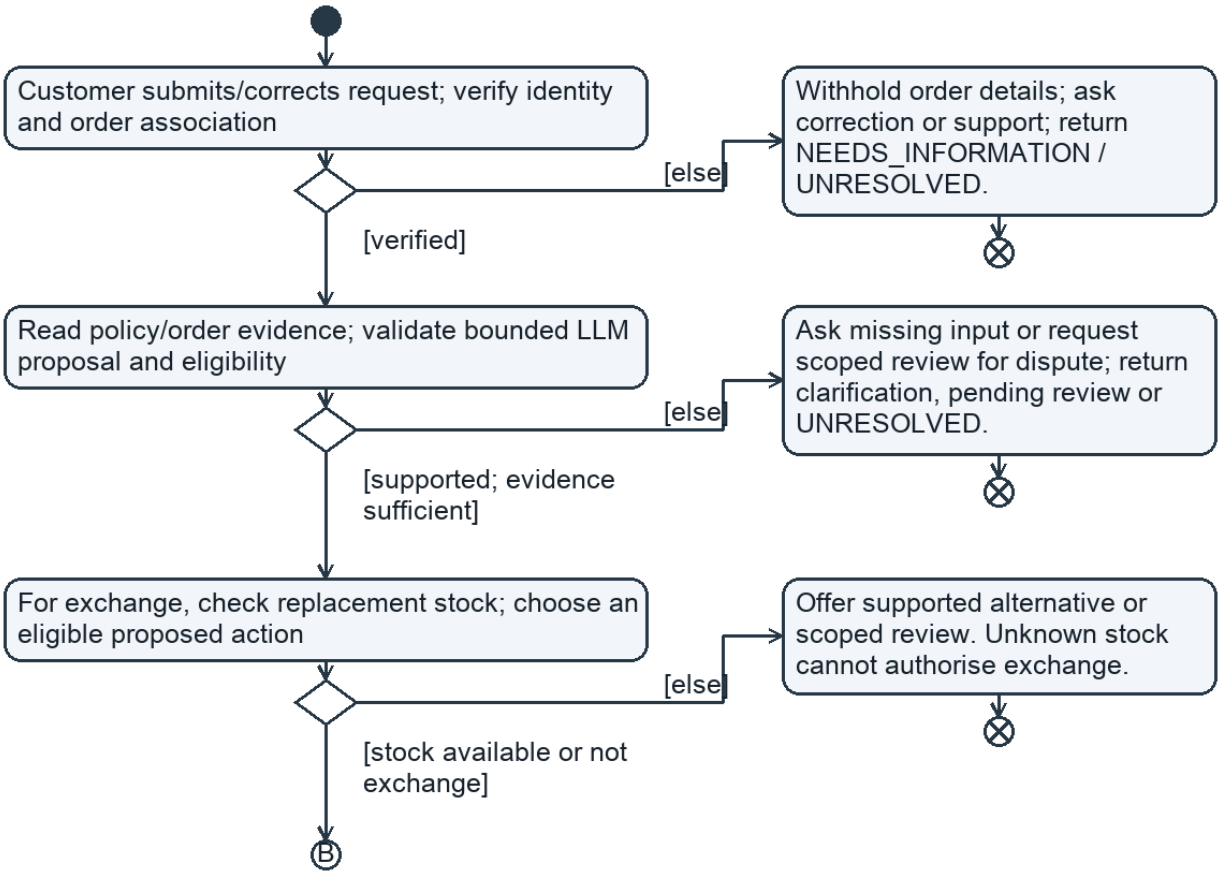


Figure 33 CS-ACT-01A Intake, evidence and routing.

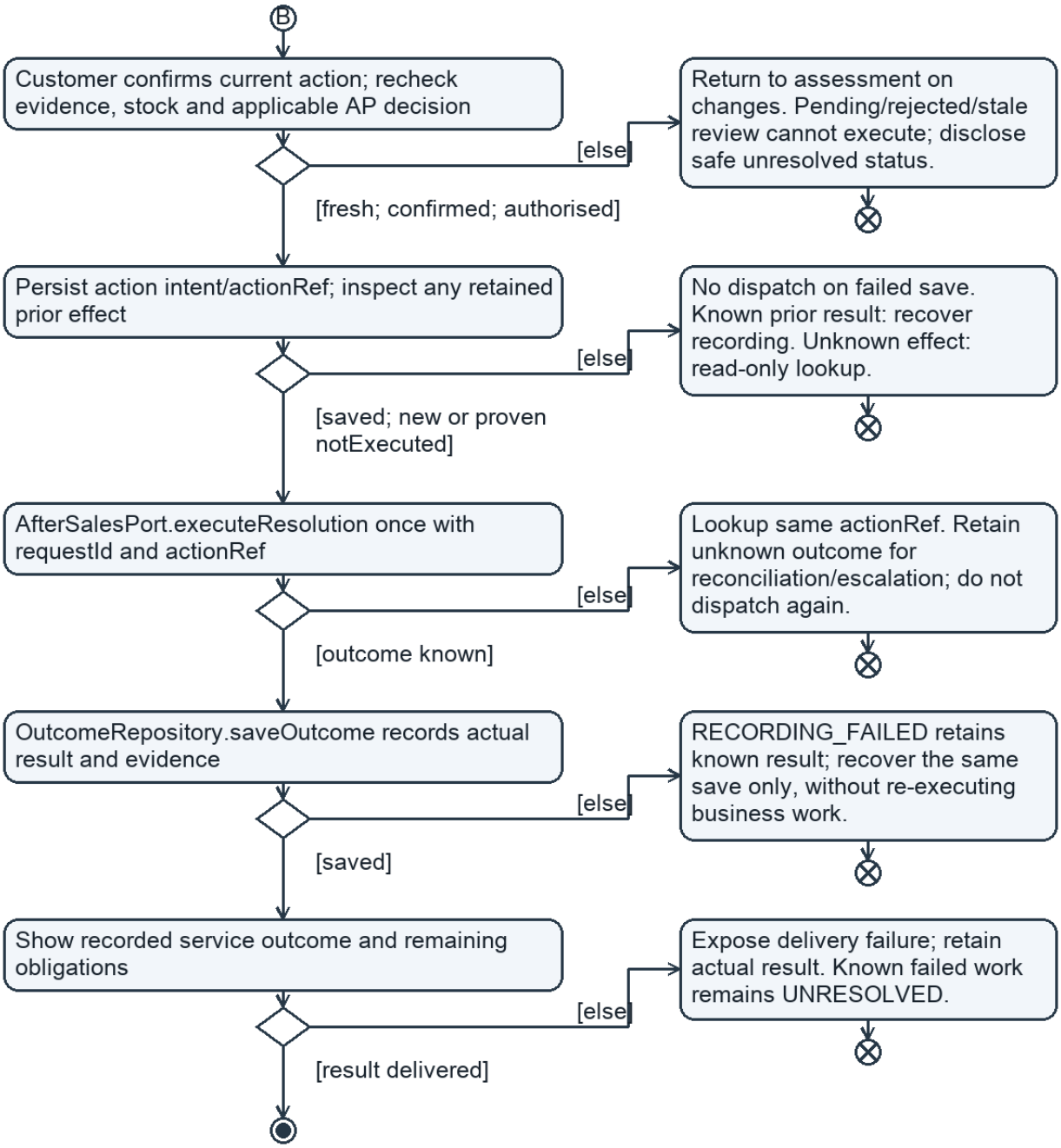


Figure 34 CS-ACT-01B Human review, confirmed execution and safe failure handling.

10.1.3 MC-ACT-01 Draft generation and review

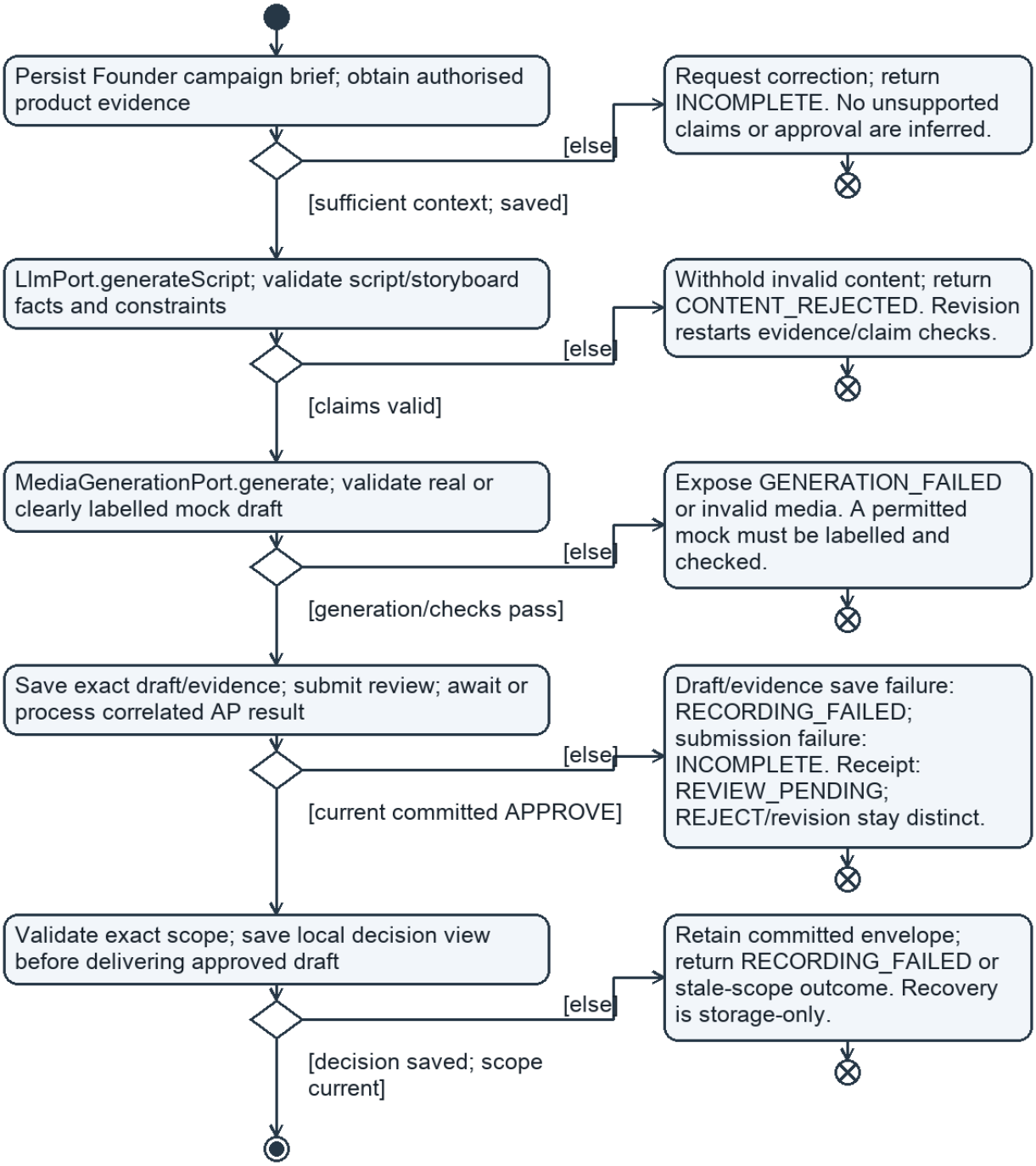


Figure 35 MC-ACT-01 Evidence/media checks precede review; only current committed approval plus local save permits APPROVED.

MC correction/revision repeats evidence, claim and media checks before a new review. Draft-save recovery resumes submission; decision-save recovery retains the committed envelope (MC-STM-01). Cancellation suppresses late content, and optional variants do not control core success.

10.1.4 AP-ACT-01 Approval processing

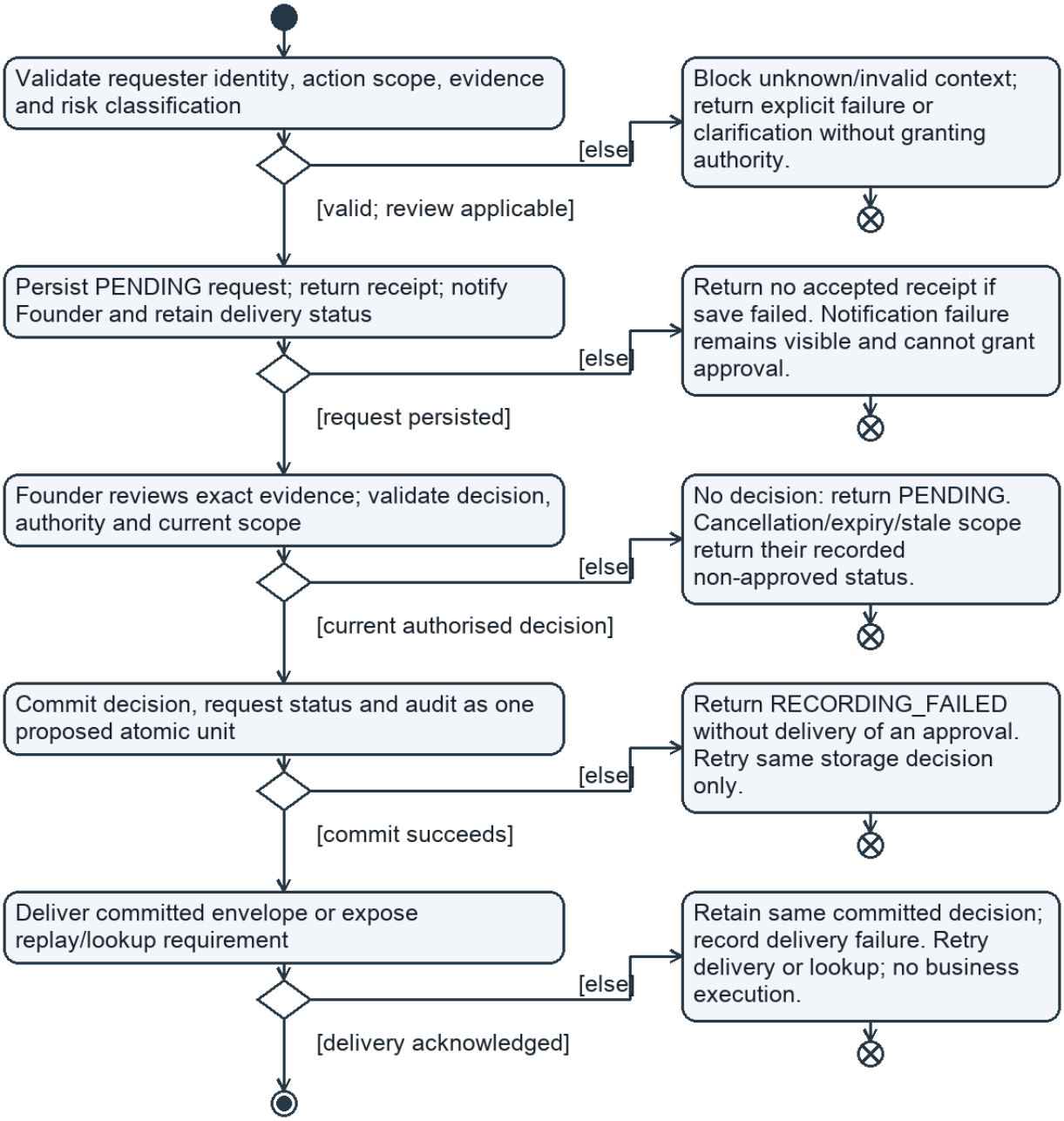


Figure 36 AP-ACT-01 No business dispatch inside approval; commit and delivery failure remain distinct.

10.1.5 CO-ACT-01 Handoff and recovery

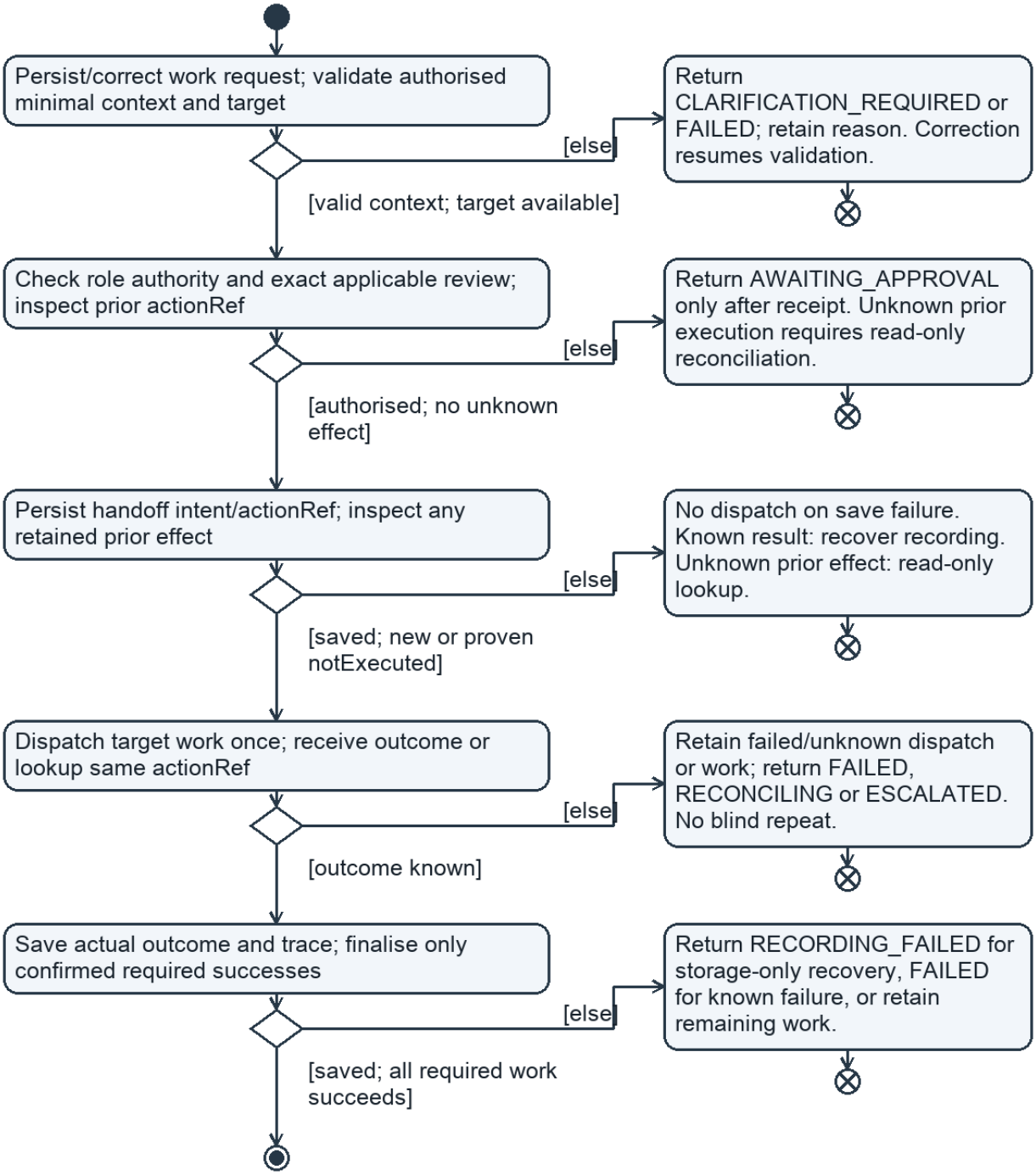


Figure 37 CO-ACT-01 Persist intent before target dispatch; unknown execution and failed save cannot complete work.

CO returns corrected context, current approval or remaining required work to validation. Its recovery paths distinguish recorded failure, uncertain execution and bounded optional retry. Unsafe recovery escalates; cancellation preserves known or unknown side effects.

10.2 Interaction models

Interaction notation: solid arrows show calls/events; dashed arrows show replies. Section 8.2 defines receiver operations and result types. Sequence branches complement the activity paths and state transitions.

10.2.1 BA-SEQ-01 Consultation interactions

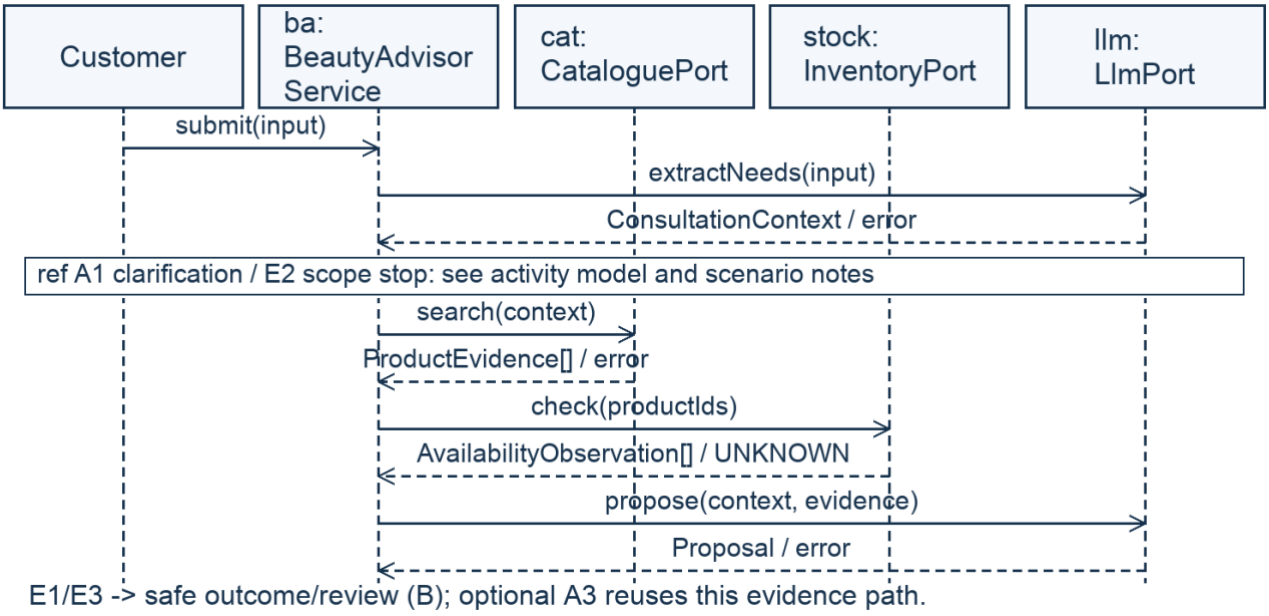


Figure 38 BA-SEQ-01A Evidence and LLM proposals validated before advice.

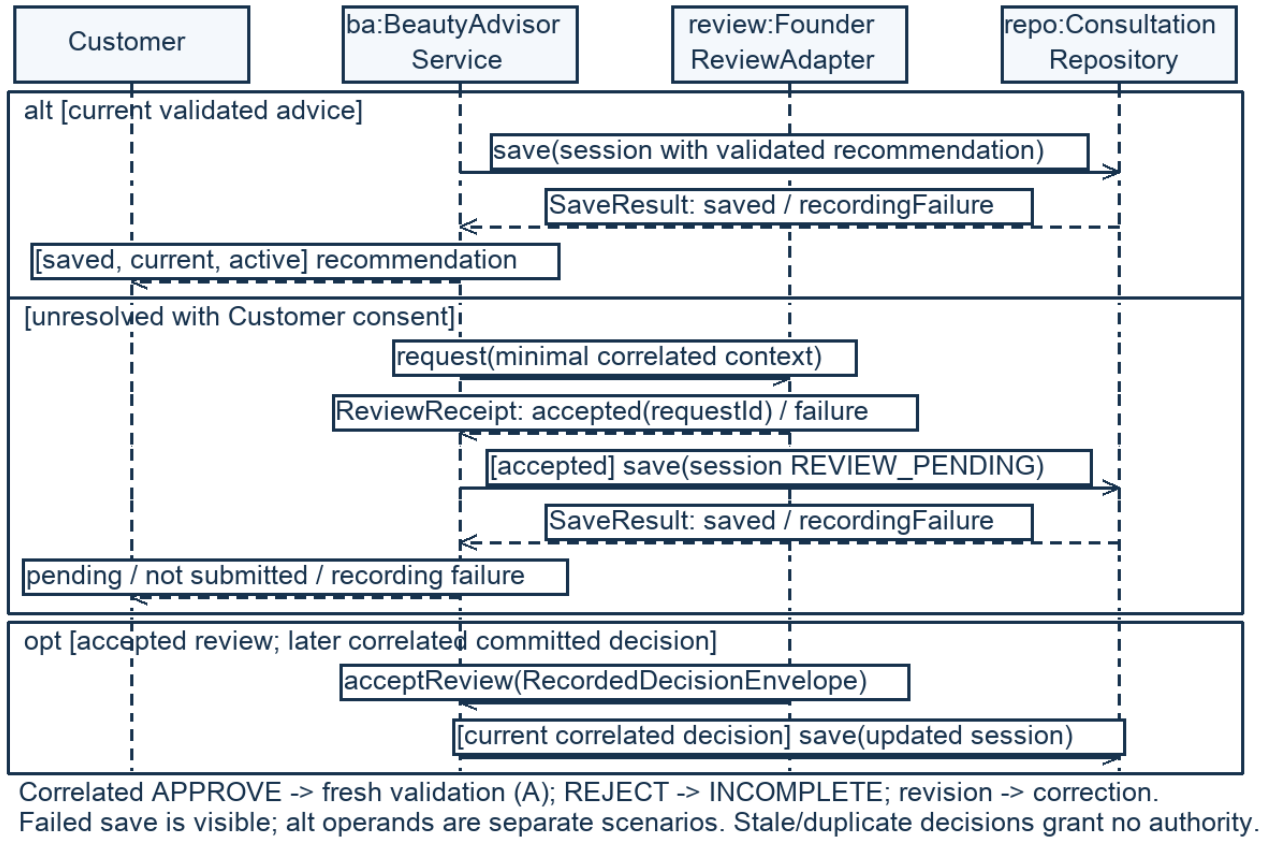


Figure 39 BA-SEQ-01B Alternative advice/review scenarios and the declared acceptReview callback; correlated approval resumes validation.

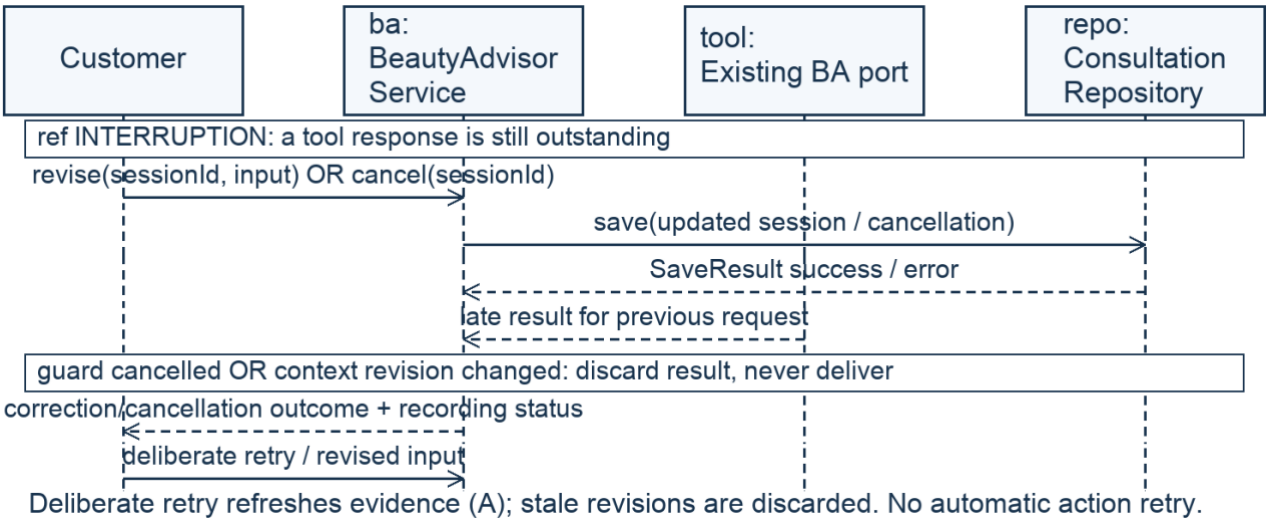


Figure 40 BA-SEQ-01C Correction/cancellation suppress stale replies; deliberate retry refreshes evidence.

10.2.2 CS-SEQ-01 After-sales interactions

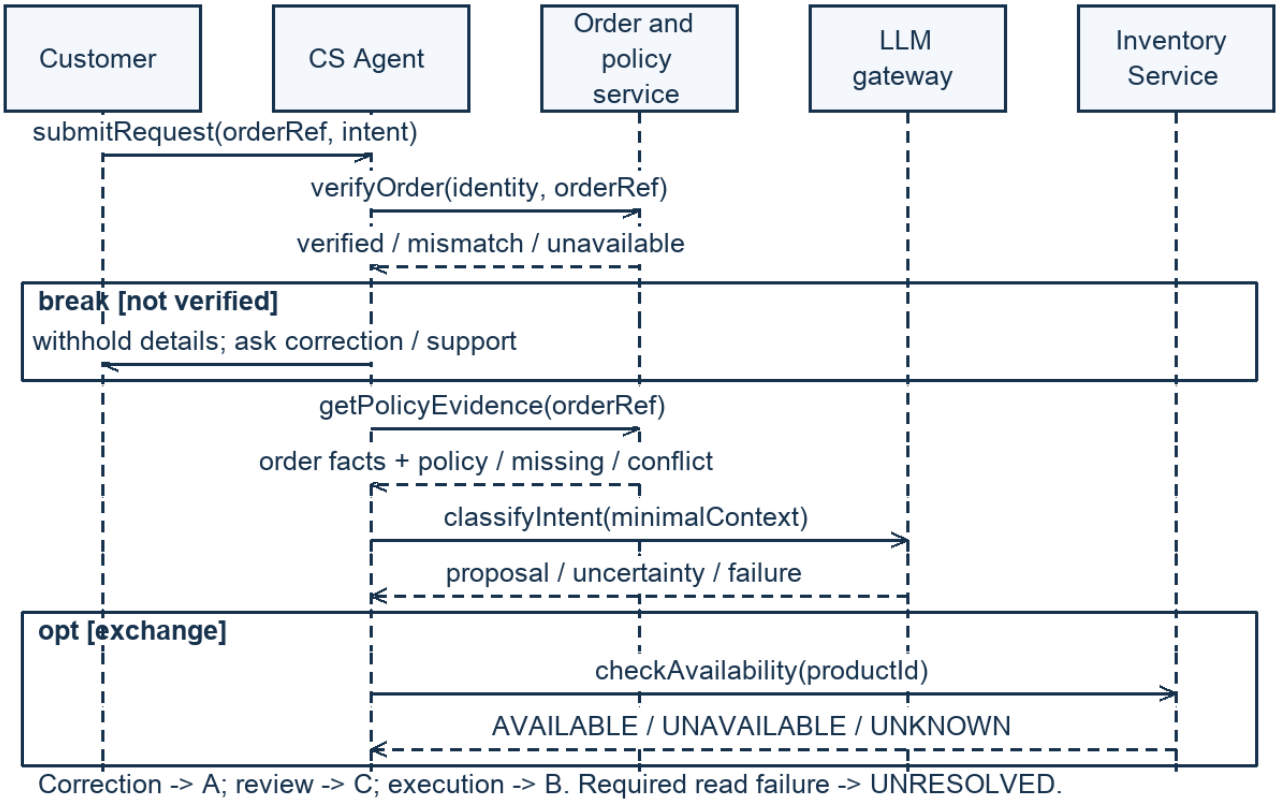


Figure 41 CS-SEQ-01A Identity/policy, bounded LLM and inventory reads.

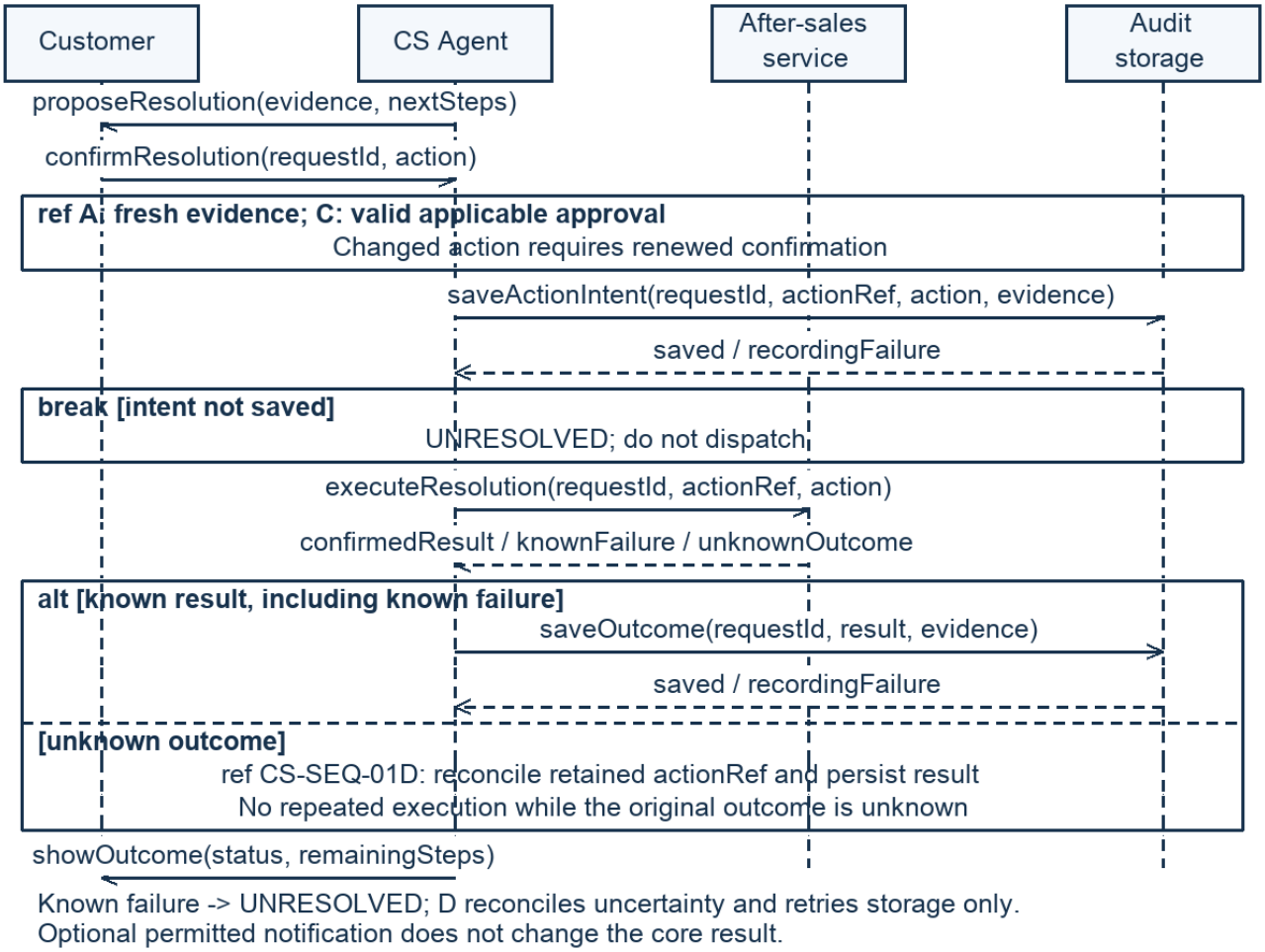


Figure 42 CS-SEQ-01B Current confirmation, saved intent, execution and recording.

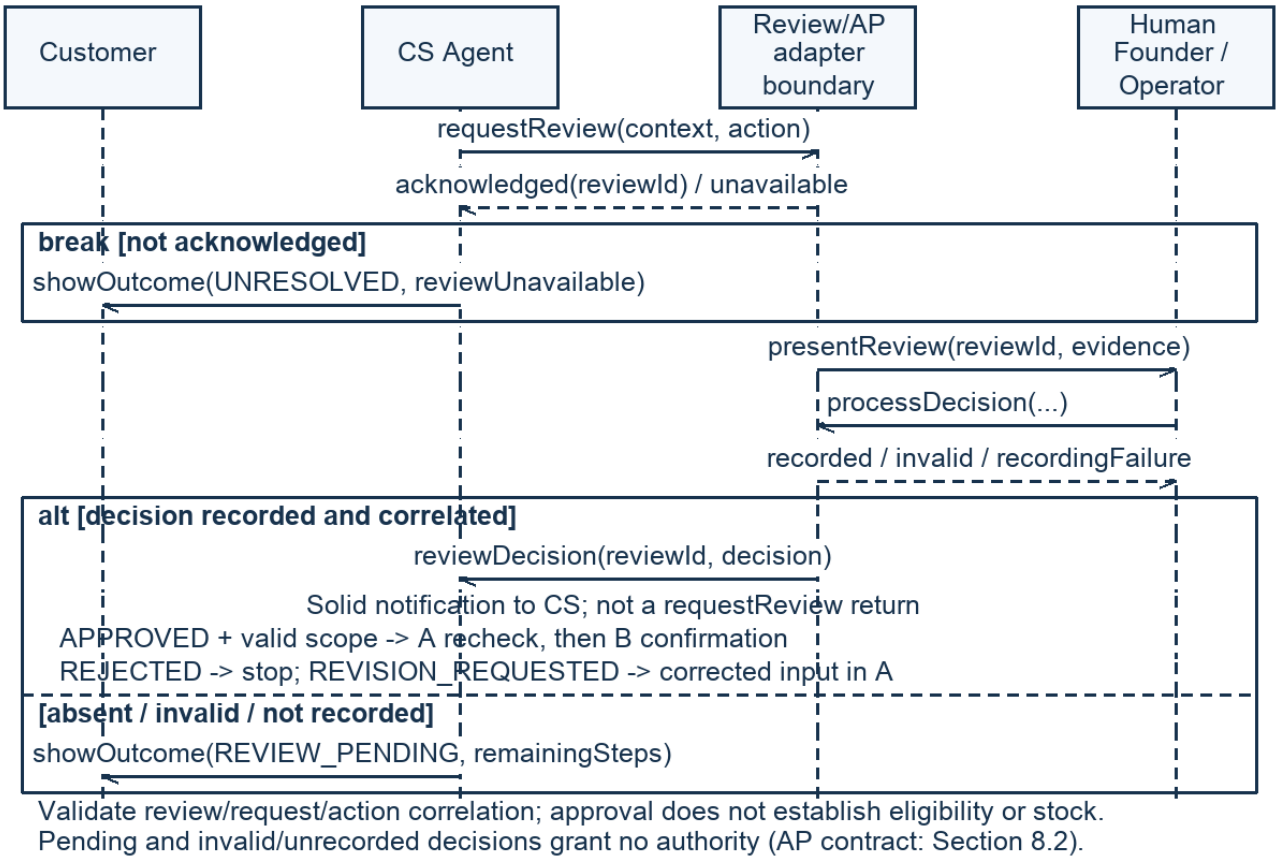


Figure 43 CS-SEQ-01C Acknowledged review and correlated decisions.

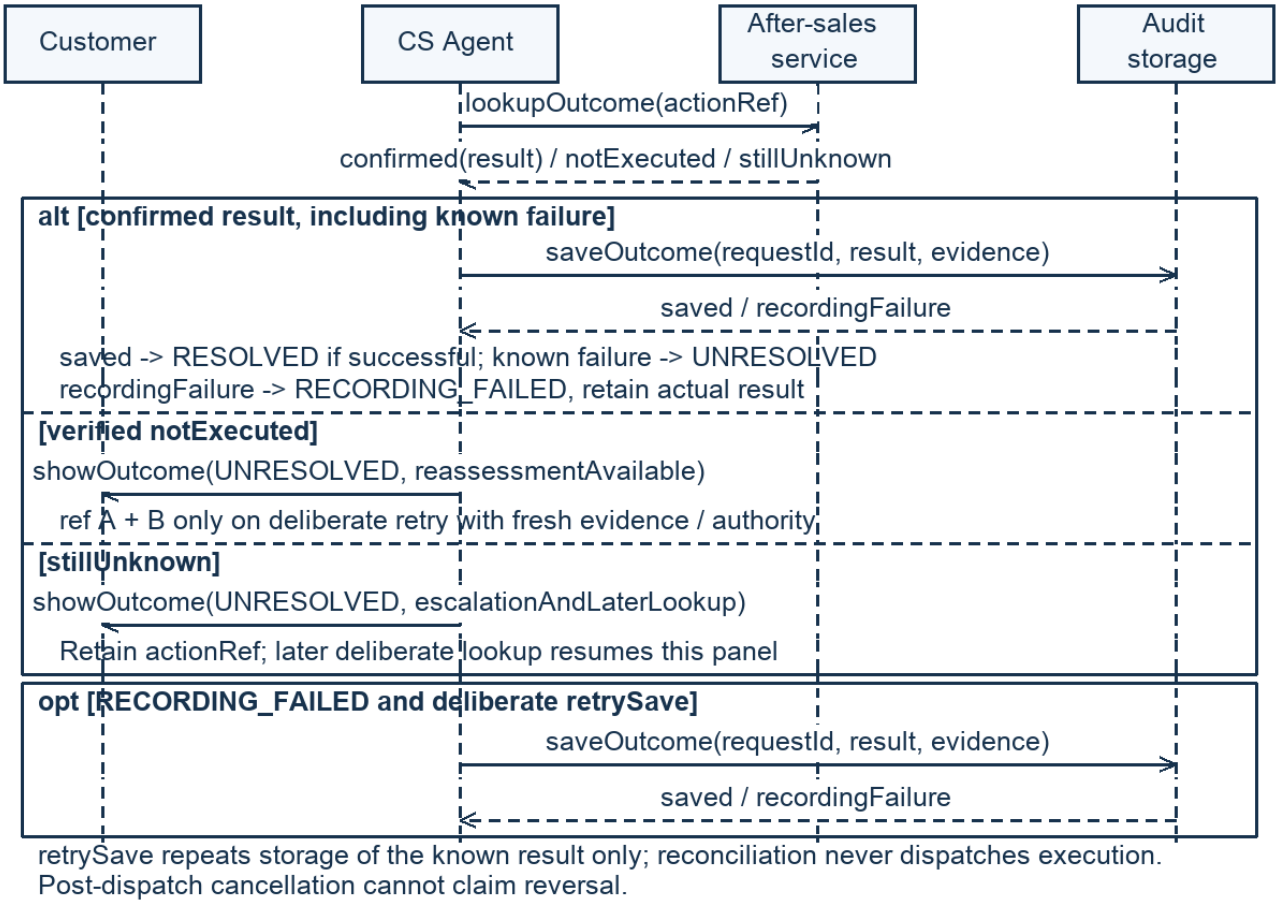


Figure 44 CS-SEQ-01D Read-only reconciliation and storage-only retry.

10.2.3 MC-SEQ-01 Draft and review interactions

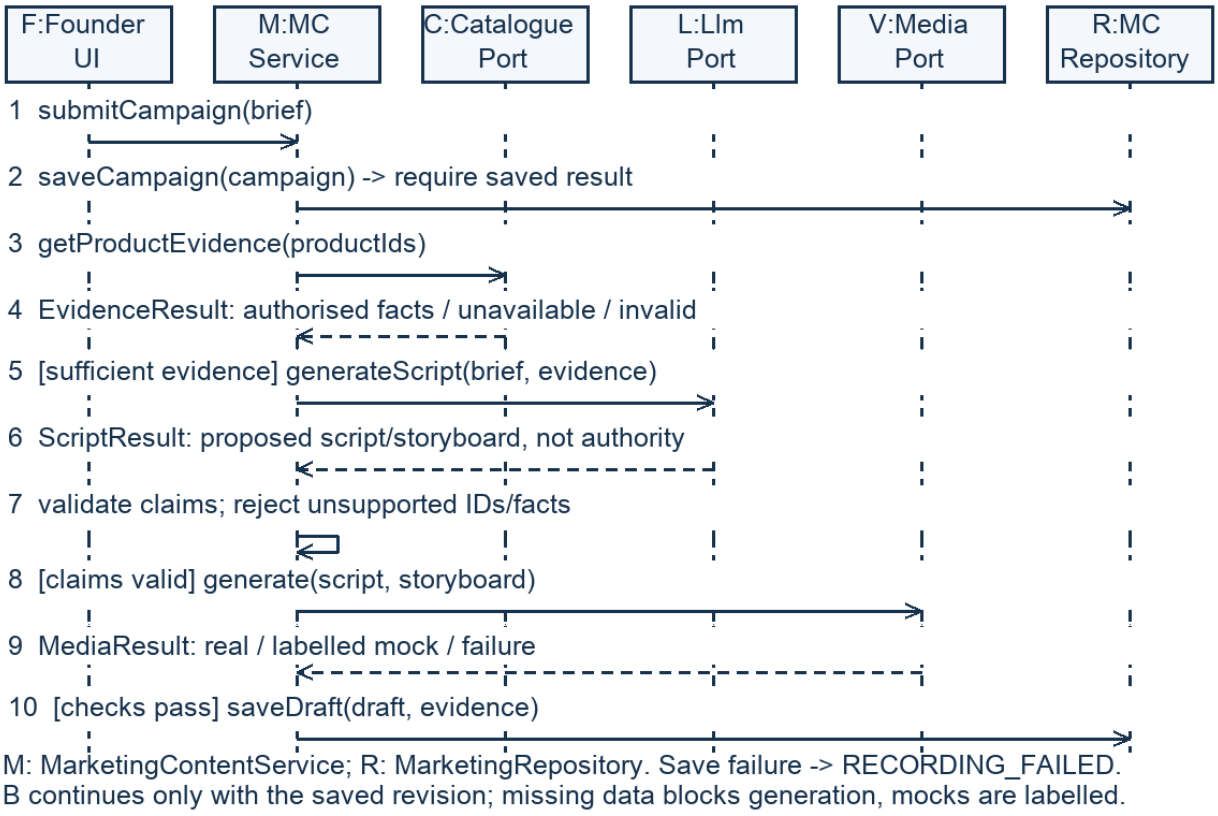


Figure 45 MC-SEQ-01A Product evidence, script validation and draft-media generation.

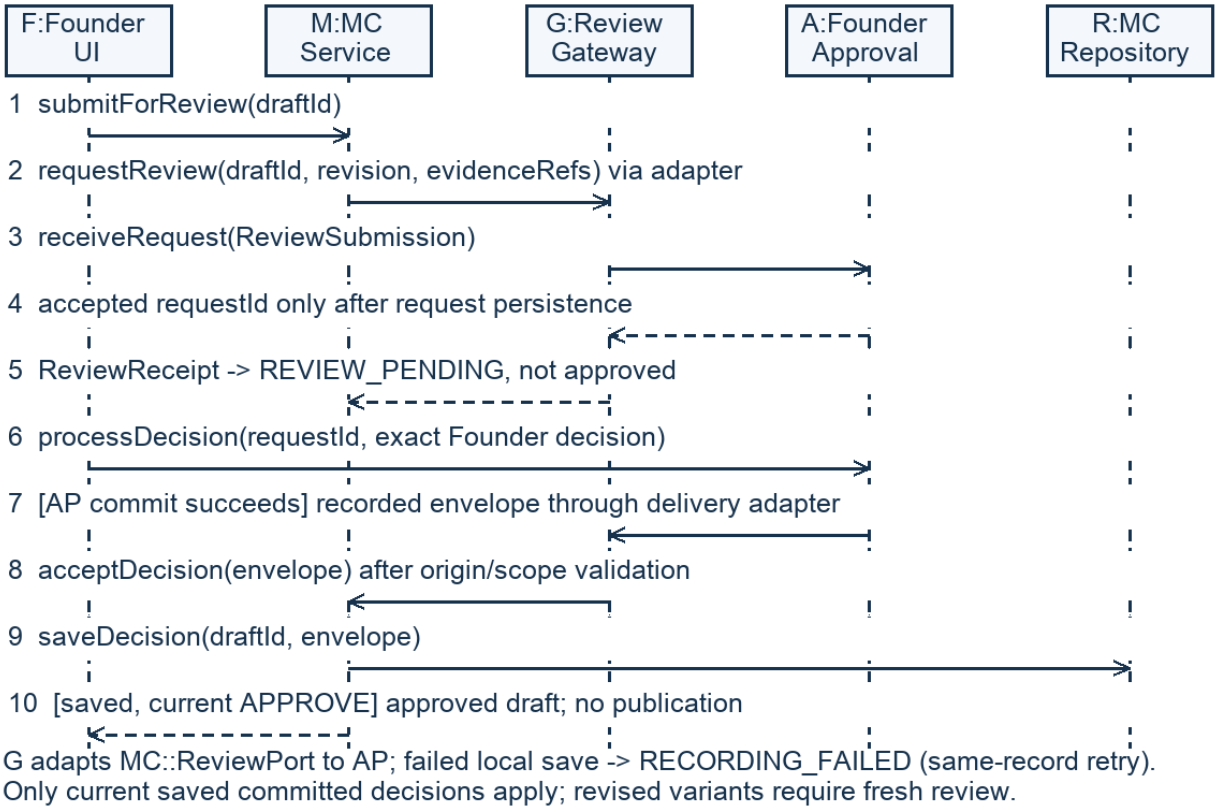


Figure 46 MC-SEQ-01B Exact-draft review, committed decision delivery and local decision recording.

10.2.4 AP-SEQ-01 Decision commit and delivery

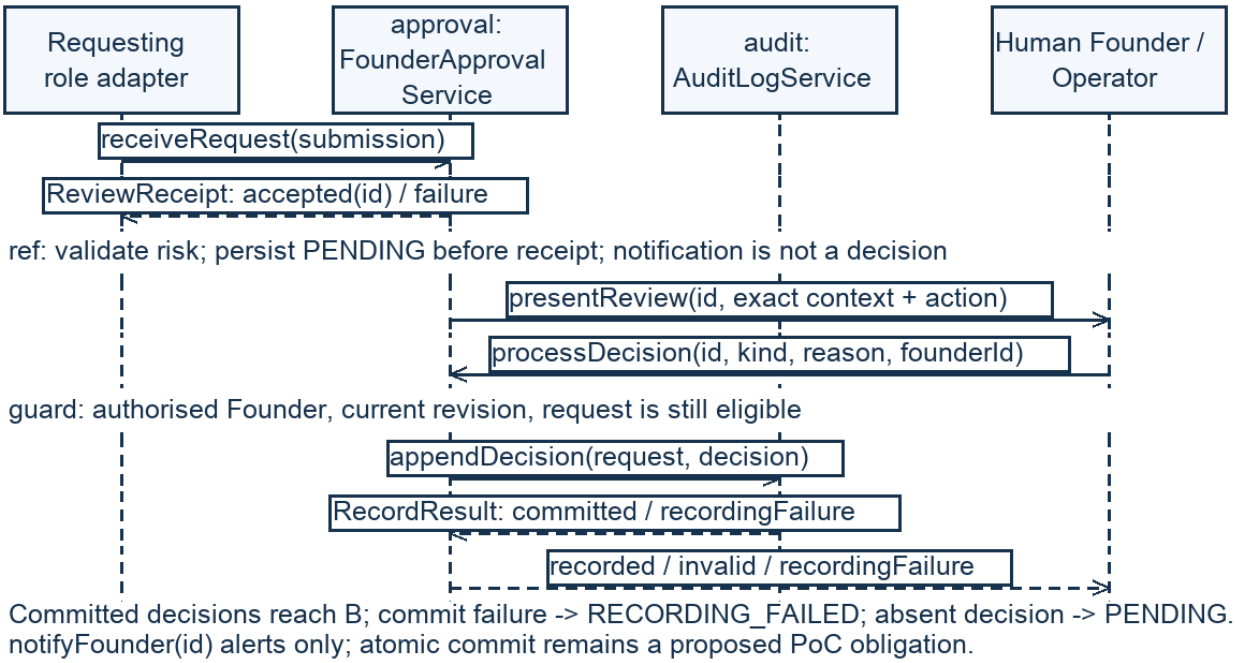


Figure 47 AP-SEQ-01A Request receipt and authorised exact-scope commit before delivery.

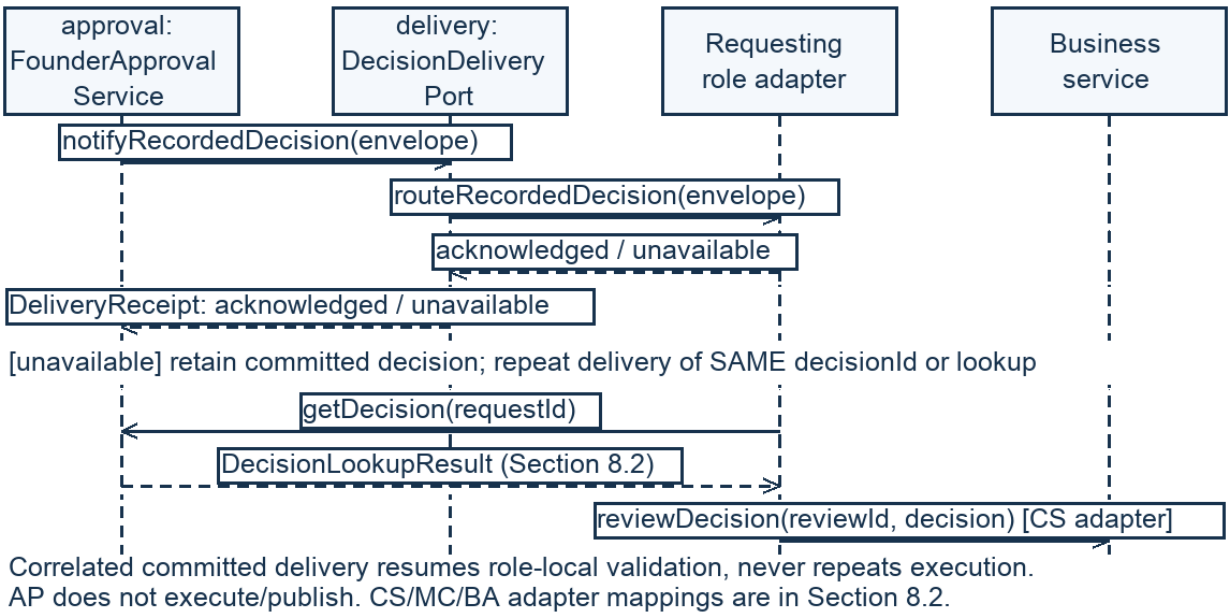


Figure 48 AP-SEQ-01B Same committed decision delivery/lookup and independent requester reassessment.

10.2.5 CO-SEQ-01 Handoff and recovery interactions

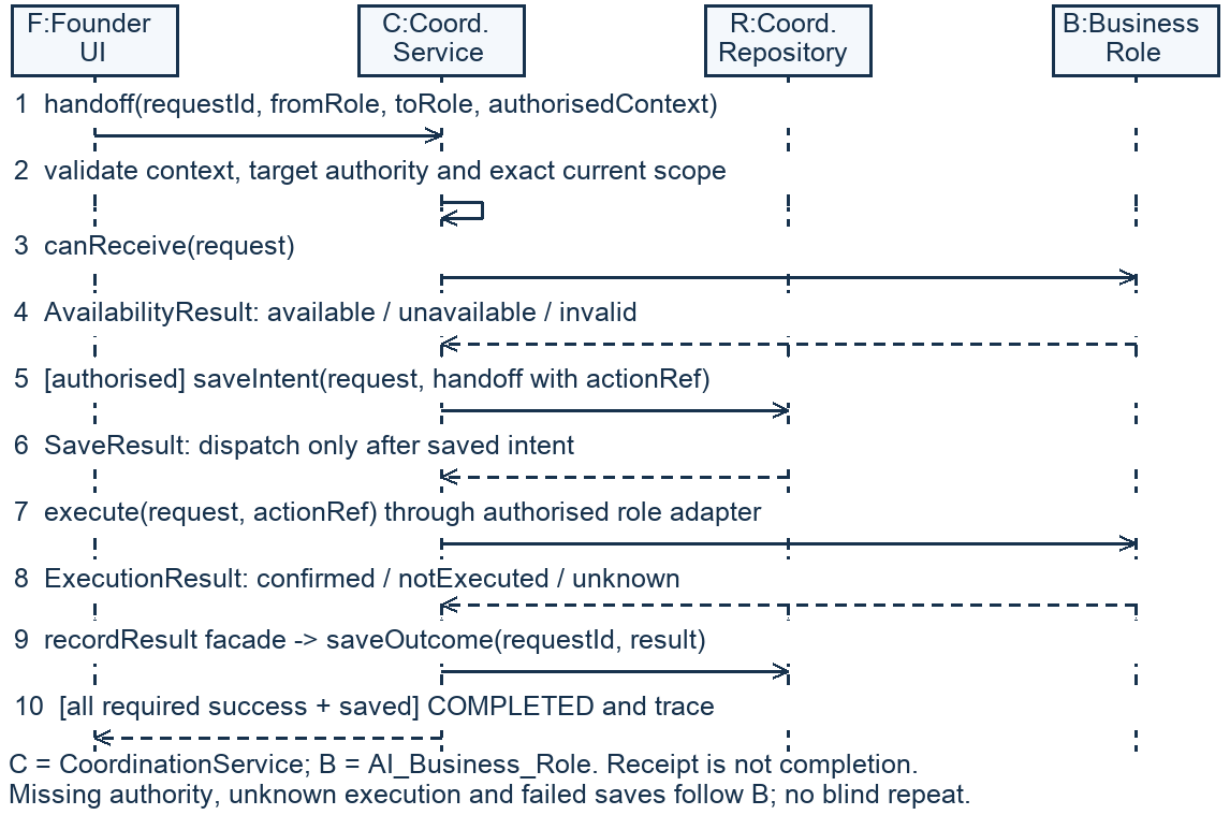


Figure 49 CO-SEQ-01A Authorised handoff, saved intent and recorded target outcome.

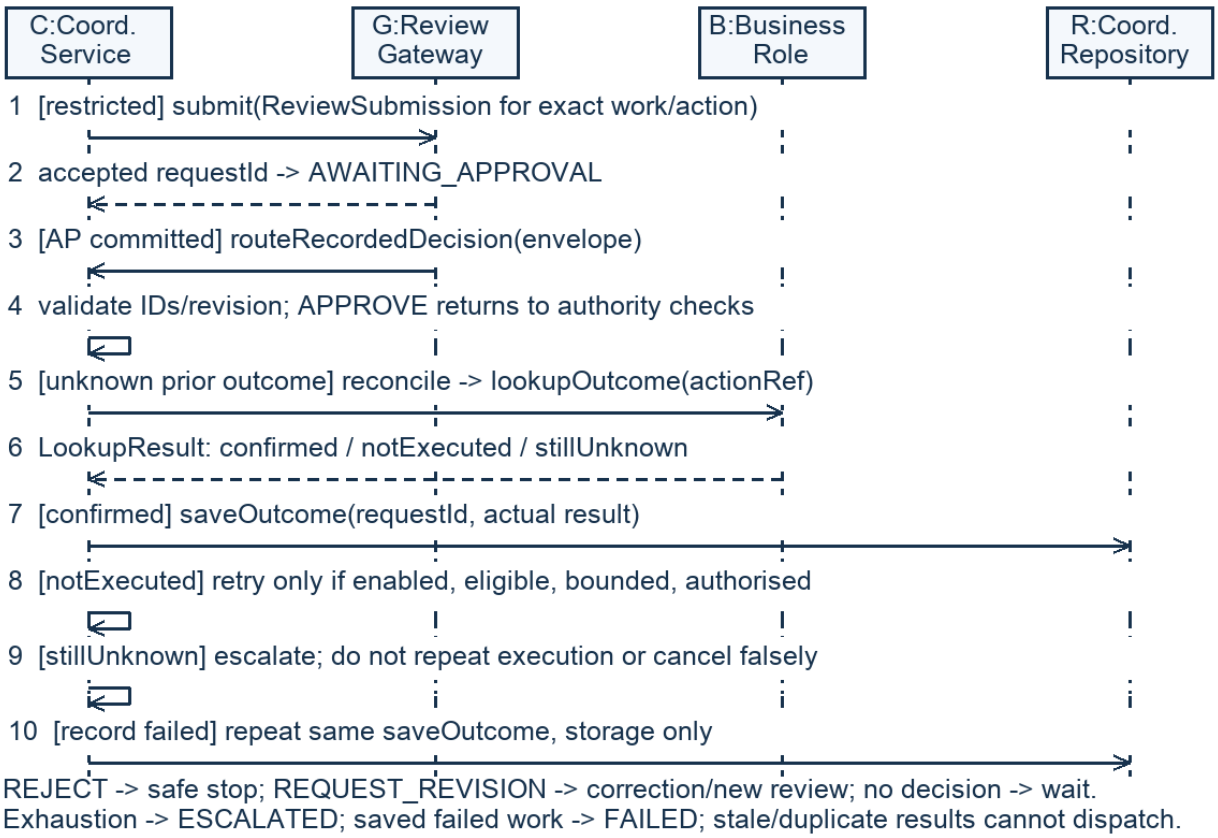


Figure 50 CO-SEQ-01B Approval validation, outcome lookup and bounded recovery.

10.3 State machine models

10.3.1 BA-STM-01 ConsultationSession lifecycle

BA-STM-01 follows ConsultationSession through its six declared states. COMPLETED requires delivery of current, validated, saved advice without cancellation. Review returns to validation; a cancelled session requires a fresh session.

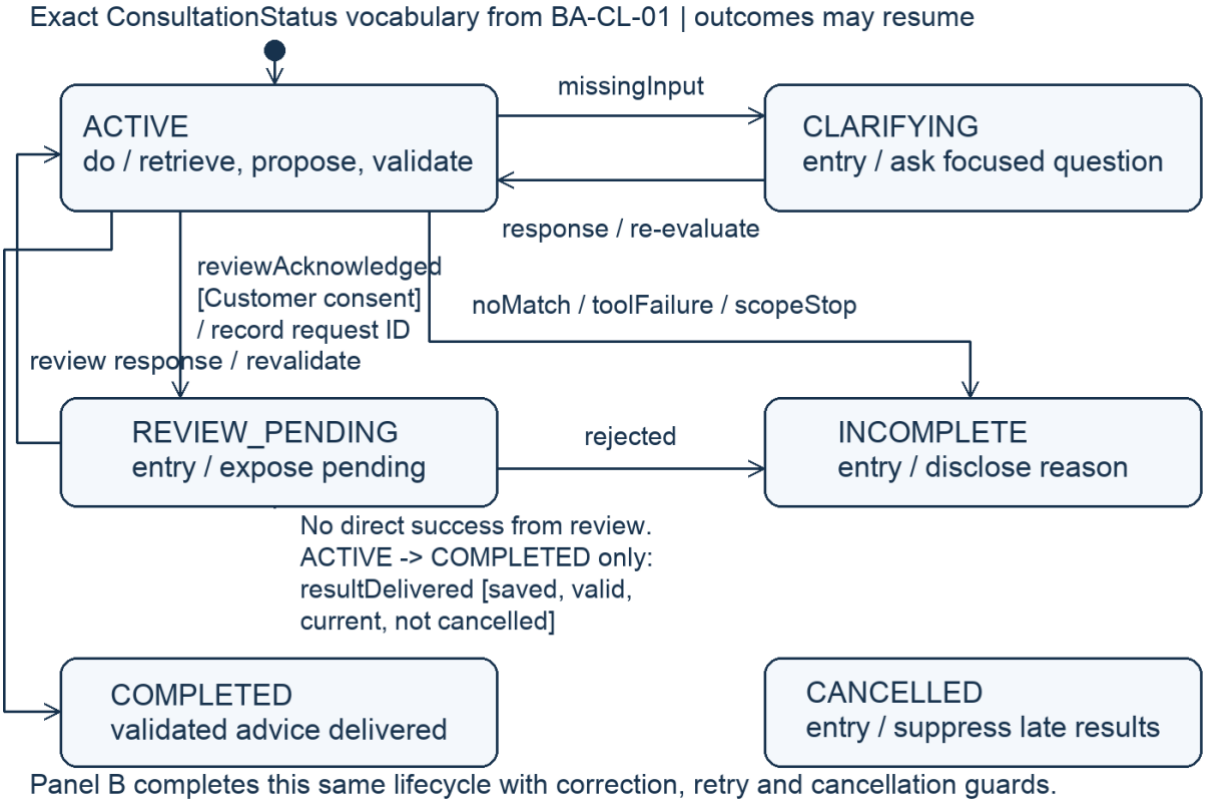


Figure 51 BA-STM-01A ConsultationSession lifecycle.

Companion table to panel A | source -- event [guard] / effect --> target

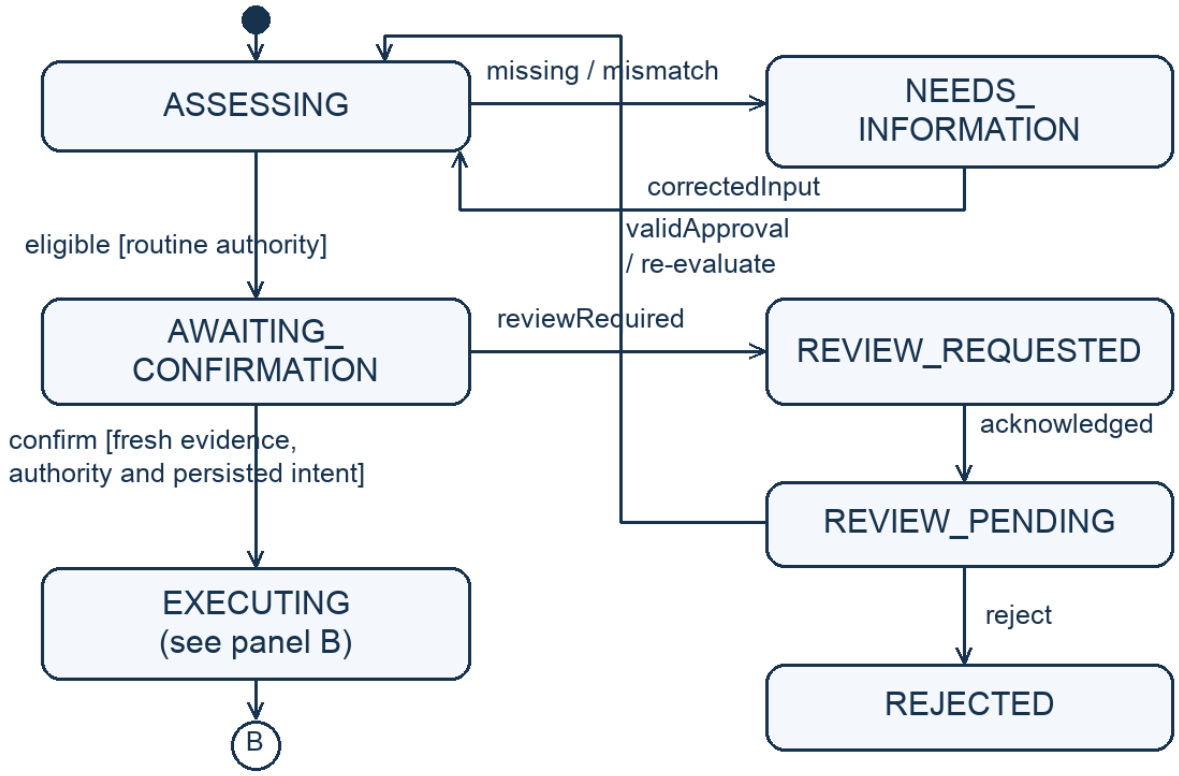
ACTIVE	missingInput / ask focused question	CLARIFYING
CLARIFYING	response / re-evaluate needs	ACTIVE
CLARIFYING	decline / disclose missing information	INCOMPLETE
ACTIVE	reviewAcknowledged [consent] / retain request ID	REVIEW_PENDING
REVIEW_PENDING	reviewResponse [relevant, current] / revalidate	ACTIVE
REVIEW_PENDING	reviewRejected / disclose rejection	INCOMPLETE
REVIEW_PENDING	no decision / disclose pending (internal)	REVIEW_PENDING
ACTIVE	noMatch | toolFailure | scopeStop | saveFailed	INCOMPLETE
ACTIVE	resultDelivered [valid, saved, current, not cancelled]	COMPLETED
Resumable states*	revise(input) / update revision, invalidate old results	ACTIVE
INCOMPLETE	retry [Customer initiated] / refresh evidence	ACTIVE
Nonterminal states*	cancel / stop; save if possible; suppress replies	CANCELLED
Any state	lateResult [cancelled or stale] / discard (internal)	same state

* Resumable: ACTIVE, CLARIFYING, REVIEW_PENDING, INCOMPLETE, COMPLETED.
* Nonterminal for cancel: ACTIVE, CLARIFYING, REVIEW_PENDING, INCOMPLETE.
Failed save is disclosed; no decision means wait. CANCELLED needs a fresh session.

Figure 52 BA-STM-01B Companion event/guard/effect specification for the same model.

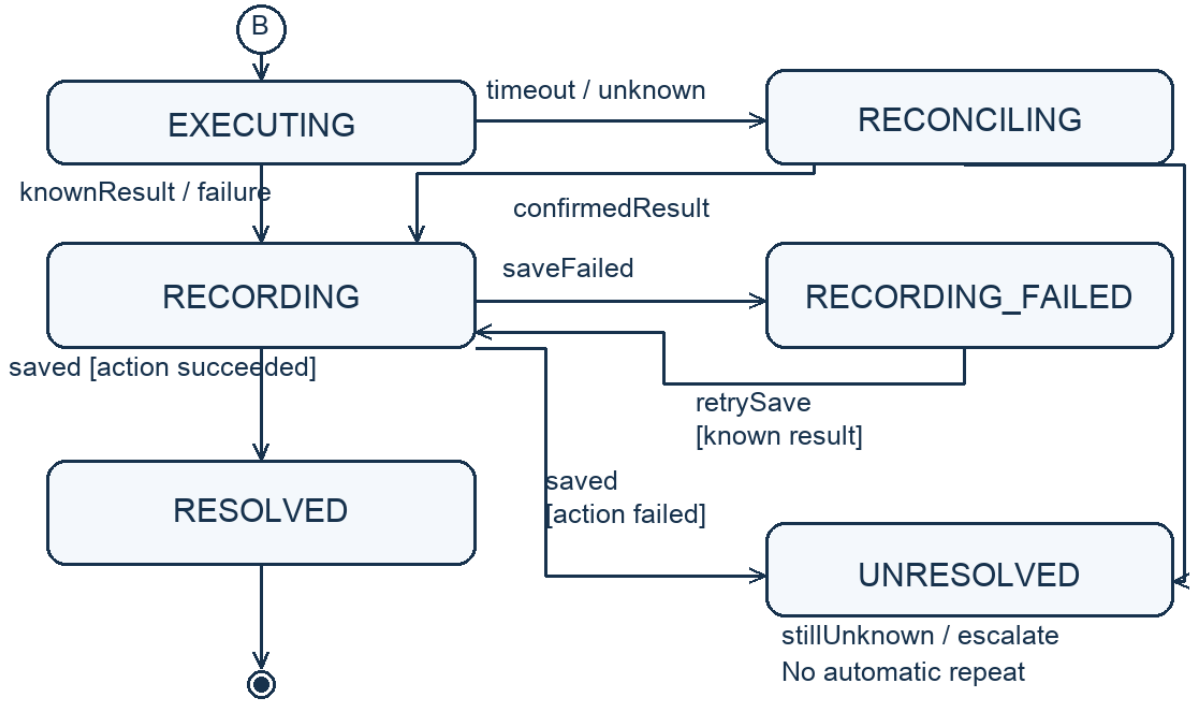
10.3.2 CS-STM-01 AfterSalesRequest lifecycle

AfterSalesRequest spans A/B. RESOLVED is a recorded service outcome, with settlement/physical completion separate; known failure remains UNRESOLVED. Unknown execution requires read-only reconciliation.



ASSESSING -> REVIEW_REQUESTED: unresolved policy/dispute/restricted action.
REVIEW_PENDING -> NEEDS_INFORMATION: revision; absent/invalid decision: stay pending.
Submission, required-read or intent-save failure -> UNRESOLVED; no dispatch.
Pre-dispatch cancellation -> CANCELLED; safe deliberate retry: UNRESOLVED -> ASSESSING.
REJECTED/CANCELLED terminate this request without financial reversal.

Figure 53 CS-STM-01A Assessment, clarification, confirmation and review states.



RECONCILING -> ASSESSING: verified notExecuted + deliberate retry.
UNRESOLVED -> RECONCILING: later lookup of retained actionRef.
RECONCILING -> RECORDING: confirmed success/failure; retrySave is storage only.
Cancel after dispatch reconciles actual outcome; optional notification changes no state.

Figure 54 CS-STM-01B Execution, recording, reconciliation and observable outcomes.

10.3.3 MC-STM-01 PromotionalContentDraft lifecycle

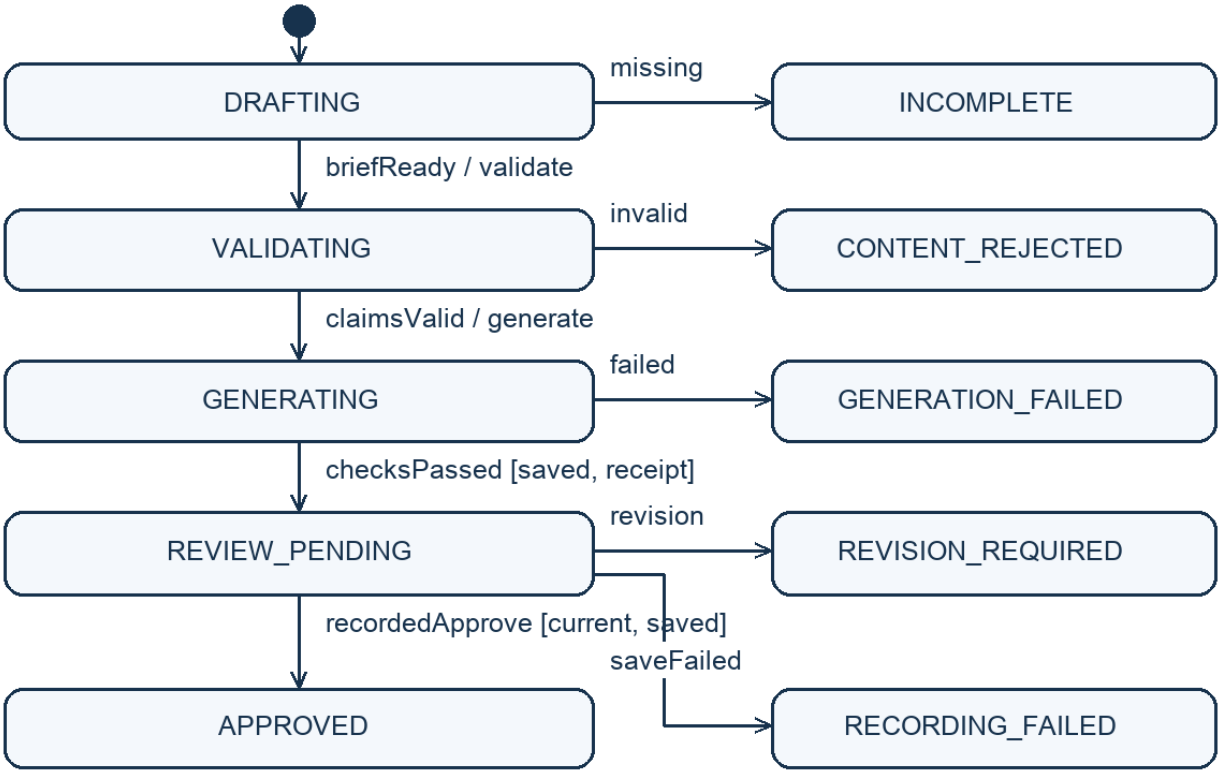


Figure 55 MC-STM-01 PromotionalContentDraft states; side states remain resumable under the transition specification.

From	Event guard and effect	To
DRAFTING	briefReady [authorised sufficient context] / validate proposed claims	VALIDATING
DRAFTING	missingContext / clarify; correction -> DRAFTING	INCOMPLETE
VALIDATING	claimsValid / request media generation; invalid output -> CONTENT_REJECTED	GENERATING
CONTENT_ REJECTED	reviseContent [authorised correction] / regenerate/recheck	DRAFTING
GENERATING	generationFailed / expose; declared retry -> GENERATING	GENERATION_ FAILED
GENERATING	draftSaveFailed / retain draft/evidence for same-record retry	RECORDING_ FAILED
RECORDING_ FAILED	retrySave [draft/evidence saved; no AP decision] / resume review submission	GENERATING
GENERATING	reviewSubmissionFailed / expose failure; deliberate resubmission uses saved draft	INCOMPLETE
INCOMPLETE	resubmitReview [draft saved, current] / repeat review submission only	GENERATING
GENERATING	checksPassed [draft/evidence saved and review acknowledged] / retain requestId	REVIEW_ PENDING
REVIEW_ PENDING	noDecision / wait; missing/stale/uncommitted is unauthorised	REVIEW_ PENDING
REVIEW_ PENDING	recordedReject [current] / save decision; failure -> RECORDING_FAILED	REJECTED

From	Event guard and effect	To
REVIEW_ PENDING	recordedRevision [current, saved] / reason; fresh exact-scope review	REVISION_ REQUIRED
REVISION_ REQUIRED	revise [new revision] / recheck evidence/script/media; no inherited approval	DRAFTING
REVIEW_ PENDING	recordedApprove [current exact scope and local decision save succeeds] / return draft	APPROVED
REVIEW_ PENDING	localSaveFailed [known AP decision] / retain envelope; no success	RECORDING_ FAILED
RECORDING_ FAILED	retrySave [same decision saved, current] / restore matching outcome	APPROVED or REJECTED or REVISION_ REQUIRED
APPROVED	contextChanged / invalidate prior applicable scope; create a new draft revision	SUPERSEDED
Any active state	cancel [no concealed effect] / suppress late/duplicate/stale events	CANCELLED

10.3.4 AP-STM-01 ApprovalRequest lifecycle

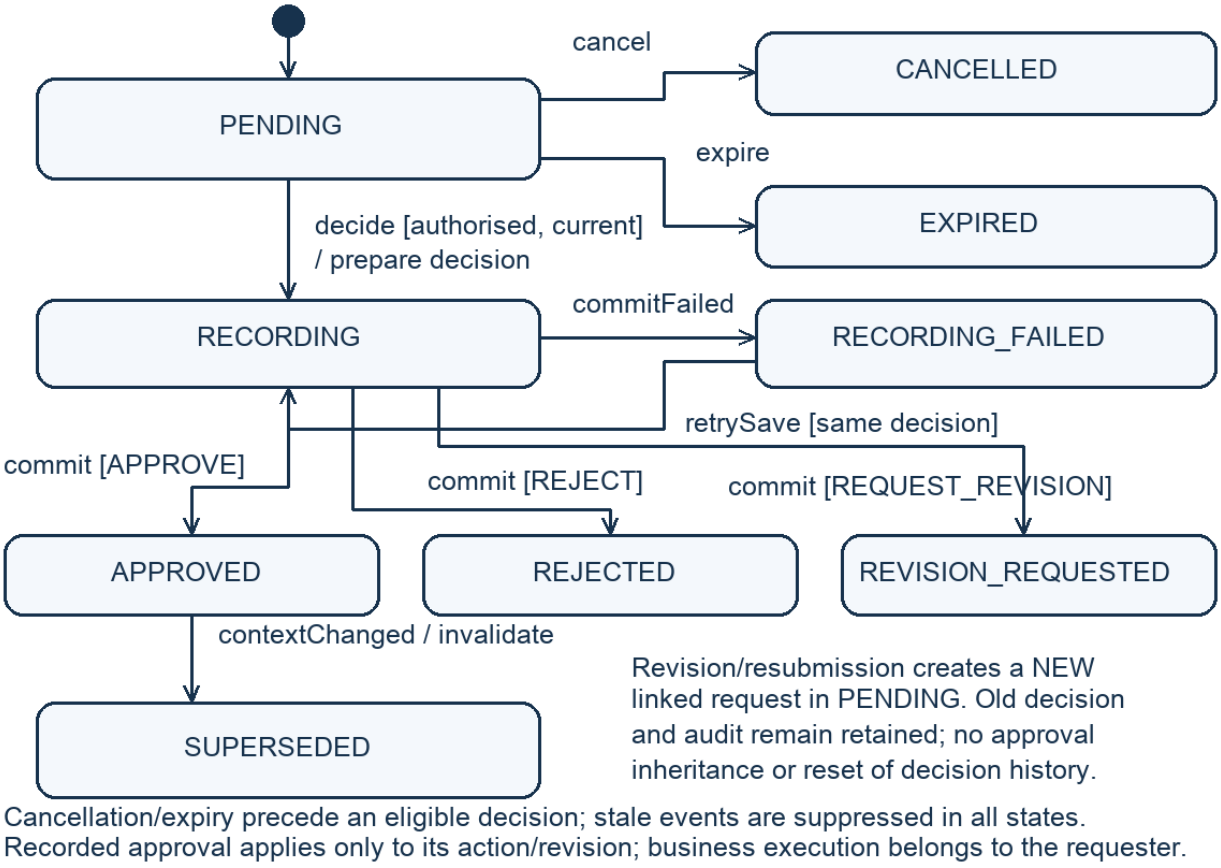


Figure 56 AP-STM-01 ApprovalRequest states exclude business EXECUTING or COMPLETED.

Transitions: initial accepted/persisted request -> PENDING. decide [authorised, current, eligible] -> RECORDING; commit(APPROVE/REJECT/REQUEST_REVISION) -> APPROVED/REJECTED/REVISION_REQUESTED respectively. commitFailed -> RECORDING_FAILED; deliberate retrySave(same decision) -> RECORDING. No decision is an internal PENDING wait. Predecision cancellation/declared expiry -> CANCELLED/EXPIRED. Context change invalidates an applicable approved scope -> SUPERSEDED. All old records remain retained; a changed action/revision creates a new linked request in PENDING. Delivery failure changes

delivery status, not a committed request state. Late, mismatched and duplicate decisions/events cannot execute business actions.

10.3.5 CO-STM-01 CrossAgentWorkRequest lifecycle

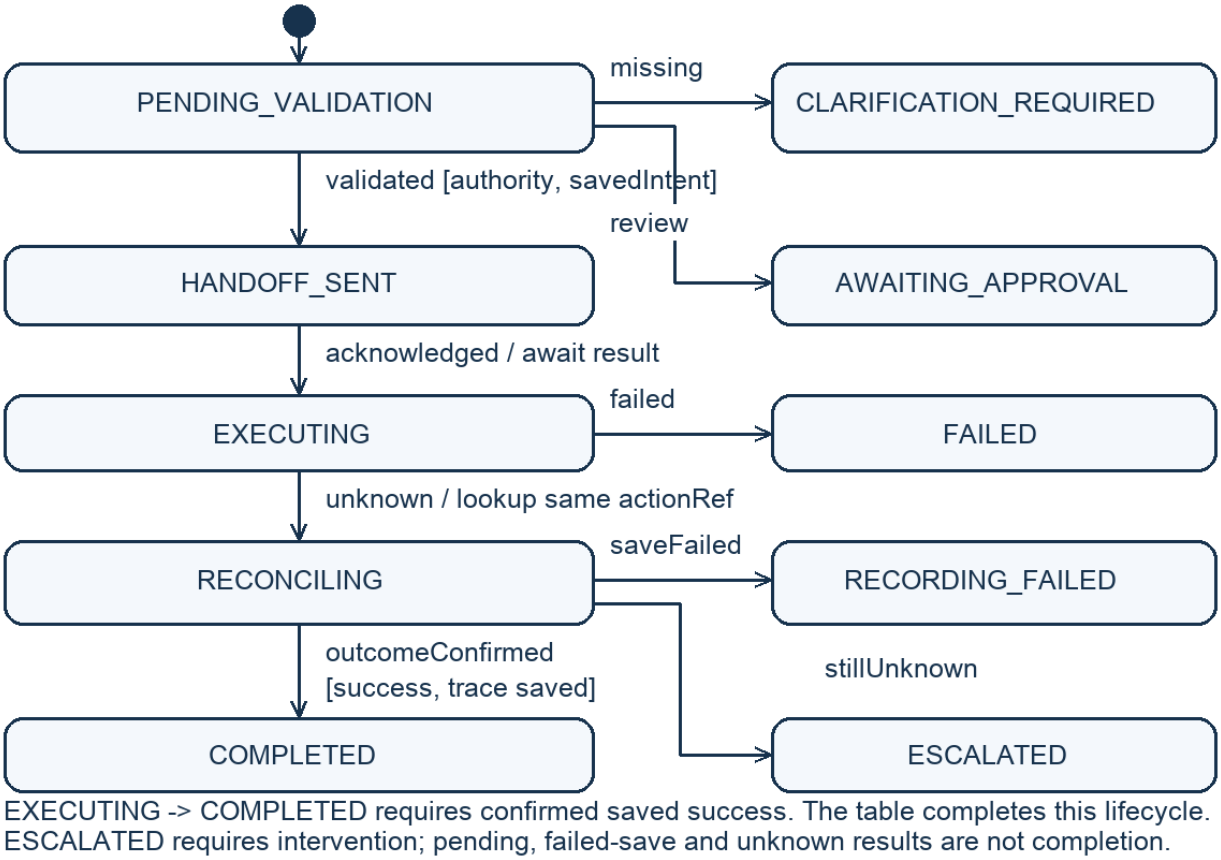


Figure 57 CO-STM-01 CrossAgentWorkRequest uses the class-declared vocabulary; escalation is durable intervention.

From	Event guard and effect	To
PENDING_ VALIDATION	missingContext / request authorised correction; correction returns to validation	CLARIFICATION_ REQUIRED
PENDING_ VALIDATION	reviewRequired [request receipt acknowledged] / retain AP reference; no dispatch	AWAITING_ APPROVAL
AWAITING_ APPROVAL	noDecision / keep waiting; valid APPROVE or corrected REQUEST_REVISION returns to validation with scope checks	AWAITING_ APPROVAL or PENDING_ VALIDATION
AWAITING_ APPROVAL	current REJECT / save safe stopped outcome; save failure -> RECORDING_FAILED	CANCELLED
PENDING_ VALIDATION	dispatch [context/target/authority current and intent saved] / execute via role adapter	HANDOFF_ SENT
HANDOFF_ SENT	acknowledged / await target result; receipt does not prove completed work	EXECUTING
HANDOFF_ SENT or EXECUTING	knownFailure / save failure details; no successful-completion claim	FAILED
HANDOFF_ SENT or EXECUTING	uncertain / lookup the same actionRef without dispatch	RECONCILING

From	Event guard and effect	To
EXECUTING or RECONCILING	confirmedSuccess [all required work + outcome saved] / publish recorded trace	COMPLETED
EXECUTING or RECONCILING	confirmedResult [more work required] / retain result, validate the next task	PENDING_ VALIDATION
RECONCILING	notExecuted / reassess retry/authority; stillUnknown -> recorded escalation	FAILED or ESCALATED
FAILED	retryEligible [CO-F-03 ON, bounded, current authority, read-only or proven no prior effect] / record attempt	RETRYING
RETRYING	retryPrepared / fresh validation and persisted attempt before dispatch; failure/exhaustion -> FAILED/ESCALATED	PENDING_ VALIDATION
Any result-recording point	saveFailed / retain known result and attempted target state; never repeat business action	RECORDING_ FAILED
RECORDING_ FAILED	retrySave [same known record succeeds] / recover its truthful recorded target state	COMPLETED or FAILED or ESCALATED or CANCELLED
FAILED or ESCALATED	authorised correction/intervention [outcome reconciled] / reassess current scope	PENDING_ VALIDATION
Active state	safeCancel [no hidden pending/unknown effect] / stop further dispatch and record actual status	CANCELLED

10.4 LLM perception decision support and adaptation

Role	Perception and tool use	Adaptation and authority
BA	Extract relevant Customer needs; retrieve authorised catalogue/stock; propose evidence-linked explanations.	Clarify missing context, withhold unsupported/medical claims and refresh stale evidence. Read-only commerce boundaries remain enforced.
CS	Interpret return/exchange/refund intent using minimum permitted context; retrieve order/policy/stock.	Deterministic eligibility, confirmation and approval gates control actions. Unknown write outcomes trigger lookup, never an LLM assurance or blind retry.
MC	Interpret Founder campaign; propose script/storyboard; validate facts and use real/mock media adapter.	Correct unsupported claims, expose generation failure and await exact-revision review. The LLM cannot approve or publish.
AP	Optional evidence summarisation assists the Founder; original evidence/action scope remains visible.	Human decision, scope validation and commit gates operate independently of generated text. AP is not another business agent.
CO	Read explicit task/context and role availability; transfer only authorised decision-relevant data.	Deterministic handoff/status/recovery controls adapt to failure or uncertainty. No generated suggestion grants target authority or approval.

11 Supporting Components

Supporting component	Purpose and future implementation boundary
Product Catalogue and Inventory Services	Read authorised facts and availability/version/observation evidence. Shared providers implement distinct role-local adapters; UNKNOWN never becomes available.
Customer/profile and identity/order services	Supply only authorised minimal context. CS verifies Customer/order association before protected disclosure; BA does not infer credentials or unrelated profile fields.
After-sales and order adapters	Execute only scoped PoC actions and provide actionRef outcome lookup. Real settlement/physical completion is separate from acknowledgement or simulated processing.
LLM Gateway	Bounded need/intent interpretation, explanations and script/storyboard proposals. Validate schema/facts and handle failure; tool results do not grant additional rights.
Media Generation Adapter	Real or labelled controlled mock image/speech/video generation. Reports result reference, simulation flag and failure; provider technology/cost/performance remain open.
Founder Approval Service and ReviewGateway	AP authoritative requests/decisions/audit; gateway origin/revision mapping and recorded delivery/lookup. Authentication and exact-scope checks surround access.
Repositories and audit	Retain request/context/source revisions, intent/actionRef, actual result and decision evidence. Storage-only retry is separate from execution. AP atomic commit is a proposed implementation obligation.
Notification and recovery policy	Queue/status delivery exposes failures without reversing committed outcomes. Retry/expiry/freshness values are explicit fixture configuration; optional automation does not replace safe manual recovery.

12 Lifecycle and Optional Views

Requirements/acceptance criteria drive feature/use-case, structure/runtime and behaviour models. Iteration remains subject to review against an identified approved requirement revision. Implementation adds adapters, persisted correlation and observable tests before demonstration/acceptance. Stage 1 models and planned cases do not prove implementation.

Package/deployment diagrams are not selected optional outputs. Sections 7/11 explain development boundaries; GR-CMP-01 shows interconnections. Future deployment must use agreed hosting, access and provider constraints.

13 Traceability and Acceptance

13.1 End to end traceability

Chain	Feature and use case path	Structure behaviour and planned acceptance
BA-M-01	BA-FR-01 -> BA-F-01 -> BA-UC-01 intake/clarification; BA-FR-02/03/04 -> BA-F-02 evidence/validation/failures; BA-O-01 -> optional BA-F-03.	ConsultationSession/Context, Recommendation/evidence and BA service/ports -> BA-OBJ/ACT/SEQ/STM -> BA-T-01 through BA-T-07.
CS-M-01	CS-F-01 -> CS-UC-01 verification/eligibility; CS-F-02 -> confirmation, review, execution/recovery; optional CS-F-03 -> notification.	AfterSalesRequest, evidence/review/resolution/actionRef -> CS-CL/OBJ/CMP and ACT/SEQ/STM -> CS-T-01 through CS-T-07.

Chain	Feature and use case path	Structure behaviour and planned acceptance
MC-M-01	MC-F-01 -> MC-UC-01 steps 1-5; MC-F-02 -> steps 6-12 and review/failure; optional MC-F-03 -> fresh variant review.	MarketingCampaign/drafts, exact revision, MC ports/local decision view -> MC-CL/OBJ/CMP and ACT/SEQ/STM -> MC-T-01 through MC-T-07.
AP-M-01	AP-F-01 -> AP-UC-01 submission/risk; AP-F-02 -> decision, commit/delivery and failures; optional AP-F-03 -> assisted successor request.	ApprovalRequest, ApprovalDecision, AuditRecord and FounderApprovalService -> AP-CL/OBJ/ACT/SEQ/STM -> AP-T-01 through AP-T-08.
CO-M-01	CO-FR-01/02 -> CO-F-01 -> CO-UC-01 validation/handoff; CO-FR-03/04 -> CO-F-02 result/recovery; CO-O-01 -> optional CO-F-03.	Work/context/hand-off/failure/retry/escalation, ApprovalReference and role adapters -> CO-CL/OBJ/CMP and ACT/SEQ/STM -> CO-T-01 through CO-T-08.
Group and NFRs	GR-C-01 through GR-C-06 and twenty role NFRs govern all five chains. GR-S-01 combines existing UCs rather than creating a sixth.	GR-RQ/FD/UCD/CL/OBJ/CMP/COL and shared adapter invariants -> GR-T-01 through GR-T-04. Stage 2 adds code, fixture, transcript and execution evidence.

13.2 Change assessment basis

Action/context changes affect requirements/use cases, evidence, Customer confirmation, review envelopes and dependent classes/interaction/state guards. Revised MC media/variants need fresh exact-scope review; CO/CS reassess authority and uncertain effects. Preserve history without inheriting permission.

Signature/result changes update callers, receivers, adapters and class/composite/communication/sequence views together. Retry changes preserve actionRef lookup and separate execution, storage and delivery. Optional changes cannot create core dependencies; Section 13.3 supplies regression specifications.

13.3 Planned acceptance specifications

These cases are planned, not executed. Use synthetic/anonymised fixtures, labelled mocks and injected failures; inspect outputs, records and call traces. Formal approval and execution evidence remain separate.

13.3.1 BA acceptance cases

Case	Input and observable expected result
BA-T-01	Clarification: Supply a request missing a decision-relevant constraint. Expect a focused question, no guessed value, then re-evaluation after the Customer responds (BA-AC-01).
BA-T-02	Grounded success: Supply a sufficient request and a matching authorised in-stock product. Expect its ID, availability evidence, linked reason and saved outcome; no order/payment call (BA-AC-02; BA-NFR-01/02).
BA-T-03	No match and service failures: Repeat with no matching product, unavailable stock, missing stock status, catalogue timeout and inventory error. Expect an explicit incomplete outcome without an available-product claim; retry refreshes evidence (BA-AC-03/06).
BA-T-04	Unreliable generation: Inject an invented product ID, unsupported product claim, conflicting evidence and malformed/failed LLM response. Expect validation to withhold the output and clarification, review offer or safe stop (BA-AC-04).
BA-T-05	Scope and review: Submit a diagnosis request; expect no diagnosis. For an unresolved product question, exercise acknowledged review, unavailable review service, pending decision and rejection. No branch invents approval or treats pending review as success (BA-AC-05/06).

Case	Input and observable expected result
BA-T-06	Optional alternatives: With BA-F-03 enabled, request an alternative to an unavailable preferred item; expect verified availability and explicit differences. With it disabled, expect a safe no-match result or ordinary matching recommendation without dependence on alternatives (BA-O-01).
BA-T-07	Revision, cancellation and persistence: Revise/cancel while results are pending; old results must not be delivered. Inject repository failure; completion must not be claimed. Inspect LLM/review payloads for unrelated data or credentials (BA-AC-06; BA-NFR-04).

13.3.2 CS acceptance cases

Case	Input and observable expected result
CS-T-01	Identity and intake (CS-AC-01): Supply a matching order and then a mismatched identity/order. Expect permitted intake in the first case; no protected disclosure or resolution call in the second. Inspect request and service-call records.
CS-T-02	Policy evidence (CS-AC-02): Exercise sufficient policy, missing return window and contradictory policy evidence. Expect referenced eligibility only with sufficient evidence; otherwise clarification or escalation without an invented rule.
CS-T-03	Routine resolution (CS-AC-03): Supply authorised evidence and Customer confirmation. Expect only the permitted service call, a saved confirmed result and remaining steps. An acknowledgement or mock refund must not be described as settled funds.
CS-T-04	Exchange availability (CS-AC-04): Exercise AVAILABLE, UNAVAILABLE and UNKNOWN service results. Expect no successful-exchange claim for the latter two and no unconfirmed substitute.
CS-T-05	Human checkpoint (CS-AC-05): Exercise disputed/high-impact cases with acknowledged pending review, approval, rejection, revision request and failed review service. Expect no restricted action without a valid decision; revalidate the proposed action after approval.
CS-T-06	Failure and reconciliation (CS-AC-06): Inject read failure, post-write timeout and audit failure. Expect an explicit unresolved/recording-failure outcome; inspect that no blind duplicate action occurs and no rollback is invented.
CS-T-07	Optional notifications (CS-AC-07): Enable and disable the feature; inject missing recipient permission and delivery failure. Expect permitted notifications only, with the core recorded resolution still accessible and unchanged.

13.3.3 MC acceptance cases

Case	Input and observable expected result
MC-T-01	Campaign preparation: Supply a clear objective and sufficient authorised product information. Expect saved brief and grounded script/storyboard; inspect claims/source references for traceability (MC-AC-01).
MC-T-02	Missing/conflicting information: Remove, contradict or leave unsupported required product information. Expect clarification, available authorised retrieval or incomplete outcome; no invented claim (MC-AC-02).
MC-T-03	Unsupported generation: Inject invented facts, unsupported claims or contradictions. Expect withheld output and revision/regeneration/clarification, never a valid forwarded draft (MC-AC-02).
MC-T-04	Media: Use validated script/storyboard with available, unavailable, failed and timed-out services. Success saves draft media/generation result; failure is explicit. Any permitted mock/fallback is labelled simulated (MC-AC-03/04).

Case	Input and observable expected result
MC-T-05	Checks: Inject a failing script, storyboard or media content/quality check. Expect recorded issue and revision, never approval-ready status. Use configured/reviewer-identified criteria; invent no numerical threshold.
MC-T-06	Review: Exercise valid Founder approval, rejection, revision, pending/missing decision and service failure. Only recorded valid approval establishes approved state; every other outcome remains unapproved for publication (MC-AC-06/07).
MC-T-07	Variants: With MC-F-03 ON, expect campaign/source traceability and no unsupported claims. The new draft variant needs its own covering Founder approval; no inheritance. With it OFF, MC-F-01/02 remain usable.

13.3.4 AP acceptance cases

Case	Input and observable expected result
AP-T-01	Valid submission/risk: Submit a declared restricted action with authorised evidence; expect persisted PENDING request/accepted ID, Founder queue notification and no dispatch. Repeat with missing/unauthorised context and unknown risk; expect explicit blocked/clarification outcome (AP-AC-01; AP-NFR-02).
AP-T-02	Pending/invalid decision: Inject absent, rejected, mismatched and unauthorised decisions; expect no restricted execution and truthful distinct states (AP-AC-02; AP-NFR-02).
AP-T-03	Recorded decision: Authorised Founder decides on current exact scope; expect atomic decision/request/audit commit with actor, reason, IDs and revision before event delivery (AP-AC-03; AP-NFR-01).
AP-T-04	Revision/stale context: Request revision or alter action/context after decision; expect preserved history and fresh linked review, never inherited approval (AP-AC-04; AP-NFR-02).
AP-T-05	Recording failure: Fail decision commit; expect RECORDING_FAILED, no approval event/action and storage-only same-decision retry (AP-AC-05; AP-NFR-03).
AP-T-06	Delivery recovery: Fail queue/recorded-event delivery; distinguish persisted request from committed decision. Retry same event/lookup; ignore duplicate/stale decision IDs without repeated business execution (AP-AC-06; AP-NFR-03/04).
AP-T-07	Independent constraints: Record approval, then make stock unavailable, policy ineligible, Customer confirmation absent or execution uncertain; expect the requesting role to block/reconcile, never infer business success (AP-AC-07).
AP-T-08	Optional OFF: Disable assisted resubmission; expect core APPROVE/REJECT/REQUEST_REVISION and manual fresh corrected request to work. Enable it and verify linked history, fresh approval and identical safety gates (AP-AC-08).

13.3.5 CO acceptance cases

Case	Input and observable expected result
CO-T-01	Valid hand-off: Supply sufficient authorised context. Expect the appropriate receiving role, minimal permitted transfer, saved hand-off/status and a Founder-visible trace (CO-AC-01).
CO-T-02	Invalid context: Remove/invalidate decision-relevant fields. Expect clarification or safe incomplete record, with no silent inference or completed hand-off claim (CO-AC-02).
CO-T-03	Unavailable role: Expect recorded failed/unavailable hand-off and applicable recovery/escalation/cancellation, never COMPLETED (CO-AC-03).
CO-T-04	Tool failure: Inject timeout, service unavailability and failed tool results. Expect recorded visible failure/uncertainty; dependent work cannot complete (CO-AC-04).

Case	Input and observable expected result
CO-T-05	Automatic retry: With CO-F-03 ON, inject eligible failure; expect controlled retry and saved state/attempt. With it OFF, core escalation/safe cancellation remain usable (CO-AC-05; CO-O-01).
CO-T-06	Exhaustion/uncertainty: Exhausted retry escalates to Founder or safely cancels. Unknown prior execution must be reconciled before duplicative retry or COMPLETED (CO-AC-05/07).
CO-T-07	Review boundary: Exercise approved, pending, rejected and unavailable review. Restricted action proceeds only with recorded valid approval; all other outcomes grant no authority (CO-AC-06).
CO-T-08	Recording: Fail persistence/audit of required workflow evidence. Expect visible recording failure and no fully traceable completion claim (CO-UC-01 E6; CO-NFR-01/03).

13.3.6 GR acceptance cases

Case	Input and observable expected result
GR-T-01	Coverage: Verify five owned chains, three business services, five primary UC specifications and 15 features; disabling all five optional features preserves mandatory safe outcomes.
GR-T-02	Correlation: Walk GR-S-01 across CS, CO, BA and AP using synthetic IDs; prove hand-off receipt, completed recommendation, pending AP review and CS business result remain distinct.
GR-T-03	Failure boundaries: Inject review commit/delivery and after-sales uncertain-execution failures; require no action on uncommitted approval, same-event redelivery and actionRef lookup before any repeat.
GR-T-04	Structural/runtime consistency: Type each composite/object participant against GR-CL-01 or preserved local class model; check each communication/sequence receiver and adapter map; reject stale scope and duplicated business dispatch. Verify the MC/CO local contracts and state vocabularies against their use cases and current group adapters.

The traceability table and planned cases link requirement coverage to structure, behaviour and observable outcomes. Stage 2 supplies implementation and execution evidence; Section 15 records the contribution evidence boundary.

14 Assumptions Risks and Open Decisions

Open decision or risk	Safe design treatment and evidence gap
Requirement approval and tutor interpretation	The design is proposed. CS review covers the modelling basis only; requirements approval and review by all affected module owners remain unconfirmed. Tutor clarification of requirement-view notation is pending.
Shared schema/provider capability	This report specifies proposed adapter operations/results. External atomicity, lookup, idempotency and availability must be verified or supplied by explicit controlled mocks.
Order policy, risk thresholds and expiry	No real return window, monetary threshold, risk rule or validity period is invented. Declared fixtures control the PoC; missing/unknown policy blocks restricted action.
Stock freshness and evidence quality	Record authorised references/observations and recheck before action; absence/conflict prevents supported success. Freshness duration needs agreement.
LLM claims and media quality/provider	Validate factual claims and agreed criteria, label mocks and expose failure. Provider cost/performance and any quantitative quality threshold are open.

Open decision or risk	Safe design treatment and evidence gap
Privacy, retention and production security	Minimum authorised context and role checks apply. Deployment access-control policy, retention duration and production assurance need separate evidence.
Unknown effects and recording/delivery failure	Persist correlated intent before dispatch; reconcile uncertainty; storage/delivery retry does not repeat business work. Escalation remains resumable; cancellation cannot imply rollback.
Independent contribution and ownership	Module responsibilities are summarised in Section 15. Independent completion, attribution details and participation in all group tasks remain subject to member confirmation.
Presentation/interview and Stage 2	The presentation is limited to eight minutes. Presentation and interview readiness, prototype implementation and executed test results require separate evidence.

15 Contributions and Evidence

15.1 Member contributions

Member	Module responsibility and source contribution
Danyang Wang redacted	BA (Beauty Advisor): requirements, core/optional features and consultation use case; architecture, class/object and behaviour models, with acceptance specifications and traceability.
Siyuan Lei redacted	CS (Customer Service and Order): requirements, features and after-sales use case; architecture, structural/runtime and behaviour drafts, with non-functional requirements, supporting components and acceptance specifications.
Zhengdu Wei redacted	MC (Marketing Content): requirements, features and promotional-content use case; architecture, structural/runtime and behaviour models, with supporting components, traceability and acceptance specifications.
Jiaxuan Ren redacted	AP (Approval and Governance): requirements, approval-routing and decision/audit features; use-case concepts and architecture, class/object and behaviour descriptions used in the integrated approval module.
Guiyu Liu redacted	CO (Coordination and Recovery): requirements, features and coordination/recovery use case; architecture, structural/runtime and behaviour material, with supporting components, traceability and planned acceptance scenarios.

Danyang Wang handled report integration and consistency revisions; model sources are recorded in Appendix A.

15.2 Group-model contribution basis

Group model	Attributed input used in the integrated model
GR-RQ / GR-FD	Five requirement/feature chains: BA Danyang, CS Siyuan, MC Zhengdu, AP Jiaxuan source text and CO Guiyu.
GR-UCD	Five member use-case inputs; AP specification completed during integration.
GR-CL	BA/CS, MC and CO/AP structure inputs with shared adapters. Appendix A records source attribution and local revisions.
GR-OBJ / GR-CMP / GR-COL	BA consultation, CS after-sales, CO hand-off and AP review inputs; MC in the overall part model. Figures assembled during integration.

The contribution register and preserved sources support these attributions. Individual completion, participation in every group task and acceptance of integrated revisions still require confirmation; no contribution percentages or formal approvals are inferred.

15.3 AI and tool use

OpenAI Codex assisted wording, UML review, drawing code, integration and export checks. Added/reconstructed diagrams use editable code/vector shapes, not AI-generated raster artwork. Runtime values are synthetic and tests planned. AI assistance establishes no human approval, independent contribution or executed testing; other members' tool use is reported only from preserved evidence.

References

ELEC5620 Teaching Team. Project Description. Supplied course PDF. Primary submission scope and One-Person AI Company theme.

ELEC5620 Teaching Team. Project Stage 1 Marking Criteria. Supplied course PDF. Report, individual/group model obligations, presentation and interview requirements.

ELEC5620 Teaching Team. Project Requirement Example. Supplied course PDF. Three business roles, two core/one optional feature per member and meaningful improvement.

ELEC5620 Teaching Team. Lab Notes 1-7. Supplied course PDFs; Notes 5-7 support design-time, runtime and behaviour-model development.

Appendix A Model Catalogue

This catalogue identifies module sources and final model preparation. Section 15 summarises member responsibilities, report integration and AI assistance.

Model family	Source and final preparation
GR-RQ/FD/UCD/CL/OBJ/CMP/COL	Five members' requirement and model inputs (Section 15.2); final group diagrams prepared during integration.
BA-ARCH/CL/OBJ/ACT/SEQ/STM	Danyang Wang: BA requirements and source models.
CS-CL/OBJ/CMP/ACT/SEQ/STM	Siyuan Lei: source models; CL-A, OBJ and CMP diagrams prepared during integration, with CL-B and behaviour revisions.
MC-CL/OBJ/CMP/ACT/SEQ/STM	Zhengdu Wei: source models; final model revisions completed during integration.
AP-ARCH/CL/OBJ/ACT/SEQ/STM	Jiaxuan Ren: requirements, design concepts and source descriptions; final specifications and diagrams prepared during integration.
CO-CL/OBJ/CMP/ACT/SEQ/STM	Guiyu Liu: source models; final revisions and an additional runtime diagram completed during integration.